



WHO INFLUENZA CENTRE LONDON

**Report prepared for the WHO annual consultation
on the composition of influenza vaccine for the
Northern Hemisphere**

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**Northern Hemisphere Vaccine Recommendation Meeting for 2011-12 season.
Summary of NIMR Results
Final report for WHO meeting on 14-16 February 2011.**

Influenza Activity, September 2010 – February 2011.

Influenza activity increased in several countries of Western Europe in the last four weeks of 2010 and showed evidence of West to East spread in terms of within-country intensity and geographic spread of activity (**Figure 1, page 8**). The season has to date been relatively mild compared with the 2009-10 season with peak activity, in terms of percentage of sentinel specimens testing positive for influenza, occurring in week 52/2010 (**Figure 2, page 9**). Unusually, compared with recent previous seasons where the bulk of influenza B circulation has occurred after influenza A circulation has declined, both virus types have co-circulated with similar timings relative to peak activities. Overall, type A viruses have predominated (72% in the WHO EURO Region [WR] and 71% in the ECDC Region [ER]) over type B; of the type A viruses subtyped, pandemic A(H1N1) 2009 viruses have predominated (95.4% WR, 97.4% ER) over A(H3N2) viruses (4.6% WR, 2.6% ER) with there being reports of just two seasonal A(H1N1) detections, one each in the Russian Federation and France (these have not been received for confirmation at the London WHO CC). Between countries there has been considerable variation in the proportions of type A and type B in circulation with A being predominant in most, B in some (e.g. Norway, Ukraine), while some have reported equivalent circulation of both types (e.g. Portugal, Luxembourg) (**Figure 3, page 10**). In a number of the Eastern countries of the WR influenza B dominated at the start of the season but has now been overtaken by pandemic A(H1N1) 2009 viruses. This is illustrated by the situation in the Russian Federation where virus detections by three methods (isolation, IFA, RT-PCR) all show dominance of type B early on, with some A(H3N2) in circulation, that has now been largely overtaken by the circulation of pandemic A(H1N1) 2009 viruses although there are significant differences in the relative proportions of each virus circulating in different regions of the Federation (**Figure 4, page 11**).

To date, disease severity, as related to stresses imposed on within-country health systems, has been relatively mild with the situation in the UK being a notable exception (**Figure 5, page 12**). England, Wales and Scotland have reported higher levels of ILI consultations than in the previous season, while Northern Ireland have reported similarly high numbers- notably so in December 2010 and early January 2011. Within England and Wales the bulk of ILI consultations have been in the 1- to 64-year old age groups; the number of consultations has been higher than the previous three seasons but are now showing declining trends. Based on Q surveillance that covers 20 million people in the UK, pandemic A(H1N1) 2009 viruses have dominated over other influenza viruses in what has been the third wave of A(H1N1) 2009 viruses in the UK, with all three waves showing higher peaks of ILI activity than those observed since 2003. Associated with the third wave, there have been 395 influenza-related deaths reported since week 36/2010. Of the 347 cases where influenza viruses have been typed/subtyped, 20 have been type B, seven A untyped and 320 pandemic A(H1N1) 2009 viruses.

For Europe as a whole, similar proportions of type A and type B influenza viruses (~2.5:1) have been reported to the WR and ER surveillance systems and among type A viruses pandemic A(H1N1) 2009 viruses have predominated over A(H3N2) viruses (~7.8:1), while B-Victoria lineage viruses have predominated over B-Yamagata lineage viruses (~16.5:1) (**Figure 6, page 13**).

Antigenic and genetic characteristics of influenza viruses collected between 1 August 2010 and 31 January 2011.

Table 1 (page 14) lists the clinical specimens and virus isolates received for specimens collected between 1 August 2010 and 31 January 2011. Viruses have been received from Hong Kong, the Middle East (Iran and Iraq), North Africa (Algeria, Egypt, Morocco and Tunisia), Africa (Cameroon, Cote d'Ivoire, Ghana, Madagascar and South Africa), Argentina and from many countries of Eastern Europe and Western Europe. Many viruses are still being propagated and for some of those characterised antigenically, HA and NA sequencing is ongoing. In subsequent antigenic characterisation tables, viruses for which HA and NA sequences are represented in phylogenetic analyses (Figures) are highlighted in green and yellow.

Influenza A(H1N1) Virus analysis.

Tables 2-8 (pages 15-21) show HI results for viruses analysed based on isolates derived from specimens collected between August 2010 and January 2011, mostly in the AFRO, EMRO and EURO regions of the WHO. The vast majority of viruses show good reactivity with the panel of post-infection ferret antisera used, including that raised against the current vaccine virus A/California/7/2009. Viruses with reduced activity (8-fold or greater against the serum raised against the vaccine virus) have been derived from specimens collected in Northern Ireland, Iran, Madagascar, France, Sweden, Latvia, USA, England, Egypt, Georgia and recently received viruses from Germany and Slovenia. These represented 15% of the viruses tested by HI. For viruses whose HA genes have been sequenced to date, the great majority of the viruses showing reduced HI titres carry amino acid substitutions at positions 153-155 in the HA1 protein and the HA genes of these viruses do not cluster in phylogenetic analyses (**Figures 7 & 8, pages 22 & 23**).

Figure 7 (page 22) shows a phylogenetic tree for the HA1-coding region of selected H1N1 viruses. Our September 2010 report highlighted two emerging genetic groups within the HA: the Southern Hemisphere (SH) group (key changes N125D, some with V250A and others) and the Hong Kong group (key changes S128P, V199A, I295V and others – K163T and P271S - not marked in the Figure). Sequence analysis of the viruses characterised in the London CC has revealed only limited geographic spread of viruses in the Hong Kong group but substantial spread of those in the SH group. In addition, three other genetic groups have emerged in recent months characterised by amino acid substitutions: (i) A134T, S183P – now seen in at least six countries; (ii) R205K, I216V, V249L – now seen in at least three countries; (iii) S185T- the group showing the greatest expansion and geographic spread. December isolates, associated with a peak in ILI activity, from the UK reported by HPA Colindale (Eurosurveillance January 7th 2011) and antigenically characterised at the London CC (**Table 8, page 21**) fall within genetic groups (i) and (iii). The substitution D97N is also associated with viruses in groups (ii) and (iii) and some viruses in group (iii) carry a S143G substitution. A further group characterised by amino acid substitutions N31D, S162N, A186T and V272I and represented by viruses from Denmark and South Africa has not expanded in recent months. To date, none of the genetic groups has shown evidence of significant difference in antigenicity (as assessed by HI results) compared with the current vaccine virus. An HA1 amino acid alignment for representative viruses is shown in **Figure 8 (page 23)** and the location of amino acid positions that define specific genetic groups are shown on an H1-HA structure (**Figure 9, page 25**). Some of the substitutions are prominently exposed on the surface of the HA trimer (125, 134, 183, 185, 216) but others are buried.

Considerable heterogeneity can be seen to have accumulated in the phylogenetic analysis of the N1 NA gene (**Figure 10, page 26**). The HA and NA phylogenetic trees are generally congruent: viruses from the S185T cluster in the HA generally group

together into a cluster defined by V241I in the NA. This NA cluster contains a number of sub-groups with additional substitutions. Residue 241 is buried and is not close to the sialic acid binding (active) site. Other clusters are present on the tree but these are generally populated by 'older' viruses with the exception of that represented by viruses from France, Latvia, Spain and Sweden and characterised by amino acid substitutions V264I, N397K and S442I. An NA amino acid alignment for representative viruses is shown in **Figure 11 (page 27)**.

Conclusion: Viruses that show differences in reactivity to post-infection ferret antisera raised against the vaccine virus A/California/7/2009 and against the prototype virus A/California/4/2009 have been detected. The majority of these viruses 25/29 showed good reactivity with sera raised against viruses that have substitutions between 154 and 156 of the HA (A/Bayern/69/2009 and A/Lviv/N6/2009) and it is known that many of the 25 showed similar changes that were not present in the clinical sample and were therefore associated with the substrate for isolation (MDCK-SIAT cells). Four viruses that showed low reactivity were from Slovenia. At the time of preparation of this report, the sequences of the HA and NA genes of these viruses and their corresponding clinical samples were not known.

Influenza A(H3N2) Virus analysis.

Polymorphism in the gene sequence encoding position 151 of the N2 neuraminidase within a virus isolate can result in agglutination of red blood cells through the NA. To circumvent the effect of position 151 polymorphism or any other changes in the NA within a virus isolate affecting the agglutination of red blood cells, virus neutralisation (VN) assays, measured by plaque reduction, have been carried out at the London CC. In addition HA and HI assays have been performed in the presence of oseltamivir to overcome any NA-mediated agglutination of erythrocytes.

Table 1 (page 14) lists the clinical specimens and virus isolates received for specimens collected between 1 August 2010 and 31 January 2011. Over 200 A(H3N2) viruses have been received from more than 20 countries predominantly in Europe but also with viruses from Africa, the Indian Ocean, Asia, Hong Kong and the Middle East. Many viruses have been received very recently and, at the time of preparation of this report, had yet to be propagated and analysed.

The results of virus neutralisation tests are shown in **Tables 9-13 (pages 29-33)**, and for HI tests carried out in the presence of oseltamivir in **Tables 14-16 (pages 34-36)**. The genetic groups of the viruses are also described in the tables. **Table 9 (page 29)** illustrates that both A/Victoria/208/2009 and A/Victoria/210/2009 showed elevated VN titres against all the sera; the corresponding antisera resulted in high titres against the two viruses and these sera did not differentiate A/Brisbane/10/2007 from A/Perth/16/2009 in this VN assay. Also note that the reference virus A/Hong Kong/34430/2009 was not used in the VN assay because after raising antiserum the virus lost viability on culture- it was propagated on MDCK cells and then MDCK-SIAT cells. Only a single virus analysed (A/Netherlands/9/2010) showed notably reduced reactivity with antiserum raised against A/Perth/16/2009.

Table 10 (page 30) presents additional results for antisera that displayed better differentiation between strains. Several test viruses showed low reactivity to some of the sera: several reacted with a lower titre to the A/Victoria/208/2009 antiserum, even accounting for the unusually high homologous titre; however, A/Baleares/RR6848/2010 (Spain) displayed low titre reactivity with the antiserum raised against A/Perth/16/2009. **Tables 11-13 (pages 31-33)** show VN assay results for a further 26 test viruses only two of which, A/Bretagne/2248/2010 and A/Gothenburg/1/2010, show low titre reactivity with the antiserum raised against A/Perth/16/2009.

Tables 14-15 (pages 34 & 35) show results from HI assays using guinea pig erythrocytes while results generated for **Table 16 (page 36)** used human red blood cells. There are several viruses that show low titre reactivity with antiserum raised against A/Perth/16/2009 but higher titres with antisera raised against A/Alabama/5/2010 and A/Rhode Island/1/2010 (X199).

Genetic analysis of the HA and NA genes of viruses are illustrated in **Figures 12 and 15 (pages 37 & 41)** respectively, and amino acid sequence alignments of representative examples are shown in **Figures 13 and 16 (pages 38 & 42)**. Several HA genetic groups, observed previously have been maintained. Viruses still fall into the Perth/16 and Victoria/208 genetic clades with genetic groups emerging within each clade. Within the Perth/16 clade genetic groups can be defined by: key substitutions I260M & R261Q, combined with P162S and also E50K (group B); the key substitutions N133D (loss of glycosylation site), R142G & V213A (group A). Within the Victoria/208 clade genetic groups can be defined by: key substitutions N312S, with sub-groups carrying T48A, K92R; a group defined by S45N (gain of glycosylation site); a group defined by N145S, V223I with N144D (loss of glycosylation site) or V323I; and a large group carrying substitutions D53N, Y94H, I230V, E280A with some viruses within this group additionally carrying S199A. These subgroups are shown on the structures of the HA trimer in **Figure 14 (page 40)**.

For viruses within the Perth/16 clade genetic groups A and B (n=7) all but one virus (A/Hessen/5/2010) showed good inhibition with the serum panel. Within the Victoria/208 clade genetic groups a sizable proportion of viruses showed low reactivity against antiserum raised against A/Perth/16/2009 but fewer did so with antisera raised against A/Wisconsin/15/2009. Viruses in the Victoria/208 clade within the genetic group defined by D53N, Y94H, I230V and E280A generally showed a lower level of reactivity to antiserum raised against A/Perth/16/2009 but had higher HI titres with the antisera raised against either A/Wisconsin/15/2009 or A/Alabama/5/2010 or A/Rhode Island/1/2010 (X199).

There were several viruses that showed low titre reactivity in the HI assay carried out on human red blood cells for which, at the time of preparation of this report, sequence data were not available.

The amino acid substitutions within each clade (**Figure 14, page 40**) and genetic group are located on the surface of the HA trimer in many cases, notably so in the Victoria/208 clade genetic group defined by D53N, Y94H, I230V, E280A and S199A.

The NA gene also shows evidence of genetic change with emergence of virus groups. As noted in the previous meeting the NA gene can be divided, like the HA, into three main groups: Perth/16 clade groups A and B, and the Victoria/208 clade (**Figure 15, page 41**). An NA amino acid sequence alignment for representative viruses is shown in **Figure 16 (page 42)**. Amino acid substitutions have been marked on the neuraminidase monomer and tetramer (**Figure 17, page 44**). No antigenic analysis of the NA has been carried out. The majority of the amino acid substitutions that define the genetic groups are on the surface of the NA tetramer but away from the sialic acid binding site located in the space between residues 119 and 274. The NA gene and the HA gene of viruses show infrequent gene reassortment.

Conclusion: The antigenic analysis of viruses in the Victoria/208 clade is difficult to clearly interpret. Several show reduced or low reactivity with antiserum raised against the prototype virus and the vaccine virus A/Perth/16/2009, and yield better titres with antisera raised against viruses isolated more recently. However, this is not a consistent pattern seen within viruses of the same genetic groups. Possible differences in the dynamics of the assays carried out in the presence of oseltamivir needs to be considered. Results from neutralisation assays lead to similar conclusions to those from

HI; these results indicate that recently collected viruses show better reactivity with antisera raised against Victoria/208 clade viruses that have circulated more recently than the A/Perth/16/2009 vaccine virus.

Influenza B Virus Analyses.

The vast majority of influenza B viruses were from the B/Victoria lineage (88%) and viruses of both lineages have been predominantly, but not exclusively, received from countries in Europe. Viruses were also received from Africa, the Middle East, the Indian Ocean and Asia.

B/Victoria lineage viruses.

The results of antigenic analysis of influenza B/Victoria lineage viruses by HI assay using turkey red blood cells are shown in **Tables 17-20 (pages 45-48)**. As expected the majority of viruses showed low reactivity with sera raised against viruses propagated in hens' eggs, although several examples of test viruses propagated in hens' eggs can be seen on **Table 19 (page 47)**; these viruses showed an inconsistent pattern of HI which was not associated with substitutions at the N-linked glycosylation site located at residues 197-199 of HA1, loss of which is associated with propagation in hens' eggs. **Table 19 (page 47)** also shows several cell-propagated viruses that have good levels of reactivity with antisera raised against the egg-propagated viruses but not associated with the loss of glycosylation at 197 for viruses in which the HA gene sequences were determined. Some of this group of viruses show unusually low reactivity to antisera raised against more recently isolated viruses but the sequence of none of the HA or NA genes has been carried out for this minor set of viruses. All cell grown viruses analysed in the test of 09/02/2011 (**Table 20, page 48**) consistently showed low reactivity against the antiserum raised against egg-grown B/Brisbane/60/2009 and better reactivity against those antisera raised against more recently isolated cell grown viruses; sequencing of representative viruses from this set has yet to be performed.

Two viruses, B/England/81/2010 and B/England/121/2010 (**Table 17, page 45**), showed low reactivity with all antisera raised against the most recent reference strains. These two viruses were exceptional: all other viruses reacted well with sera raised against cell-propagated viruses (e.g. sera raised against B/Paris/1762/2008, B/Hong Kong 514/2009 and B/Odessa/3886/2010) that are closely related genetically and show close amino acid identity to the vaccine virus (B/Brisbane/60/2008).

A phylogenetic tree of influenza B Victoria-lineage HA genes is shown in **Figure 18 (page 49)** and an HA1 amino acid sequence alignment of representative viruses is shown in **Figure 19 (page 50)**. The phylogenetic tree illustrates that the vast majority of viruses belong to the Brisbane/60 genetic clade. Most recently collected viruses carry the amino acid substitution I146V and a significant number also carry the substitution L58P. B/England/81/2010 and B/England/121/2010, the viruses that showed low reactivity with all antisera raised against the most recent reference strains, can be seen to group into two clades at the bottom of the phylogenetic tree- one in clade 4 and one in clade 5, both of which are genetically more closely related to the B/Malaysia/2506/2004 prototype virus.

A phylogenetic tree of influenza B/Victoria lineage NA genes is shown in **Figure 20 (page 52)** and an NA amino acid sequence alignment for representative viruses is shown in **Figure 21 (page 53)**. The phylogenetic tree illustrates that the vast majority of viruses belong to the Brisbane/60 genetic clade. There are numerous amino acid substitutions mapped on the NA phylogenetic tree but the significance of these is unclear. No evidence for reassortment between clades of the influenza B/Victoria lineage is apparent from the HA and NA phylogenetic trees (**Figures 18 & 20, pages 49 & 52**).

B/Yamagata lineage viruses.

Less than 30 influenza B/Yamagata lineage viruses have been received, in small numbers, from Europe, Hong Kong, Madagascar, Morocco and Ghana (**Table 1, page 14**).

The results of antigenic analysis of influenza B/Yamagata lineage viruses by HI assay using turkey red blood cells are shown in **Tables 21-24 (pages 55-58)**. All test viruses were propagated in tissue culture cells. Most viruses showed good reactivity with antisera raised against viruses of the B/Bangladesh/3333/2007 clade (these include B/Algeria/G-486/2010 and B/Wisconsin/1/2010) but some viruses have shown lowered reactivity. Two viruses from Hong Kong (B/Hong Kong/1832/2010 and B/Hong Kong/1895/2010) showed markedly reduced titres against the panel of sera shown in **Table 21 (page 55)**; B/Niedersachsen/2/2010 similarly in **Table 22 (page 56)**; and three viruses from Finland (B/Finland/33/2010, B/Finland/34/2010 and B/Finland/39/2010) in **Table 23 (page 57)**. All test viruses in **Table 24 (page 58)** showed good reactivity with the panel of antisera used, notably so for the new antiserum raised against B/Niedersachsen/1/2010.

A phylogenetic tree of influenza B/Yamagata lineage HA genes is shown in **Figure 22 (page 59)** and an HA1 amino acid sequence alignment for representative viruses is shown in **Figure 23 (page 60)**. The phylogenetic tree showed that all viruses fell into the B/Bangladesh/3333/2007 genetic clade. The viruses that displayed unusual behaviour in the HI assays, for which sequence was available at the time of writing this report, did not cluster in the phylogenetic tree and they showed few amino acid substitutions from the reference viruses (2 or 3 substitutions at residues 121, 183, 202, 251). When this report was written, no nucleotide sequences were available for the group of viruses from Finland that showed low reactivity.

A phylogenetic tree of influenza B/Yamagata lineage NA genes is shown in **Figure 24 (page 62)** and an NA amino acid sequence alignment for representative viruses is shown in **Figure 25 (page 63)**. The NA phylogenetic tree seems largely congruent with the HA phylogenetic tree with few amino acid substitutions marking shallow nodes; this contrasts with the pattern seen in the NA genes of the B/Victoria lineage viruses (**Figure 20, page 52**).

Conclusion: Influenza B/Victoria lineage viruses have predominated. Genetically the vast majority of viruses of the B/Victoria lineage were from the B/Brisbane/60/2008 clade. Generally, the viruses from this clade showed good titres with antisera raised against cell-propagated reference viruses genetically closely related to the vaccine strain but a proportion of viruses (~2%) showed reduced titres against a more recently isolated reference virus propagated in mammalian cell culture.

Antiviral resistance testing.

Antiviral resistance testing was performed using a sialidase inhibition assay with MUNANA as substrate and zanamivir and oseltamivir as inhibitors. A summary of the viruses tested is shown in **Table 25 (page 65)**. Seasonal influenza A(H3N2) and influenza B viruses from ECDC-affiliated countries in Europe are tested by the HPA Colindale Laboratory as a long standing agreement within EISN and ECDC; none of these viruses have shown resistance to either drug.

No altered sensitivity of A(H3N2) viruses or influenza B viruses to zanamivir or oseltamivir was observed but the numbers tested by us were low (~70). For the A(H1N1)2009 viruses, 185 have been propagated to sufficient titre and assayed for resistance to zanamivir and oseltamivir. Two viruses from Serbia were sensitive to

oseltamivir but had reduced sensitivity to zanamivir (IC_{50} s of 6.8nM & 3.7nM) which correlated with G147R substitution in their neuraminidases.

Full NA gene sequencing for influenza A and B viruses have confirmed the lack/low levels of neuraminidase inhibitor resistance markers among circulating viruses. In terms of resistance to amantadine and rimantadine, M-gene sequencing has shown all pandemic A(H1N1) 2009 and A(H3N2) viruses analysed to be resistant due to S31N substitution in the M-2 ion channel protein.

Conclusion: Very few viruses resistant to oseltamivir and/or zanamivir have been detected in circulating viruses. Some viruses with a level of decreased susceptibility to the antiviral compounds have been found but the significance of these variant viruses is unclear.

Antigenic and Genetic Characterisation of Viruses recovered from Zoonotic Infections.

Two zoonotic infections, the result of the transfer of virus from pigs to humans, were detected in Switzerland. The viruses (A/Switzerland/2649356/2009 and A/Switzerland/3885165/2010) were analysed genetically and antigenically. Full genome analysis of the two viruses has been carried out and a phylogenetic tree of the HA gene is shown in **Figure 26 (page 66)**. All genes of the virus were closely related to the avian-lineage swine influenza viruses commonly found in European pigs since their initial transfer from the avian pool to pigs at some period prior to 1979. The most closely related viruses were collected in recent years from infected pigs in Europe (this is unlike the H1N1 human infection in China which is more closely related to avian-lineage viruses collected from pigs in Asia).

Antigenic analysis was carried out with post-infection ferret antisera to a panel of avian-lineage swine influenza viruses and turkey erythrocytes in an HI assay (**Table 26, page 67**). The reference panel of sera and viruses showed a variable pattern of cross inhibition by the ferret sera but it was apparent that the viruses from the zoonotic infections showed good reactivity with serum from viruses collected in Western France (A/swine/Côtes-d'Armor/324/2007) and Switzerland (A/swine/Switzerland/7492/2005), viruses closely related genetically to the viruses from the two recent zoonotic transfers.

No H5N1 viruses, H9N2 viruses or other animal-origin viruses have been received at the London CC over the past 6 months.

Conclusion: Despite the presence of A(H1N1) 2009 viruses in pigs, the risk of zoonotic infection from established swine influenza viruses remains.

Figure 1. 2010-2011 Influenza Season WHO Euro Region: Intensity and Geographic Spread by Week.

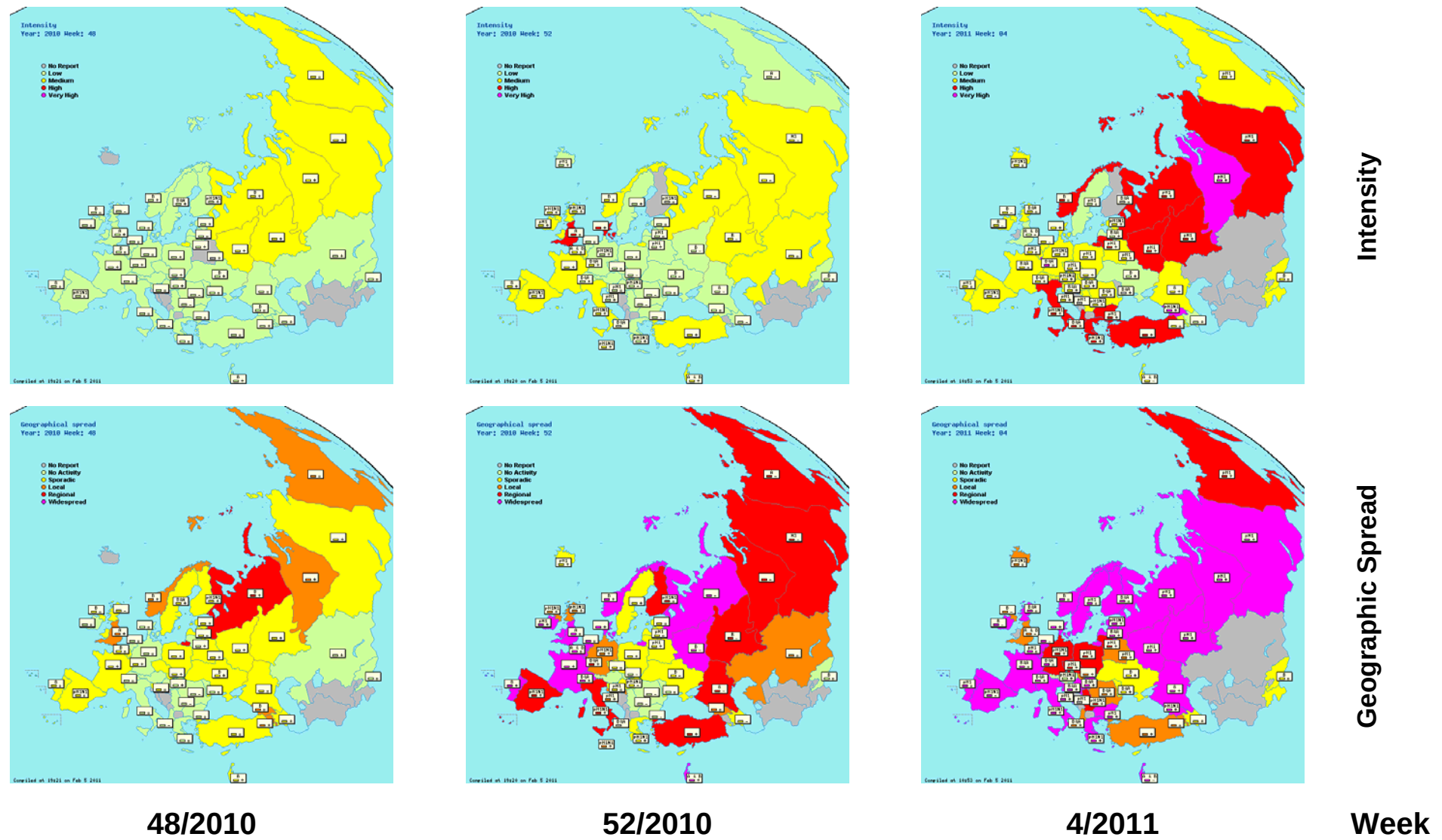
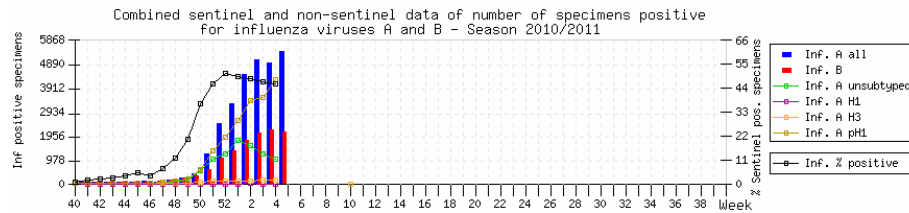
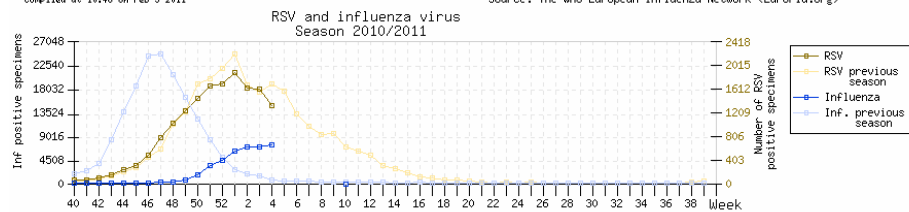


Figure 2. 2010-2011 Influenza Season: Detections/Characterisations - WHO Euro & ECDC Regions.

Combined Sentinel and Non-Sentinel Influenza Detections



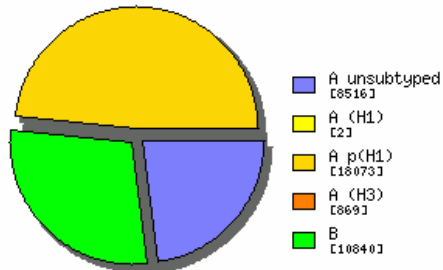
Source: The WHO European Influenza Network (EuroFlu.org)



Source: The WHO European Influenza Network (EuroFlu.org)

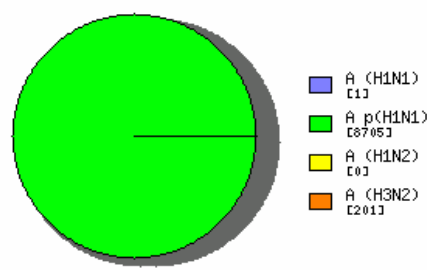
Cumulative influenza virus typings and subtypings* Season 2010/2011

Total number - [N = 38300]
(includes N-subtyped and N-unsubtyped)



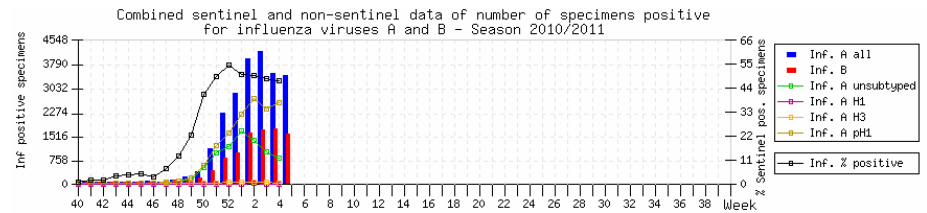
Compiled at 10:48 on Feb 5 2011

Number of N-subtyped - [N = 8907]

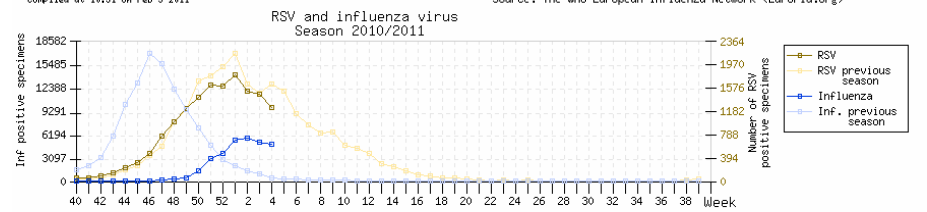


* Sentinel and non-sentinel respiratory specimens combined

WHO Euro Region



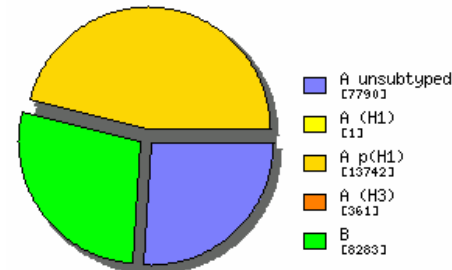
Source: The WHO European Influenza Network (EuroFlu.org)



Source: The WHO European Influenza Network (EuroFlu.org)

Cumulative influenza virus typings and subtypings* Season 2010/2011

Total number - [N = 30177]
(includes N-subtyped and N-unsubtyped)



Compiled at 10:51 on Feb 5 2011

Number of N-subtyped - [N = 8691]



* Sentinel and non-sentinel respiratory specimens combined

ECDC EU/EAA Countries

Figure 3. 2010-2011 Influenza Season: Detections/Characterisations – Individual Countries.

Combined Sentinel and Non-Sentinel Influenza Detections

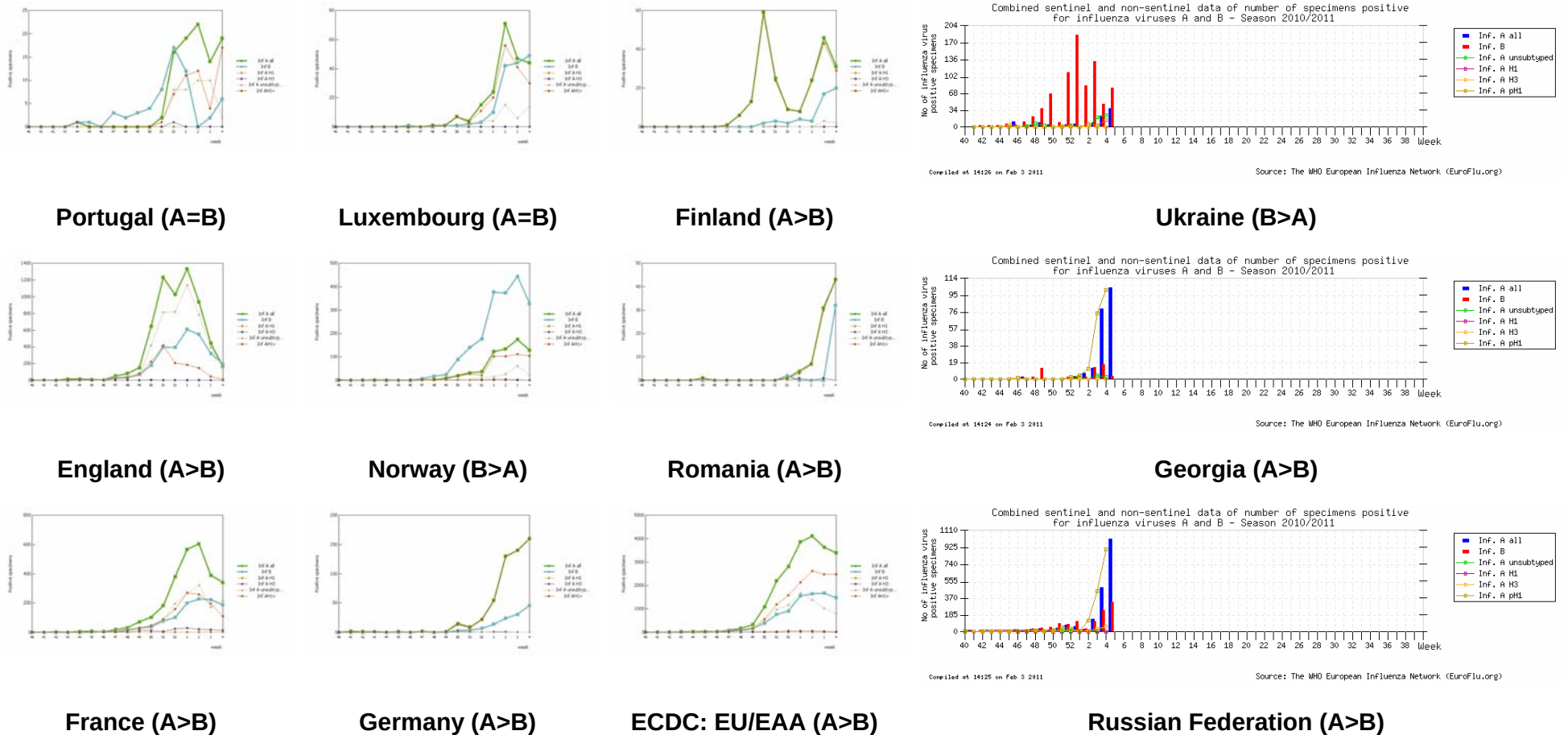


Figure 4. 2010-2011 Influenza Season Russian Federation: Regional variation (type/subtype).

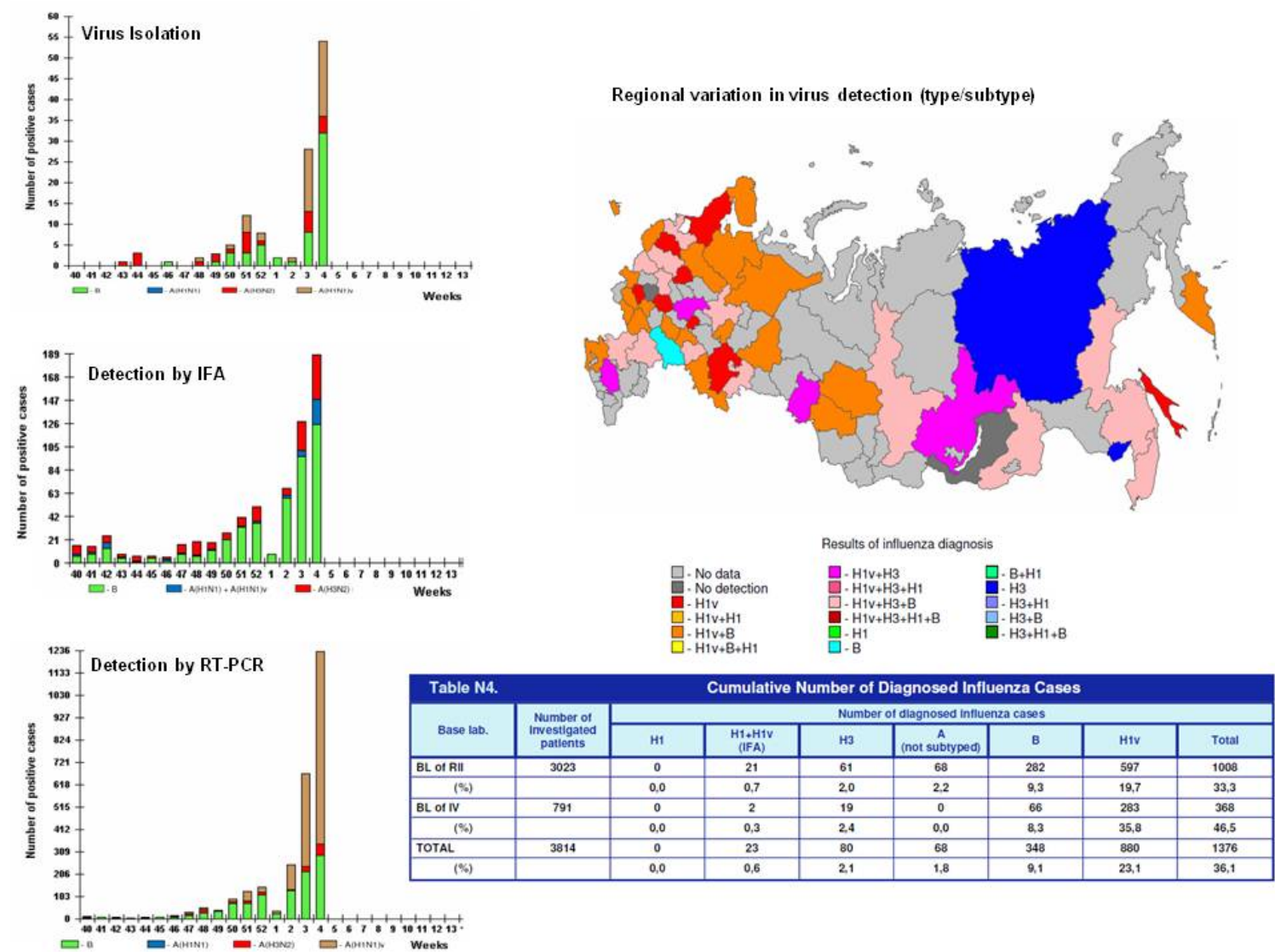
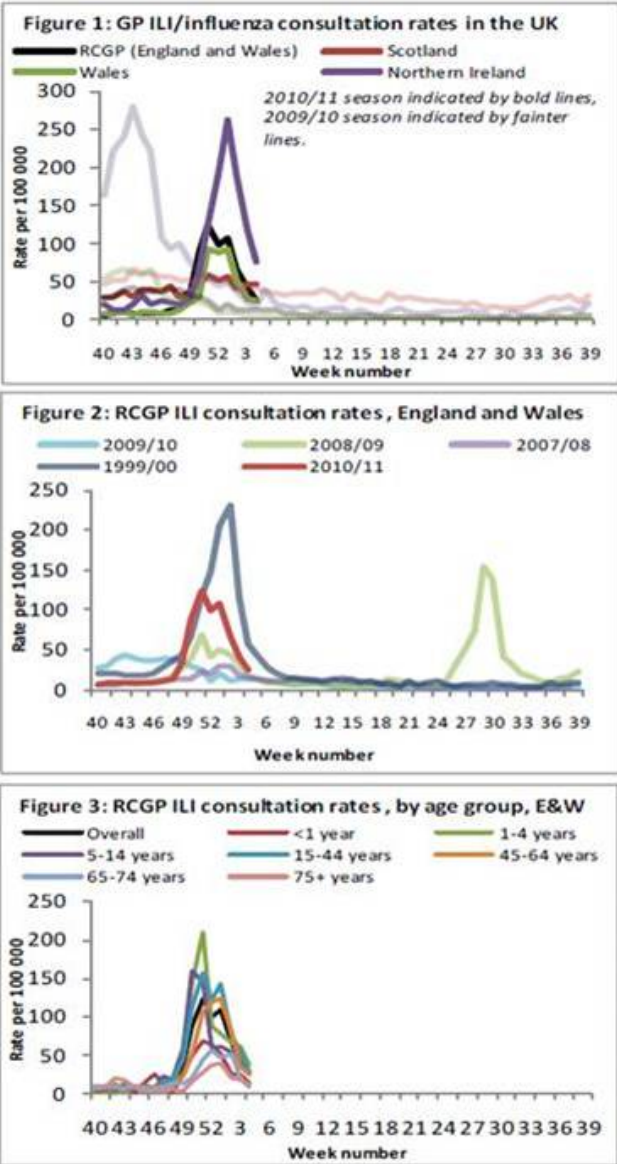


Figure 5. 2010-2011 Influenza Season in the UK.



Information from
HPA and RCGP
weekly reports

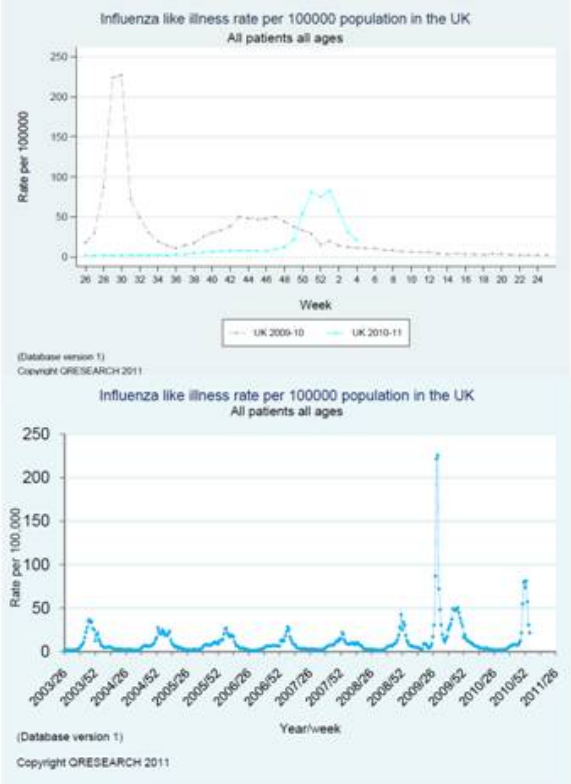


Table 1: Sentinel virological surveillance in the UK

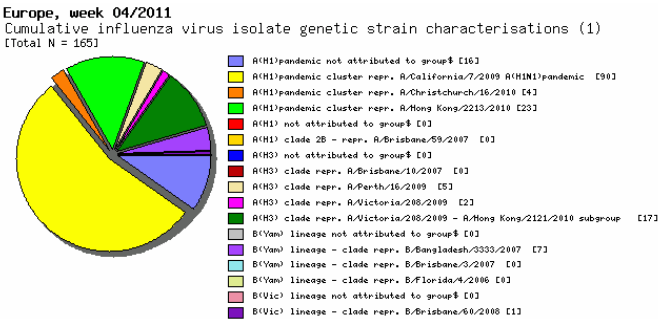
Week	England	Scotland	Northern Ireland	Wales
52	233/368 (63.3%)	130/186 (69.9%)	16/21 (76.2%)	14/19 (73.7%)
1	181/442 (41%)	99/181 (54.7%)	25/52 (48.1%)	15/22 (68.2%)
2	146/452 (32.3%)	104/242 (43%)	17/40 (42.5%)	7/21 (33.3%)
3	63/290 (21.7%)	68/165 (41.2%)	17/33 (51.5%)	3/8 (-)
4	28/128 (21.9%)	20/45 (44.4%)	8/19 (42.1%)	1/7 (-)

NB. Proportion positive omitted when fewer than 10 specimens tested

Since wk36/2010 to Feb 02/2011: 395 influenza-related deaths (281 England).
347 typed/subtyped: 320 H1pdm, 20 FluB, 7 A untyped.
Mainly middle-aged and younger adults.
Majority in CMO-defined risk groups for vaccination – but low uptake.

Figure 6. 2010-2011 Influenza Season WHO Euro Region: Antigenic & Genetic Characterisation.

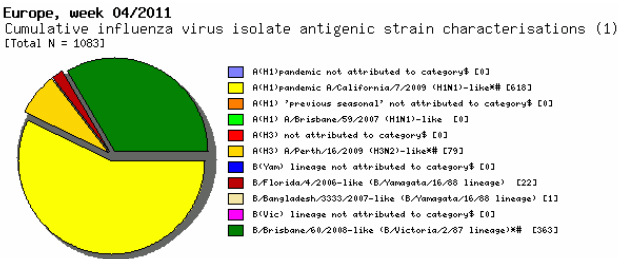
Combined Sentinel and Non-Sentinel Influenza Detections



(1) Sentinel and non-sentinel specimens combined

Characterisations are based on the hemagglutinin gene (genetic characterisation).
Influenza A virus isolates include both neuraminidase not subtyped and neuraminidase subtyped isolates.

Compiled at 10:56 on Feb 5 2011



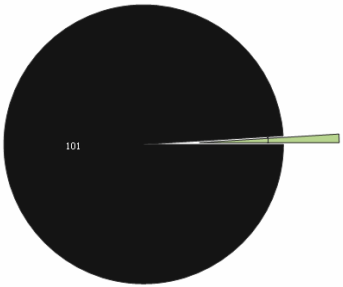
(1) Sentinel and non-sentinel specimens combined

Characterisations are based on the hemagglutinin protein (antigenic characterisations).
Influenza A virus isolates include both neuraminidase not subtyped and neuraminidase subtyped isolates.

Compiled at 10:56 on Feb 5 2011

Typing

38300	No	30407
72%	A	71%
28%	B	29%



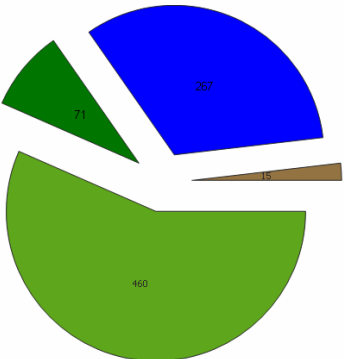
Legend:

- A(H1)v California/7/2009-like
- A(H1) A/Brisbane/59/2007-like
- A(H1) A/Hong Kong/2652/2006-like
- A(H3) A/Brisbane/10/2007-like
- B(Yam) B/Brisbane/3/2007-like
- B(Yam) B/Florida/4/2006-like
- B(Vic) B/England/393/2008-like

Genetic
Characterisation

Subtyping

1083	No	1009
57.1%	H1p	56.5%
7.3%	H3	7.2%
33.5%	BVic	34.3%
2.1%	BYam	2.0%
2	H1	1



Legend:

- A(H1)v California/7/2009-like
- A(H3) A/Brisbane/10/2007 (H3N2)-like
- A(H3) A/Perth/16/2009 (H3N2)-like
- B/Brisbane/60/2008-like (B/Victoria/2/07 lineage)
- B/Florida/4/2006-like (B/Yamagata/16/98 lineage)

Antigenic
Characterisation

WHO Euro

ECDC

Table 1. Summary of clinical samples and isolates received, collected between August 2010 and January 2011

MONTH	A Untyped	H1N1pdm		H3N2		B Untyped	B Yamagata lineage		B Victoria lineage	
Country		Number received	A/California/7/2009-like	Number received	A/Perth/16/2009-like*		Number received	B/Florida/4/2006-like	Number received	B/Brisbane/60/2008-like
AUGUST										
Egypt	5	1	1							
Ghana		21	8							
Hong Kong				8	8					
Netherlands									1	1
South Africa		5	5						3	3
Switzerland		2	2							
United Kingdom				1	0					
SEPTEMBER										
Ghana		4	3	3	in process					
Madagascar		3	2	5	in process				1	1
Reunion		2	2							
Sweden				2	2					
United Kingdom				2	1					
OCTOBER										
Argentina				1	in process					
Belgium				2	in process					
Cote d'Ivoire		10	in process	10	in process	1				
France		1	1			25				
Germany				2	in process		1	1	4	4
Ghana				7	in process					
Greece				1	in process					
Madagascar				2	in process					
Norway		1	1	1	in process					
Portugal									4	0
Spain				1	in process					
Switzerland									2	1
United Kingdom		2	2	1	in process				2	2
NOVEMBER										
Argentina				9	in process					
Belgium	3	1	in process			4			2	2
Cote d'Ivoire		2	in process	23	in process	6				
Denmark				1	1					
Egypt	3	1	1						2	2
Finland		1	1				1	1		
France		2	2	6	6		1	1	5	5
Germany		1	1							
Ghana				12	5					
Hong Kong		1	in process	3	in process		2	2		
Iran							1	1	1	1
Italy		3	3						3	2
Madagascar	2	3	3	12	in process				4	4
Norway		1	1				3	3		
Portugal				1	in process				16	2
Slovenia									1	1
Spain		5	5						1	0
Sweden		2	2	2	2		1	1	1	1
United Kingdom		10	10	2	in process		1	1	8	8
DECEMBER										
Austria		6	4							
Belgium	3	20	14	2	1	12	3	3	15	15
Croatia		8	in process							
Denmark		5	in process							
Egypt	1	14	14							
Finland		2	in process	1	1		2	2		
France		20	12	19	11		2	2	9	9
Georgia		1	1							
Germany		8	8	3	in process		4	4	7	7
Ghana									1	1
Hong Kong		4	2	4	in process				2	2
Iraq		2	in process							
Iran		11	8	1	in process					
Ireland		10	in process			1			1	in process
Italy		3	in process	1	in process	4			2	1
Latvia		5	5						2	2
Luxembourg		5	3						1	1
Madagascar		1	0	7	in process				8	6
Morocco		5	1	1	1				5	1
Netherlands				1	1					
Norway		1	1		1				7	6
Portugal		1	0						14	8
Romania		1	in process	2	2					
Slovenia		6	5						2	2
Spain		14	in process			2			2	2
Sweden		6	6						4	in process
Switzerland	2	7	in process			1			1	in process
United Kingdom		14	12	1	1		3	3	8	8
JANUARY										
Belgium	2					1			1	1
Germany									2	1
France							1	1	3	3
Georgia	6	6	6			2			5	5
Greece	1	28	in process	2	1	3				
Croatia		2	0							
Iraq	2	6	in process	1	in process					
Ireland				2	in process					
Italy		8	in process	1	in process	3				
Latvia	1	3	3							
Morocco		1	0	1	1				2	1
Netherlands		1	in process	1	1	1				
Romania		2	1	1	in process					
Slovenia		14	14						3	3
Spain		14	in process			1				
Sweden				1	1					
Switzerland		20	in process						2	in process
United Kingdom		1	in process							
UNKNOWN										
Received after 01/09/2010										
Argentina				24	in process				15	10
Belgium		1	1							
France		1	1							
Reunion		2	2	1	in process				1	1
South Africa		2	1							
Switzerland		6	2							
United Kingdom				2	in process					
Total Received = 893	31	382	182	201	48	67	26	26	186	136

* Although the bulk of these viruses have been isolated, based on NA activity, due to problems related to Oseltamivir sensitivity of red blood cell agglutination, limited HI data has been generated. A portion of the viruses have been assessed by plaque reduction neutralisation and sequencing is ongoing.

Table 2 . Antigenic analyses of pandemic influenza A(H1N1) viruses (22/12/2010)

			Haemagglutination inhibition titre ¹								
			Post infection ferret sera								
Viruses	Collection date	Passage History	A/Cal 4/09 C4/F14/09	A/Cal 7/09 F16/09	A/Eng 195/09 F06/10	A/Auck 3/09 C4/17/09	A/Bayern 69/09 C4/33/09	A/Lviv N6/2009 C4/34/09	A/HK 2212/2010 F18/10	A/HK 2212/2010 F21/10 Egg	A/HK 2200/2010 F22/10 Egg
REFERENCE VIRUSES											
A/California/4/2009	2009-04-01	C1,E3	2560	1280	5120	2560	1280	2560	2560	2560	2560
A/California/7/2009	2009-04-09	E2/E6	1280	640	640	640	640	1280	640	640	1280
A/England/195/2009	2009-04-28	MDCK1/MDCK4	1280	1280	2560	2560	1280	1280	1280	1280	2560
A/Auckland/3/2009	2009-04-25	Ex/E3	2560	2560	2560	2560	1280	2560	2560	1280	2560
A/Bayern/69/2009	2009-07-01	MDCK4/SIAT1	40	80	80	80	320	320	80	160	160
A/Lviv/N6/2009	2009-10-27	MDCK4/SIAT2	320	160	160	160	640	1280	160	640	320
A/Hong Kong/2212/2010	2010-07-16	MDCK2/MDCK4	1280	1280	1280	1280	640	640	1280	1280	1280
A/Hong Kong/2212/2010	2010-07-16	E3	2560	2560	5120	5120	1280	2560	5120	5120	5120
A/Hong Kong/2200/2010	2010-07-14	E3	1280	2560	2560	2560	1280	2560	2560	2560	5120
TEST VIRUSES											
A/La Reunion/823/2010		MDCK2/MDCK1	2560	5120	5120	5120	1280	2560	2560	ND	ND
A/La Reunion/838/2010		MDCK2/MDCK1	1280	1280	2560	2560	640	1280	1280	ND	ND
A/Christchurch/16/2010	2010-07-12	E2/E1	1280	2560	2560	2560	640	1280	1280	2560	2560
A/Johannesburg/116/2010	2010-08-10	MDCK1/MDCK2	1280	1280	2560	2560	640	1280	ND	ND	ND
A/Johannesburg/117/2010	2010-08-12	MDCK1/MDCK2	1280	1280	1280	1280	640	1280	ND	ND	ND
A/Johannesburg/119/2010	2010-08-19	MDCK1/MDCK2	2560	2560	2560	2560	1280	2560	ND	ND	ND
A/Switzerland/3628801/2010	2010-08-19	MDCK2	640	640	640	640	320	320	320	640	320
A/Switzerland/3628824/2010	2010-08-19	MDCK2	640	640	640	1280	320	640	640	1280	1280
A/Ghana/FS4228/2010	2010-08-25	SIAT1/MDCK1	2560	1280	1280	2560	640	1280	1280	ND	ND
A/Ghana/FS4241/2010	2010-08-25	SIAT1/MDCK2	640	1280	640	1280	640	640	640	ND	ND
A/Ghana/FS4249/2010	2010-08-26	SIAT1/MDCK2	1280	1280	1280	1280	640	1280	640	ND	ND
A/Ghana/FS4259/2010	2010-08-27	SIAT1/MDCK2	640	320	320	320	640	640	320	ND	ND
A/La Reunion/803/2010	2010-09-16	MDCK4/MDCK1	1280	1280	1280	640	640	1280	1280	ND	ND
A/La Reunion/815/2010	2010-09-24	MDCK2/MDCK2	2560	1280	2560	2560	1280	2560	2560	ND	ND
A/Lyon-CHU/43.28/2010	2010-10-23	MDCK2/MDCK1	640	1280	1280	1280	640	640	640	ND	ND
A/Northern Ireland/1/2010	2010-10-29	M2/MDCK1	80	80	80	80	320	1280	160	160	160
A/England/118/2010	2010-11-19	S1/MDCK1	640	640	640	1280	640	640	640	ND	ND

1. = starting serum dilution 1/40

Vaccine virus

Sequences included in HA phylogeny

Table 3 . Antigenic analyses of pandemic influenza A(H1N1) viruses (05/01/2011)

			Haemagglutination inhibition titre ¹									
			Post infection ferret sera									
Viruses	Collection date	Passage History	A/Cal 4/09 C4/F14/09	A/Cal 7/09 F05/10	A/Eng 195/09 F06/10	A/Auck 3/09 C4/17/09	A/Bayern 69/09 C4/33/09	A/Lviv N6/2009 C4/34/09	A/HK 2212/2010 F18/10	A/HK 2212/2010 F21/10 Egg	A/HK 2200/2010 F22/10 Egg	A/Chch 16/2010 F30/10
REFERENCE VIRUSES												
A/California/4/2009	2009-04-01	C1,E3	2560	1280	2560	2560	1280	2560	2560	2560	2560	1280
A/California/7/2009	2009-04-09	E2/E6	1280	1280	1280	1280	640	2560	1280	1280	1280	1280
A/England/195/2009	2009-04-28	MDCK1/MDCK4	1280	2560	5120	5120	1280	5120	2560	5120	5120	2560
A/Auckland/3/2009	2009-04-25	Ex/E3	1280	2560	2560	5120	1280	5120	2560	2560	2560	2560
A/Bayern/69/2009	2009-07-01	MDCK4/SIAT1	80	80	80	80	320	320	80	160	80	160
A/Lviv/N6/2009	2009-10-27	MDCK4/SIAT2	320	160	160	160	1280	1280	320	320	320	320
A/Hong Kong/2212/2010	2010-07-16	MDCK2/MDCK4	1280	1280	2560	2560	640	2560	2560	2560	2560	2560
A/Hong Kong/2212/2010	2010-07-16	E3	2560	2560	5120	5120	2560	5120	5120	5120	5120	5120
A/Hong Kong/2200/2010	2010-07-14	E3	2560	2560	2560	2560	1280	5120	2560	5120	5120	2560
A/Christchurch/16/2010	2010-07-12	E2/E1	2560	2560	2560	2560	2560	5120	2560	2560	5120	5120
TEST VIRUSES												
A/Ghana/FS4325/2010	2010-08-30	SIAT1/MDCK1	1280	1280	2560	2560	1280	1280	2560	2560	1280	1280
A/Ghana/FS4400/2010	2010-09-08	SIAT1/MDCK1	1280	1280	2560	2560	640	1280	1280	2560	1280	1280
A/Ghana/FS4466/2010	2010-09-28	SIAT1/MDCK1	1280	2560	2560	2560	1280	1280	2560	2560	2560	1280
A/Navarra/RR6915/2010	2010-11-04	SIAT1/MDCK1	640	1280	640	1280	640	1280	1280	1280	1280	5120
A/Navarra/RR6914/2010	2010-11-11	SIAT1/MDCK1	1280	1280	1280	1280	1280	1280	1280	2560	1280	5120
A/Madrid/SO8034/2010	2010-11-18	SIAT2/MDCK1	1280	1280	1280	1280	2560	2560	2560	1280	2560	1280
A/England/127/2010	2010-11-22	SIAT1/MDCK2	640	1280	1280	1280	640	640	1280	640	1280	640
A/England/143/2010	2010-11-23	SIAT1/MDCK3	640	640	1280	640	1280	2560	640	1280	640	640
A/England/151/2010	2010-11-23	SIAT1/MDCK2	640	1280	1280	1280	640	1280	1280	1280	1280	640
A/Milano/3/2010	2010-11-24	MDCK1/MDCK1	1280	1280	1280	1280	1280	1280	2560	2560	1280	5120
A/Perugia/1/2010	2010-11-24	MDCK3/MDCK1	2560	2560	5120	2560	1280	2560	5120	2560	2560	2560
A/England/119/2010	2010-11-25	SIAT1/MDCK1	1280	2560	2560	2560	640	1280	1280	2560	2560	1280
A/England/142/2010	2010-11-26	SIAT1/MDCK4	640	640	640	640	640	1280	640	1280	640	640
A/Milano/4/2010	2010-11-29	MDCK2/MDCK1	640	1280	1280	1280	1280	1280	2560	2560	1280	5120
A/Pais Vasco/RR6909/2010	2010-12-01	SIAT1/MDCK1	1280	1280	1280	1280	640	1280	1280	1280	1280	1280

1. = starting serum dilution 1/40

Vaccine virus

Sequences included in HA phylogeny

Table 4 . Antigenic analyses of pandemic influenza A(H1N1) viruses (13/01/2011)

			Haemagglutination inhibition titre ¹						
			Post infection ferret sera						
Viruses	Collection date	Passage History	A/Cal 7/09 F05/10	A/Eng 195/09 F06/10	A/Auck 3/09 C4/17/09	A/Bayern 69/09 C4/33/09	A/Lviv N6/2009 C4/34/09	A/HK 2212/2010 F21/10 Egg	A/Chch 16/2010 F30/10
REFERENCE VIRUSES									
A/California/7/2009	2009-04-09	E2/E5	2560	2560	2560	1280	2560	5120	1280
A/England/195/2009	2009-04-28	MDCK1/MDCK4	2560	2560	2560	640	1280	2560	1280
A/Auckland/3/2009	2009-04-25	Ex/E3	2560	2560	2560	2560	2560	5120	1280
A/Bayern/69/2009	2009-07-01	MDCK4/SIAT1	80	80	80	320	320	160	80
A/Lviv/N6/2009	2009-10-27	MDCK5	160	160	160	1280	640	640	320
A/Hong Kong/2212/2010	2010-07-16	E3	5120	2560	2560	2560	2560	5120	2560
A/Christchurch/16/2010	2010-07-12	E2/E1	2560	2560	2560	2560	2560	5120	5120
TEST VIRUSES									
A/Ghana/FS4240/2010	2010-08-27	MDCK3	1280	1280	2560	320	640	640	640
A/Ghana/FS4293/2010	2010-08-30	SIAT1/MDCK2	640	1280	1280	320	320	1280	640
A/Ghana/FS4395/2010	2010-09-01	SIAT1/MDCK3	1280	2560	2560	640	640	2560	1280
A/Madrid/RR6872/2010	2010-11-17	SIAT2/MDCK3	640	1280	1280	320	320	1280	1280
A/Madagascar/7178/2010	2010-11-19	MDCK1/MDCK1	1280	2560	2560	640	1280	1280	640
A/Madrid/SO8057/2010	2010-11-24	SIAT2/MDCK3	640	1280	640	320	320	1280	640
A/Tehran/5336/2010	2010-12-01	MDCK2/MDCK1	2560	1280	2560	1280	1280	2560	1280
A/Tehran/5370/2010	2010-12-01	MDCK2/MDCK1	640	640	1280	1280	1280	1280	640
A/Tehran/5652/2010	2010-12-01	MDCK3/MDCK1	160	320	640	1280	1280	640	640
A/Tehran/5660/2010	2010-12-01	MDCK3/MDCK1	2560	2560	5120	1280	2560	2560	1280
A/Tehran/5853/2010	2010-12-01	MDCK1/MDCK1	2560	1280	2560	1280	1280	2560	1280
A/Tehran/5858/2010	2010-12-01	MDCK1/MDCK1	640	1280	640	320	640	640	320
A/Tehran/5987/2010	2010-12-01	MDCK1/MDCK1	2560	2560	2560	1280	2560	2560	1280
A/Alborz/5607/2010	2010-12-01	E2/E1	1280	2560	2560	640	1280	1280	1280
A/Valladolid/53/2010	2010-12-10	MDCK1/MDCK2	1280	2560	2560	640	640	2560	1280
A/Luxembourg/548/2010	2010-12-28	C1/MDCK1	2560	2560	2560	640	1280	2560	1280
A/Luxembourg/562/2010	2010-12-29	C1/MDCK1	1280	2560	1280	640	2560	1280	640
A/Luxembourg/567/2010	2010-12-30	C1/MDCK1	2560	2560	2560	1280	2560	2560	2560

1. = starting serum dilution 1/40

Vaccine virus

Sequences included in HA phylogeny

Table 5. Antigenic analyses of pandemic influenza A(H1N1) viruses (18/01/2011)

			Haemagglutination inhibition titre ¹						
			Post infection ferret sera						
Viruses	Collection date	Passage History	A/Cal 7/09 F16/09	A/Eng 195/09 F06/10	A/Auck 3/09 F29/10	A/Bayern 69/09 C4/33/09	A/Lviv N6/2009 C4/34/09	A/HK 2212/2010 F21/10 Egg	A/Chch 16/2010 F30/10
REFERENCE VIRUSES									
A/California/7/2009	2009-04-09	E2/E5	1280	2560	2560	1280	2560	1280	1280
A/England/195/2009	2009-04-28	MDCK1/MDCK4	2560	2560	2560	1280	1280	2560	1280
A/Auckland/3/2009	2009-04-25	Ex/E3	2560	2560	2560	1280	2560	2560	1280
A/Bayern/69/2009	2009-07-01	MDCK4/SIAT1	80	80	80	320	320	160	160
A/Lviv/N6/2009	2009-10-27	MDCK5	160	160	160	1280	1280	640	320
A/Hong Kong/2212/2010	2010-07-16	E3	2560	5120	2560	2560	2560	5120	2560
A/Christchurch/16/2010	2010-07-12	E2/E1	2560	2560	2560	2560	5120	2560	5120
TEST VIRUSES									
A/Madagascar/5681/2010	2010-07-28	MDCK2	2560	2560	2560	1280	1280	2560	1280
A/Madagascar/6460/2010	2010-09-20	MDCK2	2560	2560	2560	2560	2560	5120	2560
A/Madagascar/6515/2010	2010-09-24	MDCK2	1280	1280	1280	640	640	1280	1280
A/England/87/2010	2010-10-26	SIAT1/MDCK1	2560	2560	2560	640	2560	2560	2560
A/England/106/2010	2010-11-06	SIAT1/MDCK1	320	320	320	1280	1280	640	640
A/England/124/2010	2010-11-12	SIAT1/MDCK1	640	1280	1280	640	640	1280	2560
A/England/139/2010	2010-11-14	SIAT1/MDCK1	2560	5120	2560	1280	2560	2560	2560

1. = starting serum dilution 1/40

Vaccine virus

Table 6. Antigenic analyses of pandemic influenza A(H1N1) viruses (25/01/2011)

			Haemagglutination inhibition titre ¹						
			Post infection ferret sera						
Viruses	Collection date	Passage History	A/Cal 7/09 F05/10	A/Eng 195/09 F06/10	A/Auck 3/09 F29/10	A/Bayern 69/09 C4/33/09	A/Lviv N6/2009 C4/34/09	A/HK 2212/2010 F21/10 Egg	A/Chch 16/2010 F30/10
REFERENCE VIRUSES									
A/California/7/2009	2009-04-09	E2/E5	2560	2560	2560	1280	2560	2560	1280
A/England/195/2009	2009-04-28	MDCK5	2560	2560	2560	1280	2560	2560	1280
A/Auckland/3/2009	2009-04-25	Ex+3	2560	5120	5120	1280	2560	2560	2560
A/Bayern/69/2009	2009-07-01	MDCK5	80	80	80	320	320	160	80
A/Lviv/N6/2009	2009-10-27	MDCK4/SIAT2	160	160	160	1280	2560	640	320
A/Hong Kong/2212/2010	2010-07-16	E3	2560	2560	2560	1280	2560	2560	2560
A/Christchurch/16/2010	2010-07-12	E2/E1	2560	2560	5120	2560	5120	2560	5120
TEST VIRUSES									
A/Egypt/50/2010	2010-06-12	E2/E1	2560	2560	5120	2560	2560	2560	2560
A/Egypt/52/2010	2010-06-12	E1/E1	1280	2560	2560	1280	2560	1280	1280
A/Egypt/62/2010	2010-08-12	E2/E1	1280	1280	2560	640	1280	2560	1280
A/England/87/2010	2010-10-26	SIAT1/MDCK3	1280	2560	1280	640	640	1280	640
A/England/106/2010	2010-11-06	SIAT1/MDCK3	640	640	640	640	640	640	640
A/England/107/2010	2010-11-13	MDCK1/MDCK2	1280	2560	1280	640	2560	1280	640
A/Baden-Wuerttemberg/11/2010	2010-11-19	C3/MDCK1	1280	640	640	1280	2560	1280	640
A/Madagascar/7473/2010	2010-11-23	MDCK3	1280	1280	1280	320	640	1280	640
A/Madagascar/7519/2010	2010-11-30	MDCK3	160	80	320	640	640	320	320
A/Hong Kong/6922/2010	2010-12-02	MDCK2/MDCK2	640	5120	1280	320	1280	640	320
A/Hong Kong/6938/2010	2010-12-04	MDCK2/MDCK1	1280	1280	2560	640	1280	1280	640
A/Egypt/75/2010	2010-12-14	E1/E1	1280	2560	2560	640	2560	1280	1280
A/Latvia/12-35825/2010	2010-12-14	MDCK1/MDCK1	640	640	640	640	2560	640	2560
A/Nordrhein-Westfalen/11/2010	2010-12-16	C3/MDCK1	80	160	160	320	320	320	80
A/Latvia/12-39098/2010	2010-12-19	MDCK1/MDCK1	1280	1280	1280	320	1280	1280	640
A/Egypt/83/2010	2010-12-20	E1/E1	1280	2560	2560	640	2560	1280	1280
A/Baden-Wuerttemberg/13/2010	2010-12-20	C2/MDCK1	640	320	640	1280	1280	640	320
A/Nordrhein-Westfalen/12/2010	2010-12-20	C2/MDCK1	1280	1280	1280	640	2560	2560	640
A/Egypt/93/2010	2010-12-22	E1/E1	5120	5120	5120	2560	5120	5120	2560
A/Egypt/94/2010	2010-12-22	E1/E1	2560	1280	2560	640	1280	640	640
A/Egypt/94/2010	2010-12-22	C1/MDCK1	1280	1280	2560	640	1280	5120	640
A/Egypt/96/2010	2010-12-22	E1/E1	1280	2560	2560	640	2560	2560	1280
A/Egypt/96/2010	2010-12-22	C1/MDCK1	2560	5120	5120	2560	2560	2560	1280
A/Egypt/103/2010	2010-12-22	E1/E1	1280	1280	2560	640	2560	2560	1280
A/Latvia/12-756p/2010	2010-12-22	MDCK1/MDCK2	640	640	1280	640	1280	640	640
A/Nordrhein-Westfalen/10/2010	2010-12-22	C2/MDCK1	2560	2560	2560	1280	2560	1280	1280
A/Nordrhein-Westfalen/14/2010	2010-12-22	C2/MDCK1	160	160	320	640	640	320	320
A/Egypt/104/2010	2010-12-27	E1/E1	1280	1280	1280	1280	2560	1280	1280
A/Latvia/12-72162/2010	2010-12-27	MDCK1/MDCK2	5120	640	1280	640	2560	640	640
A/Nordrhein-Westfalen/13/2010	2010-12-27	C2/MDCK1	160	160	160	1280	640	320	160
A/Baden-Wuttermberg/14/2010	2010-12-27	C3/MDCK1	40	80	160	160	320	80	80
A/Berlin/12/2010	2010-12-29	C3/MDCK1	80	80	160	640	320	320	160

1. = starting serum dilution 1/40

Vaccine virus

Sequences included in HA phylogeny

Table 7. Antigenic analyses of pandemic influenza A(H1N1) viruses (02/02/2011)

			Haemagglutination inhibition titre ¹						
			Post infection ferret sera						
Viruses	Collection date	Passage History	A/Cal 7/09 F05/10	A/Eng 195/09 F06/10	A/Auck 3/09 F29/10	A/Bayern 69/09 C4/33/09	A/Lviv N6/2009 C4/34/09	A/HK 2212/2010 F21/10 Egg	A/Chch 16/2010 F30/10
REFERENCE VIRUSES									
A/California/7/2009	2009-04-09	E2/E5	2560	5120	5120	1280	5120	5120	2560
A/England/195/2009	2009-04-28	MDCK1/MDCK4	2560	2560	2560	1280	2560	2560	1280
A/Auckland/3/2009	2009-04-25	Ex/E3	2560	5120	5120	1280	5120	2560	2560
A/Bayern/69/2009	2009-07-01	MDCK4/SIAT1	80	80	80	320	320	160	80
A/Lviv/N6/2009	2009-10-27	MDCK5	320	320	160	2560	2560	640	320
A/Hong Kong/2212/2010	2010-07-16	E3	5120	5120	5120	2560	5120	5120	5120
A/Christchurch/16/2010	2010-07-12	E2/E1	2560	5120	5120	2560	5120	5120	5120
TEST VIRUSES									
A/Paris/1585/2010	2010-10-04	C1/MDCK1	320	160	160	640	640	160	160
A/Norway/1227/2010	2010-10-20	MDCK1MDCK1	1280	2560	2560	640	2560	2560	1280
A/Kentucky/9/2010	2010-11-02	E3/E1	320	320	320	2560	5120	1280	640
A/Norway/1278/2010	2010-11-22	MDCK1MDCK1	1280	2560	2560	640	1280	2560	640
A/Picardie/1814/2010	2010-11-22	C2/MDCK1	1280	2560	2560	640	1280	2560	640
A/Paris/1820/2010	2010-11-25	C1/MDCK1	1280	1280	1280	640	2560	1280	640
A/Stockholm/9/2010	2010-11-27	C4/MDCK1	2560	2560	2560	2560	2560	2560	2560
A/Stockholm/8/2010	2010-11-27	C4/MDCK1	2560	2560	1280	1280	2560	2560	2560
A/Stockholm/11/2010	2010-12-01	C0/MDCK1	1280	2560	2560	1280	1280	2560	5120
A/Stockholm/12/2010	2010-12-02	C0/MDCK1	1280	1280	1280	640	1280	2560	5120
A/Paris/2226/2010	2010-12-03	C2/MDCK1	640	1280	1280	640	640	1280	640
A/Norway/1326/2010	2010-12-05	MDCK1MDCK1	2560	2560	2560	1280	2560	2560	1280
A/Paris/1961/2010	2010-12-09	C2/MDCK1	2560	2560	2560	1280	1280	2560	640
A/Pays de Loire/1976/2010	2010-12-09	C2/MDCK1	5120	5120	5120	1280	2560	2560	2560
A/Paris/1958/2010	2010-12-10	C2/MDCK1	1280	1280	1280	640	1280	1280	640
A/Stockholm/14/2010	2010-12-12	C1/MDCK1	2560	2560	5120	1280	2560	2560	1280
A/Brussels/G0713/2010	2010-12-13	MDCK3	2560	5120	5120	1280	1280	5120	1280
A/Brussels/G0729/2010	2010-12-13	MDCK2	2560	5120	5120	640	2560	2560	1280
A/Stockholm/15/2010	2010-12-13	C3/MDCK1	160	160	160	640	1280	320	160
A/Stockholm/13/2010	2010-12-13	C0/MDCK1	2560	2560	2560	640	1280	2560	1280
A/Agadir/76/2011	2010-12-14	MDCK/MDCK1	1280	2560	2560	640	1280	2560	1280
A/Pays de Loire/2059/2010	2010-12-14	C2/MDCK1	2560	2560	2560	640	1280	1280	1280
A/Brussels/G0738/2010	2010-12-15	MDCK3	640	1280	1280	320	640	640	640
A/Brussels/G0743/2010	2010-12-15	MDCK2	640	1280	1280	1280	2560	1280	640
A/Brussels/G0755/2010	2010-12-15	MDCK2	2560	2560	2560	640	1280	1280	640
A/Paris/2227/2010	2010-12-18	C1/MDCK1	1280	1280	1280	320	640	1280	640
A/Paris/2207/2010	2010-12-20	C1/MDCK1	1280	2560	2560	640	1280	2560	1280
A/Brussels/G0769/2010	2010-12-20	MDCK3	1280	2560	1280	640	640	1280	640
A/Brussels/G0770/2010	2010-12-20	MDCK3	2560	2560	2560	640	2560	2560	1280
A/Brussels/G0798/2010	2010-12-20	MDCK3	640	640	640	1280	640	640	640
A/Brussels/G0768/2010	2010-12-21	MDCK2	2560	2560	2560	640	1280	2560	1280
A/Brussels/S0174/2010	2010-12-21	MDCK2	2560	2560	2560	640	1280	1280	1280
A/Egypt/89/2010	2010-12-22	E2/E2	5120	5120	5120	1280	2560	5120	2560
A/Brest/2244/2010	2010-12-23	C1/MDCK1	1280	2560	2560	640	1280	2560	1280
A/Brussels/S0205/2010	2010-12-24	MDCK2	1280	2560	2560	640	640	1280	640
A/Latvia/12-42632/2010	2010-12-27	MDCK/MDCK2	2560	2560	5120	1280	2560	2560	1280
A/Egypt/114/2010	2010-12-27	E1/E2	2560	2560	5120	1280	5120	2560	1280
A/Nord Pas de Calais/2289/2010	2010-12-28	C1/MDCK1	2560	2560	2560	640	1280	2560	1280
A/Centre/2305/2010	2010-12-28	C1/MDCK1	1280	1280	1280	640	1280	2560	5120
A/Pays de Loire/2308/2010	2010-12-28	C1/MDCK1	1280	2560	2560	640	1280	1280	640
A/Rabat/123/2011	2010-12-29	MDCK/MDCK1	640	1280	1280	640	1280	1280	640
A/Paris/2301/2010	2010-12-30	C1/MDCK1	160	320	160	160	1280	80	160
A/Latvia/01-30266/2011	2011-01-03	MDCK/MDCK2	1280	1280	1280	640	1280	1280	640
A/Latvia/01-31313/2011	2011-01-04	MDCK/MDCK2	2560	1280	1280	640	1280	1280	1280
A/Latvia/01-33865/2011	2011-01-10	MDCK/MDCK1	1280	2560	2560	640	1280	1280	640

1. = starting serum dilution 1/40

Vaccine virus

Sequences included in HA phylogeny

Table 8. Antigenic analyses of pandemic influenza A(H1N1) viruses (08/02/2011)

Haemagglutination inhibition titre ¹											
Post infection ferret sera											
Viruses	Collection date	Passage History	A/Cal 7/09 F05/10	A/Eng 195/09 F06/10	A/Auck 3/09 F17/09	A/Bayern 69/09 C4/33/09	A/Lviv N6/2009 C4/34/09	A/HK 2212/2010 F21/10	A/Chch 16/2010 F30/10	A/Eng 87/2010 F1/11	A/Eng 106/2010 F2/11
REFERENCE VIRUSES											New Sera
A/California/7/2009	2009-04-09	E2/E5	5120	5120	5120	2560	5120	5120	2560	5120	2560
A/England/195/2009	2009-04-28	MDCK1/MDCK4	2560	2560	5120	640	2560	2560	1280	5120	2560
A/Auckland/3/2009	2009-04-25	Ex/E3	5120	5120	5120	2560	5120	5120	2560	5120	5120
A/Bayern/69/2009	2009-07-01	MDCK4/SIAT1	80	80	80	640	640	160	160	160	320
A/Lviv/N6/2009	2009-10-27	MDCK5	320	320	160	2560	2560	640	640	160	1280
A/Hong Kong/2212/2010	2010-07-16	E3	5120	5120	5120	2560	5120	5120	2560	5120	5120
A/Christchurch/16/2010	2010-07-12	E2/E1	2560	5120	5120	2560	5120	5120	5120	5120	5120
A/England/87/2010	2010-10-26	SIAT1/MDCK2	2560	2560	2560	640	1280	1280	1280	2560	2560
A/England/106/2010	2010-11-06	SIAT1/MDCK3	640	1280	1280	1280	1280	1280	1280	1280	2560
TEST VIRUSES											
A/Brussels/G0709/2010		MDCK4	1280	1280	1280	320	640	1280	640	1280	1280
A/Brussels/G0689/2010	2010-12-06	MDCK4	2560	2560	2560	1280	2560	2560	1280	5120	2560
A/Finland/6/2010	2010-12-09	E12/E1	2560	2560	2560	640	2560	2560	1280	2560	1280
A/England/226/2010	2010-12-11	SIAT1/MDCK2	640	1280	1280	320	640	1280	320	1280	1280
A/England/277/2010	2010-12-12	MDCK1/MDCK2	640	1280	1280	320	640	1280	320	1280	1280
A/Egypt/73/2010	2010-12-14	E1/E3	320	160	320	640	1280	320	320	160	640
A/England/334/2010	2010-12-15	MDCK1/MDCK2	320	320	320	320	640	320	320	640	640
A/England/404/2010	2010-12-15	SIAT1/MDCK2	80	80	80	320	320	160	80	160	320
A/Egypt/76/2010	2010-12-15	E3/E3	2560	1280	2560	1280	2560	2560	1280	2560	1280
A/Croatia/18008-1/2010	2010-12-15	E1	2560	1280	2560	640	1280	1280	1280	2560	1280
A/Croatia/18008-2/2010	2010-12-15	E1	2560	2560	2560	640	1280	1280	1280	2560	1280
A/Austria/592212/2010	2010-12-17	C2/MDCK1	1280	2560	1280	640	1280	1280	1280	2560	2560
A/Brussels/G0766/2010	2010-12-20	MDCK4	1280	1280	1280	640	1280	1280	640	2560	1280
A/Brussels/G0771/2010	2010-12-20	MDCK4	1280	2560	2560	640	1280	2560	1280	5120	2560
A/Slovenia/1098/2010	2010-12-20	MDCK1/MDCK1	80	80	80	40	80	160	160	160	160
A/England/461/2010	2010-12-20	MDCK1/MDCK2	1280	1280	2560	640	2560	1280	1280	2560	2560
A/Austria/592642/2010	2010-12-21	C2/MDCK1	1280	2560	1280	640	1280	1280	1280	2560	2560
A/Austria/592870/2010	2010-12-22	C2/MDCK1	1280	2560	1280	640	1280	1280	1280	2560	1280
A/Slovenia/1126/2010	2010-12-22	MDCK1/MDCK1	1280	1280	1280	320	640	1280	640	1280	2560
A/England/530/2010	2010-12-22	SIAT1/MDCK1	1280	1280	2560	640	1280	1280	640	2560	2560
A/Austria/593095/2010	2010-12-23	C2/MDCK1	1280	1280	1280	320	1280	1280	640	2560	1280
A/England/526/2010	2010-12-23	MDCK1/MDCK1	2560	2560	2560	640	2560	2560	1280	5120	2560
A/England/537/2010	2010-12-24	SIAT1/MDCK1	2560	2560	2560	640	2560	2560	1280	5120	2560
A/Brussels/S0204/2010	2010-12-26	MDCK3	1280	1280	2560	1280	1280	2560	1280	2560	2560
A/England/385/2010	2010-12-27	MDCK1/MDCK2	640	1280	1280	320	640	640	320	1280	1280
A/Slovenia/1182/2010	2010-12-28	MDCK2/MDCK1	1280	1280	2560	640	1280	2560	1280	2560	2560
A/Georgia/3/2011	2010-12-29	MDCK2	320	320	320	1280	1280	640	640	640	1280
A/Zamora/62/2010	2010-12-29	MDCK2/MDCK1	1280	1280	2560	640	1280	1280	640	2560	2560
A/Slovenia/1192/2010	2010-12-30	MDCK1/MDCK1	1280	1280	1280	320	1280	1280	640	1280	2560
A/Slovenia/1195/2010	2010-12-30	MDCK2/MDCK1	40	160	160	40	80	160	320	320	320
A/England/375/2010	2010-12-31	SIAT1/MDCK1	320	320	320	160	320	320	160	640	320
A/England/415/2010	2010-12-31	MDCK1/MDCK2	1280	2560	2560	640	2560	5120	1280	2560	2560
A/Slovenia/24/2011	2011-01-03	MDCK2/MDCK1	2560	2560	1280	640	1280	2560	2560	2560	2560
A/Slovenia/30/2011	2011-01-03	MDCK2/MDCK1	1280	2560	1280	320	1280	2560	1280	2560	2560
A/Georgia/23/2011	2011-01-05	MDCK2	1280	2560	2560	1280	2560	2560	1280	2560	2560
A/Slovenia/69/2011	2011-01-05	MDCKx/MDCK1	2560	5120	2560	1280	2560	5120	1280	5120	2560
A/Georgia/38/2011	2011-01-06	MDCK3	2560	5120	5120	1280	2560	2560	2560	5120	5120
A/Lasi/47959/2011	2011-01-06	MDCK1/MDCK1	1280	2560	2560	640	2560	1280	640	2560	2560
A/Slovenia/86/2011	2011-01-07	MDCKx/MDCK1	1280	2560	2560	640	1280	2560	1280	2560	2560
A/Slovenia/89/2011	2011-01-07	MDCKx/MDCK1	1280	2560	1280	640	1280	1280	640	2560	1280
A/Georgia/51/2011	2011-01-09	MDCK2	320	640	640	320	640	640	320	640	640
A/Slovenia/114/2011	2011-01-10	MDCKx/MDCK1	1280	1280	1280	640	1280	1280	640	2560	1280
A/Slovenia/158/2011	2011-01-10	MDCKx/MDCK1	1280	2560	640	640	1280	1280	1280	2560	2560
A/Slovenia/167/2011	2011-01-10	MDCKx/MDCK1	320	640	640	160	640	640	320	640	640
A/Northern Ireland/2/2011	2011-01-10	MDCK1/MDCK1	160	160	160	2560	2560	640	320	320	2560
A/Slovenia/146/2011	2011-01-11	MDCKx/MDCK1	160	320	160	40	160	320	160	320	160
A/Slovenia/205/2011	2011-01-11	MDCKx/MDCK1	640	1280	1280	320	640	1280	320	1280	1280
A/Georgia/73/2011	2011-01-12	MDCK2	1280	2560	2560	640	1280	1280	1280	2560	1280
A/Georgia/74/2011	2011-01-12	MDCK2	1280	1280	1280	640	640	1280	640	2560	1280
A/Georgia/81/2011	2011-01-12	MDCK3	2560	2560	2560	1280	2560	2560	1280	5120	2560
A/Slovenia/210/2011	2011-01-12	MDCKx/MDCK1	640	640	640	320	640	640	320	1280	640
A/Georgia/93/2011	2011-01-13	MDCK2	2560	1280	1280	640	1280	1280	1280	2560	2560
A/Slovenia/220/2011	2011-01-13	MDCKx/MDCK1	640	1280	640	320	640	1280	320	1280	1280
A/Slovenia/234/2011	2011-01-13	MDCKx/MDCK1	80	80	80	40	80	160	80	160	320
A/Georgia/100/2011	2011-01-14	MDCK3	1280	2560	2560	640	2560	1280	1280	2560	2560

1. = starting serum dilution 1/40

Vaccine virus

Figure 7.

Phylogenetic comparison of pandemic influenza A (H1N1) HA genes (HA1 region)

Vaccine strain

Reference strains

Collection date

Oct 2010

Nov 2010

Dec 2010 - Jan 2011

See HI Tables for
antigenic characterisation

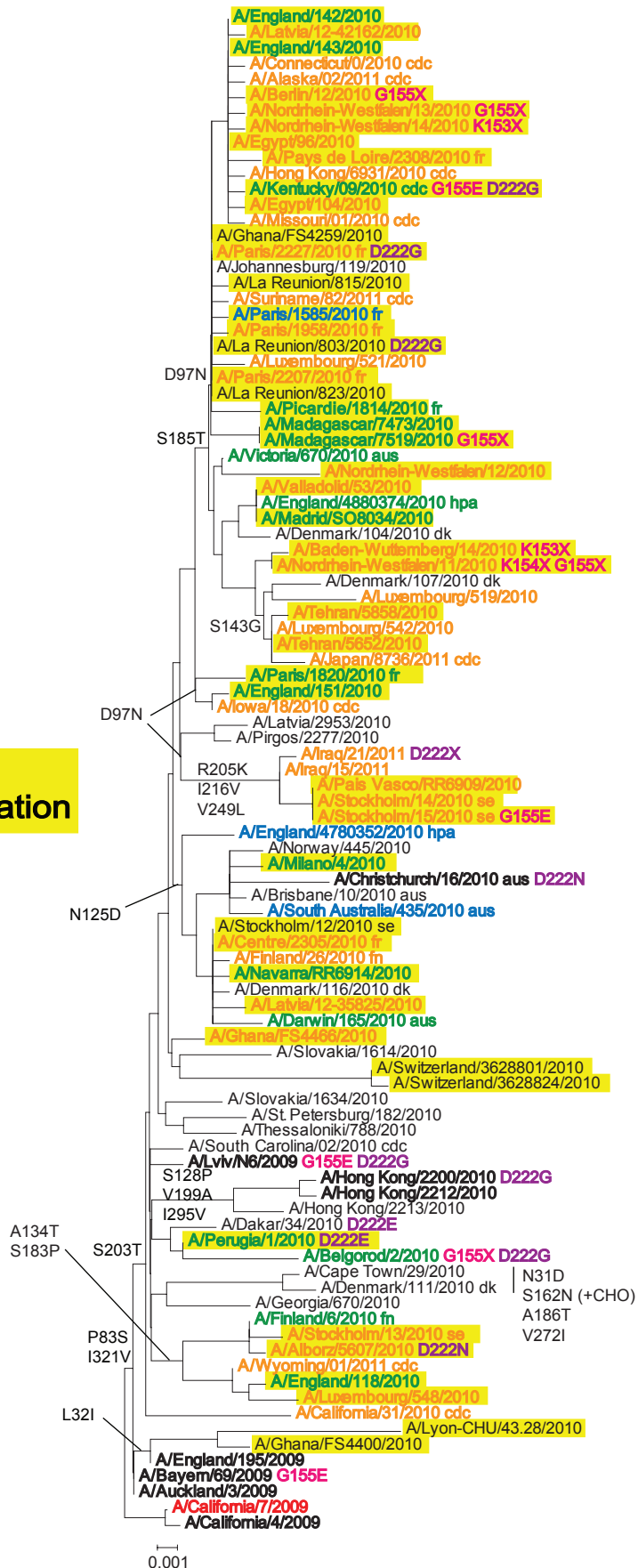


Figure 8. HA1 protein alignment for pandemic A(H1N1) viruses.

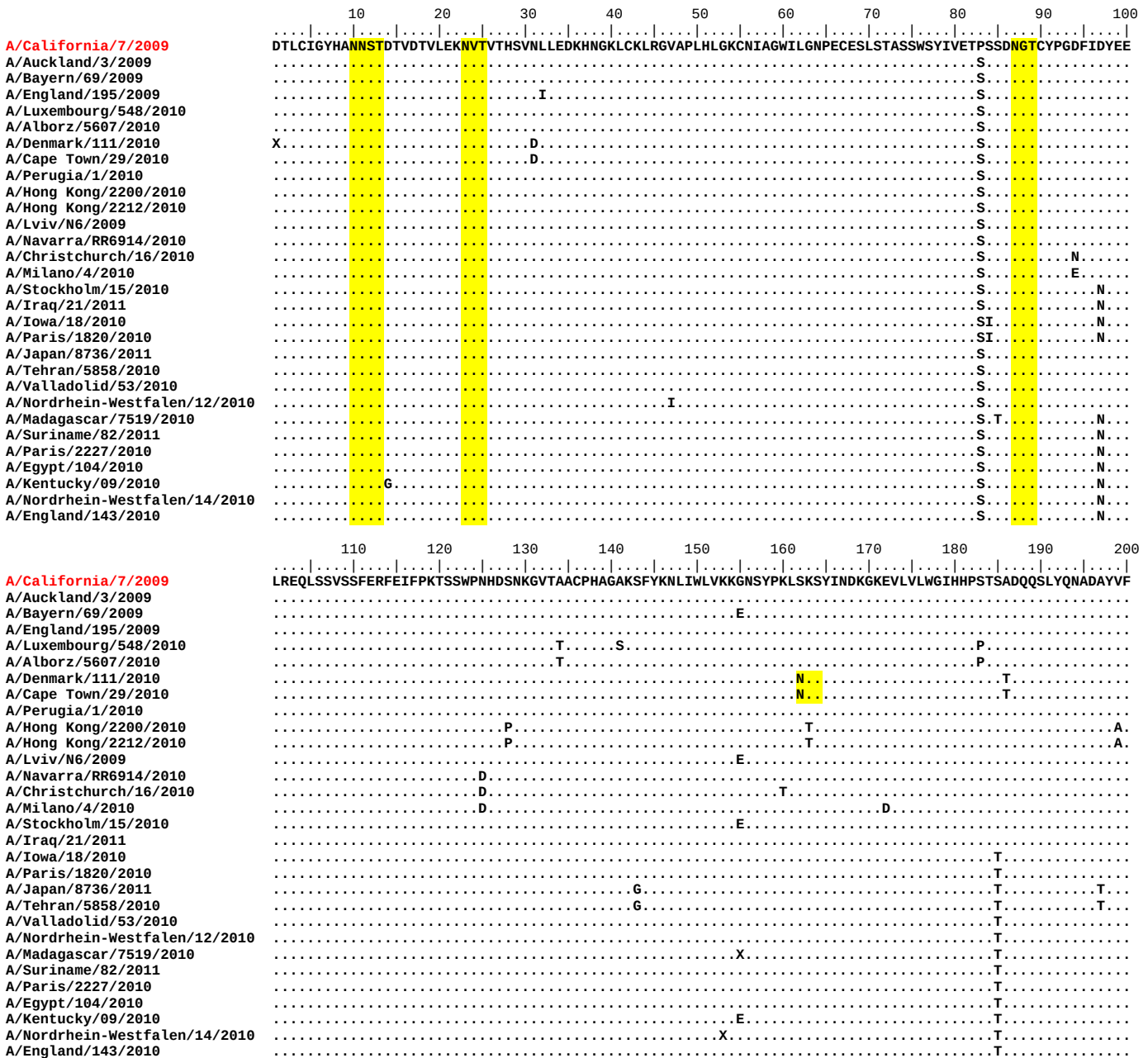


Figure 9. H1 HA location of genetic cluster defining amino acid substitutions

D97N

N125D (Southern hemisphere)

A134T, S183P (UK and elsewhere)

S185T with S143 sub-group shaded.

R205K, I216V, V249L (Iraq & Europe)

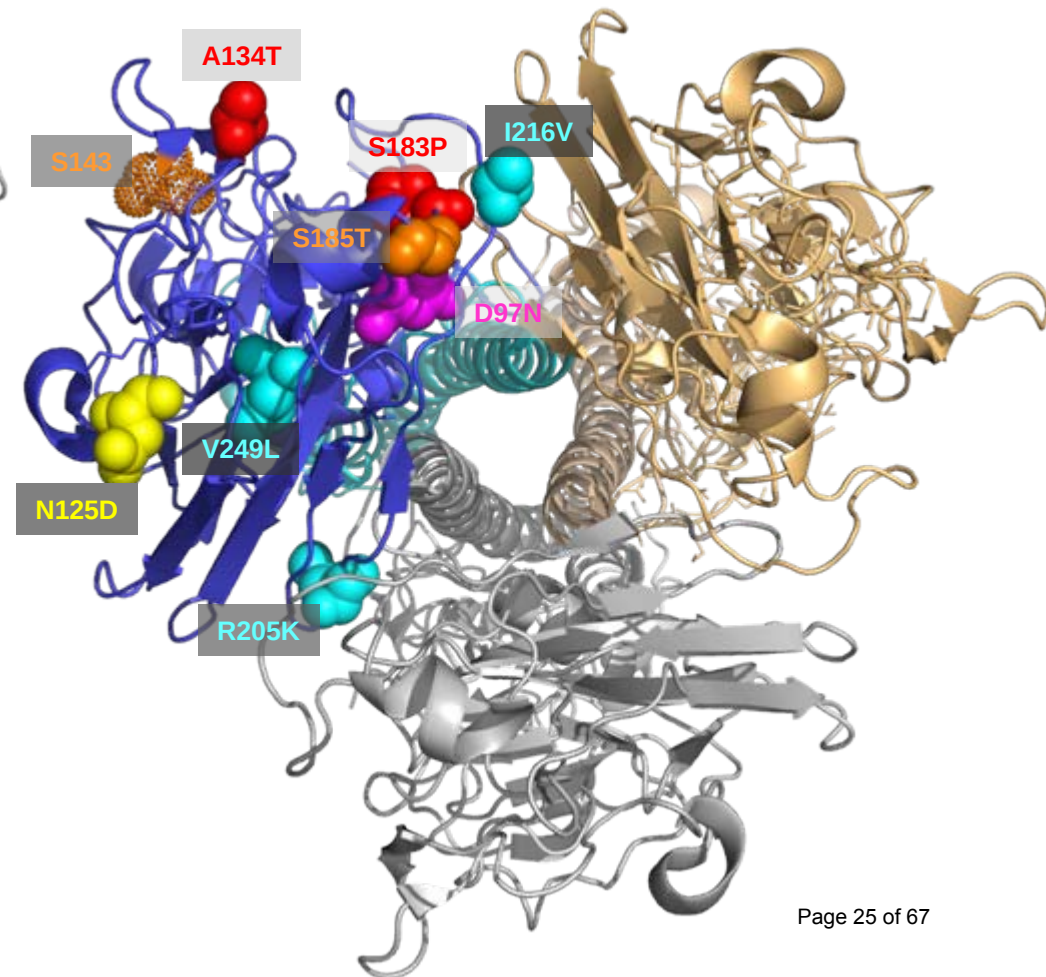
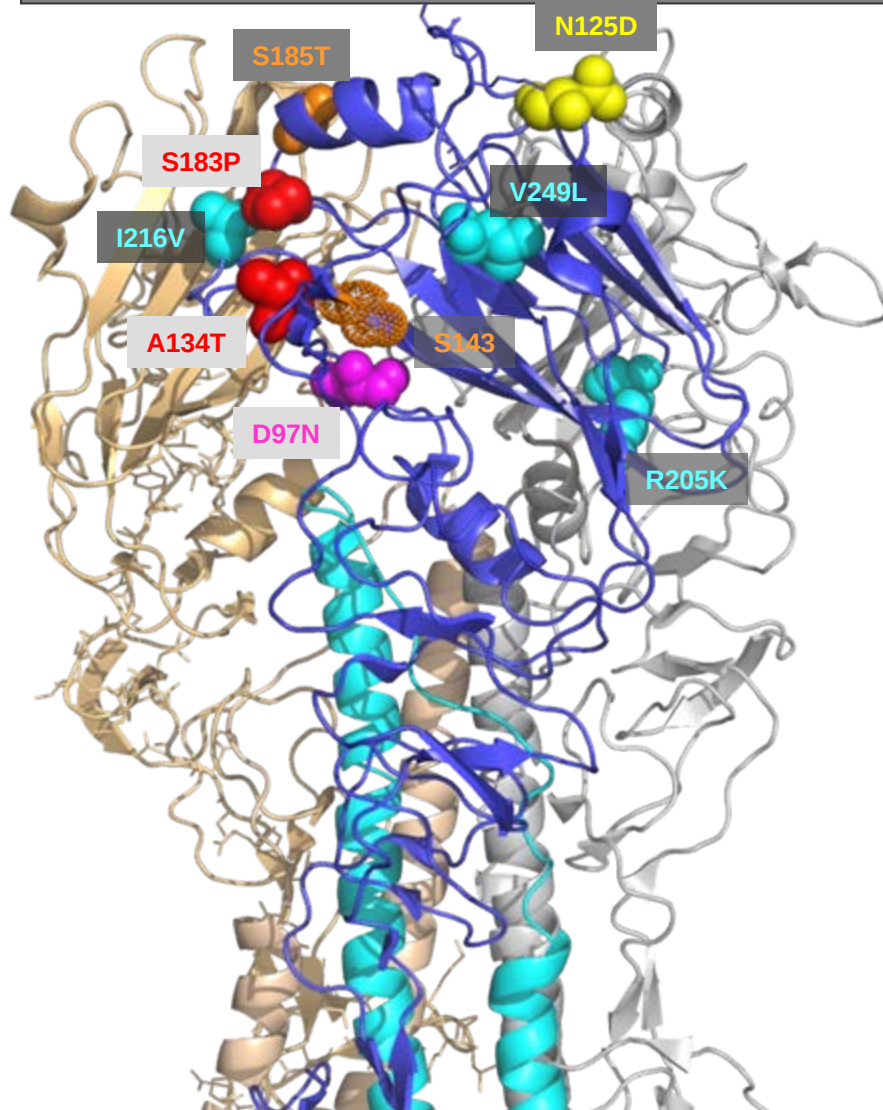


Figure 10.
Phylogenetic comparison of pandemic influenza A (H1N1) NA genes

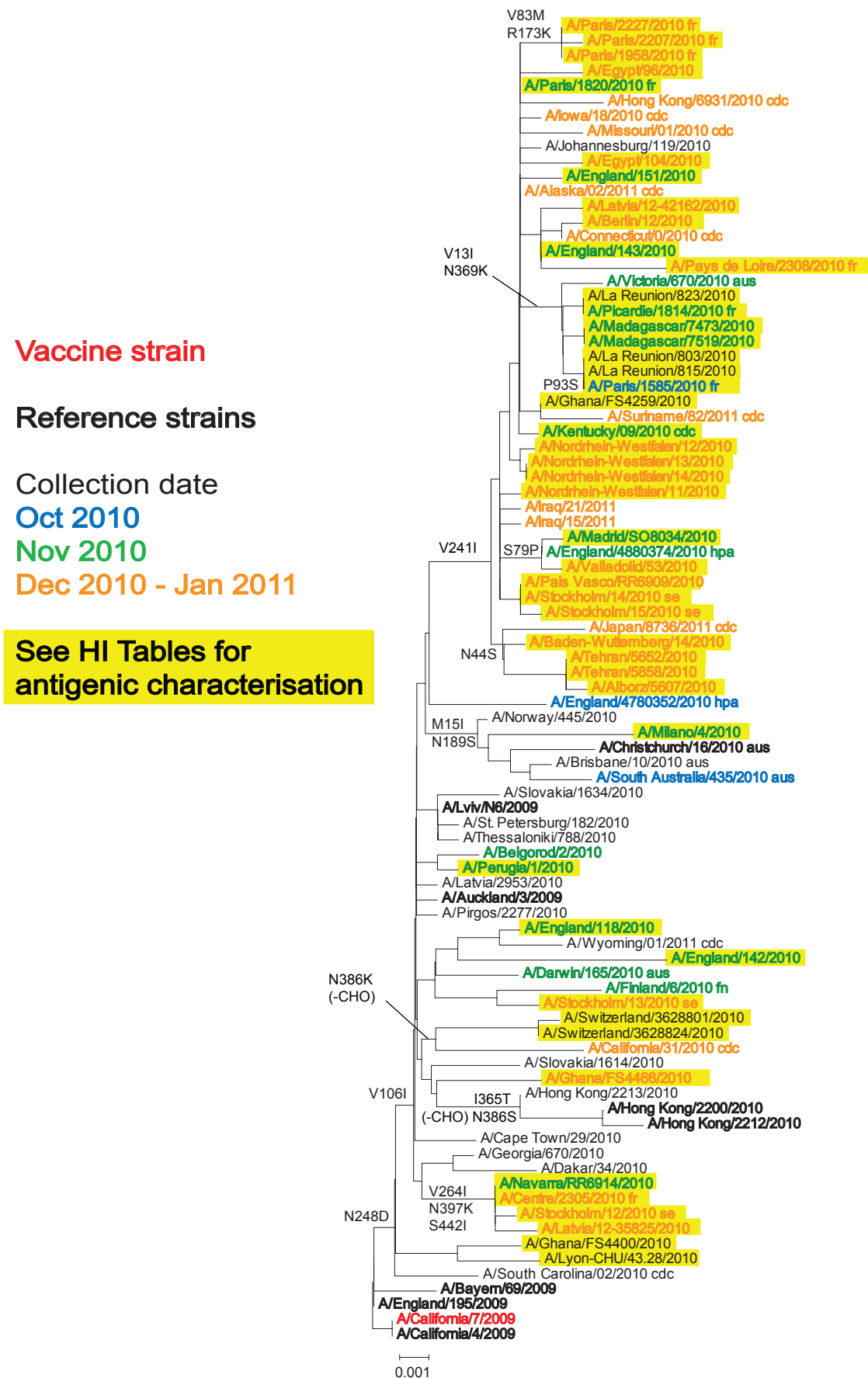


Figure 11. NA protein alignment for pandemic A(H1N1) viruses.

Potential N-linked glycosylation sites

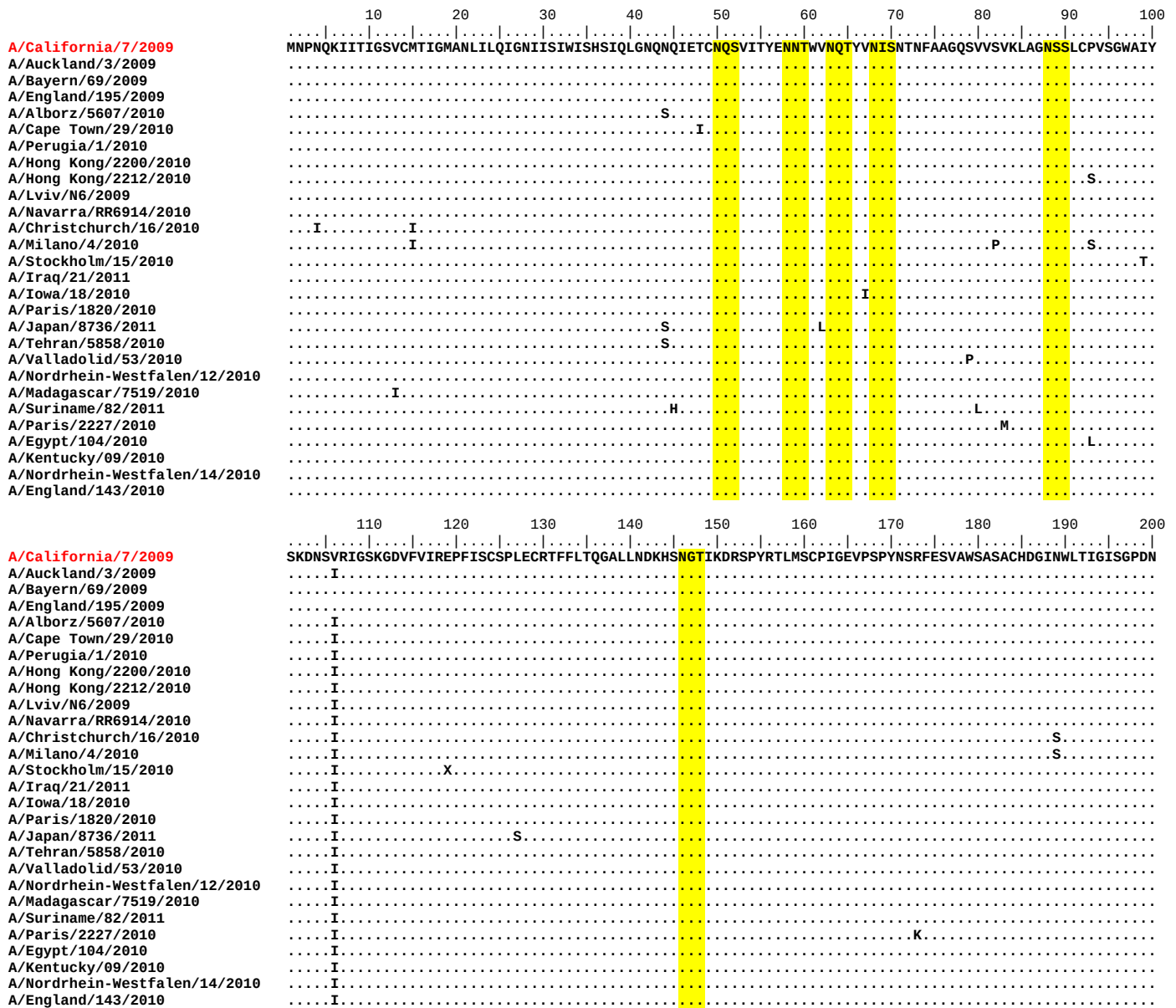


Table 9. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation - MDCK-SIAT

Viruses	Collection Date	Passage History	Neutralisation titre ¹					Amino acid substitutions compared to A/Perth/16/2009																			
			Post infection ferret sera					Defining genetic groups															Virus specific				
			A/Bris	A/Per	A/HK	A/Vic	A/Vic	K62E	K144N	N133D	R142G	V213A	I260M	R261Q	L183H	S214I	N145S	V223I	S45N	N312S	D53N	Y94H		I230V	E280A	S199A	T212A
			10/07	16/09	34430/09	208/09	210/09	Perth/16 clade	Victoria/208 clade																		
			F18/07	F30/09	F04/10	F07/10	F11/10																				
REFERENCE VIRUSES																											
A/Brisbane/10/2007	2007-02-06	E2/E5	5120	160	320	640	320																				
A/Perth/16/2009	2009-07-04	E2/E6	160	1280	640	320	320																				
A/Victoria/208/2009	2009-06-02	E3/E3	640	2560	5120	>5120	>5120	*	*					*	*											*	
A/Victoria/210/2009	2009-06-02	E2/E3	640	2560	5120	2560	>5120						*	*			*										G186V, 222X, 227X
TEST VIRUSES																											
A/Hong Kong/1985/2009	2009-04-01		80	640	2560	320	640						*	*													
A/Netherlands/9/2010	2010-04-21	MDCK2/SIAT2	40	160	640	320	160	*	*				*	*			*	*	*	*	*					*	
A/Stockholm/2/2010	2010-04-30	C1/MDCK1/SIAT1	80	320	640	160	160	*	*				*	*			*								*		F193Y, A198S
A/Johannesburg/19/2010	2010-06-26	MDCK2/SIAT2	80	640	5120	640	320																				No sequence available
A/Johannesburg/47/2010	2010-07-02	MDCK1/SIAT2	40	320	1280	320	160	*	*				*	*			*								*		
A/Johannesburg/31/2010	2010-07-08	MDCK2/SIAT2	40	320	1280	160	160	*	*				*	*			*								*		
A/Johannesburg/34/2010	2010-07-11	MDCK1/SIAT2	80	640	2560	640	1280	*	*				*	*			*								*		L59I
A/Johannesburg/60/2010	2010-07-16	MDCK3/SIAT1	80	1280	5120	640	640	*	*			*	*			*									*		
A/Cape Town/62/2010	2010-07-20	MDCK2/SIAT2	80	320	2560	320	320	*	*				*	*			*								*		L3F, L59I
A/Johannesburg/87/2010	2010-07-26	MDCK1/SIAT2	40	320	1280	320	160	*	*				*	*			*								*		

Sequences included in HA phylogeny

Vaccine virus

¹ Based on 50% plaque reduction compared to serum negative controls

Table 10. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation - MDCK-SIAT

Viruses	Collection Date	Passage History	Neutralisation titre ¹					Amino acid substitutions compared to A/Perth/16/2009																	
			Post infection ferret sera					Defining genetic groups												Virus specific					
			A/Bris 10/07 F29/09	A/Per 16/09 F30/09	A/Vic 208/09 F07/10	A/Per 10/10 F1777	A/Ala 5/10 F27/10	K62E K134N N133D R142G V213A I260M R261Q L183H S214I N145S V223I S45N N312S D53N Y94H I230V E280A S199A T212A	Perth/16 clade						Victoria/208 clade										
REFERENCE VIRUSES																									
A/Brisbane/10/2007	2007-02-06	E2/E4	5120	160	640	80	80																		
A/Perth/16/2009	2009-07-04	E3/E4	320	2560	320	320	320																		
A/Victoria/208/2009	2009-06-02	E3/E2	640	1280	>5120	1280	1280	*	*					*	*					*					
A/Perth/10/2010	2010-05-25	E2/E1	320	640	160	320	320	*	*	X		*					*	*	*	*	*	*			
A/Alabama/5/2010	2010-07-13	MK1/MDCK2/SIAT2	320	640	160	320	1280	*	*					*	*					*	*	*	*	R208K, 216X, N225D	
TEST VIRUSES																									
A/La Reunion/832/2010	unknown	MDCK2/MDCK1	160	320	80	320	320	No sequence available																	
A/Rhode Island/1/2010	2010-01-26	E4/E1	320	320	80	160	320	*	*					*	*						*	T48A, K92R, D188Y			
A/Panama/307149/2010	2010-08-20	MDCK1/MDCK2/MDCK2	80	640	160	640	1280	*	*					*	*					*	*	*	*	78X, 221X	
A/Hong Kong/3579/2010	2010-08-25	MDCKx/SIAT1	<	640	160	320	640	*	*					*	*					*	*	*	*	*	
A/Hong Kong/3570/2010	2010-08-26	MDCK1/SIAT1	160	640	320	640	1280	*	*					*	*					*	*	*	*	*	
A/Victoria/563/2010	2010-08-30	E5/E1	320	1280	5120	1280	1280	*	*					*	*					*	*	*	*	G5E, 156X, G186V, R208K, S219Y	
A/Ghana/FS-4462/2010	2010-09-28	SIAT1/SIAT1	160	320	160	160	320	*	*					*	*	*	*					*	Q80R, V323I		
A/Peleponnese/4043/2010	2010-10-11	SIAT2	320	1280	320	640	1280	* * *						*	*					*					
A/Ghana/FS-4589/2010	2010-10-22	SIAT1/SIAT1	80	320	160	160	320	*	*					*	*	*	*					*	Q80R, V323I		
A/England/4360174/2010	2010-10-26	SIAT1/SIAT1	160	640	320	320	640	*	*					*	*					*	*	*	*	*	
A/Baleares/RR6848/2010	2010-10-26	SIAT2/SIAT2	80	160	80	320	640	*	*					*	*					*	*	*	*	*	
A/Madagascar/7142/2010	2010-11-05	MDCK2/SIAT1	80	320	160	320	320	*	*					*	*	*						*	T128A		
A/Madagascar/7184/2010	2010-11-16	MDCK1/SIAT1	80	320	160	320	320	*	*	*						*	*	*						*	
A/Argentina/28342/2010	2010-11-23	SIAT1/SIAT1	160	640	320	640	1280	*	*					*	*	*						*	K238R		
A/Argentina/28370/2010	2010-11-23	SIAT1/SIAT1	80	640	320	640	640	*	*					*	*					*	*	*	*	T135A	
A/Argentina/28367/2010	2010-11-23	SIAT1/SIAT1	80	640	320	640	1280	*	*					*	*	*						*	K238R		
A/Argentina/28373/2010	2010-11-24	SIAT1/SIAT1	80	640	320	640	1280	No sequence available																	
A/Hong Kong/6894/2010	2010-11-26	MDCK2/SIAT1	80	320	80	160	320	*	*					*	*					*	*	*	*	T	*
A/Argentina/28378/2010	2010-11-30	SIAT1/SIAT1	40	320	160	320	640	*	*					*	*					*	*	*	*	*	*
A/Hong Kong/6925/2010	2010-11-30	MDCK2/SIAT1	80	1280	160	320	320	* * *						*	*					*	*	*	*	*	E50K, P162S

Sequences included in HA phylogeny

Vaccine virus

¹ Based on 50% plaque reduction compared to serum negative controls

Table 11 . Antigenic analysis of influenza A H3N2 viruses - Plaque Reduction Neutralisation - MDCK-SIAT

Viruses	Collection Date	Passage History	Neutralisation titre ¹					Amino acid substitutions compared to A/Perth/16/2009												
			Post infection ferret sera					Defining genetic groups												
			A/Bris 10/07 F29/09	A/Per 16/09 F30/09	A/Vic 208/09 F07/10	A/Per 10/10 F1777	A/Ala 5/10 F27/10	K62E K144N N133D R142G V213A I260M R261Q L183H S214I N145S V223I S45N N312S D53N Y94H I230V E280A S199A T212A	Virus specific											
								Perth/16 clade						Victoria/208 clade						
REFERENCE VIRUSES																				
A/Brisbane/10/2007	2007-02-06	E2/E4	>5120	160	640	80	80													
A/Perth/16/2009	2009-07-04	E3/E4	320	1280	640	320	320													
A/Victoria/208/2009	2009-06-02	E3/E2	640	1280	>5120	1280	1280	*	*								*			
A/Perth/10/2010	2010-05-25	E2/E1	160	640	320	320	320	*	*		X	*		*	*	*	*			
A/Alabama/5/2010	2010-07-13	VK1/MDCK2/SIAT2	80	320	160	320	640	*	*		*	*		*	*	*	*			
TEST VIRUSES																				
A/England/76/2010	2010-09-14	SIAT2/SIAT3	160	320	160	320	640	No sequence available												
A/Esfahan/6117/2010	2010-12-01	MDCK3/SIAT3	160	1280	320	320	640	E50K, P162S												
A/Madagascar/6613/2010	2010-10-05	SIAT3	160	640	320	640	1280	*	*		*	*	*		*		N31D			
A/Lisboa/Su-17+/2010	2010-11-06	SIAT3	160	640	640	640	1280	*	*		*	*		*			*			
A/Madagascar/6157/2010	2010-08-25	SIAT3	80	320	160	320	640	*	*		*	*		*			*			
A/Hong Kong/6882/2010	2010-11-23	MDCK2/SIAT3	160	320	160	320	640	*	*		*	*		*	*	*	T			
A/Madagascar/7749/2010	2010-12-10	SIAT3	40	320	160	320	640	*	*		*	*		*			*			

Sequences included in HA phylogeny

Vaccine virus

¹ Based on 50% plaque reduction compared to serum negative controls

Table 12. Antigenic analysis of influenza A H3N2 viruses - Plaque Reduction Neutralisation - MDCK-SIAT

Viruses	Collection Date	Passage History	Neutralisation titre ¹					Amino acid substitutions compared to A/Perth/16/2009																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																													
			Post infection ferret sera					Defining genetic groups												Virus specific																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
			A/Bris 10/07 F29/09	A/Per 16/09 F30/09	A/Vic 208/09 F07/10	A/Ala 5/10 CDC F38/11	A/Ala 5/10 F27/10	K62E K144N N133D R142G V213A I260M R261Q L183H S214I N145S V223I S45N N312S D53N Y94H I230V E280A S199A T212A	Perth/16 clade							Victoria/208 clade																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																					

Sequences included in HA phylogeny

Vaccine virus

¹ Based on 50% plaque reduction compared to serum negative controls

Table 13. Antigenic analysis of influenza A H3N2 viruses - Plaque Reduction Neutralisation - MDCK-SIAT

Viruses	Collection Date	Passage History	Neutralisation titre ¹					Amino acid substitutions compared to A/Perth/16/2009																			
			Post infection ferret sera					Defining genetic groups																			
			A/Bris	A/Per	A/Vic	A/Ala	A/Ala	K62E	K144N	N133D	R142G	V213A	I260M	R261Q	L183H	S214I	N145S	V223I	S45N	N312S	D53N	Y94H	I230V	E280A	S199A	T212A	Virus specific
			F29/09	F30/09	F07/10	CDC F38/11	F27/10																				
					Perth/16 clade					Victoria/208 clade																	
REFERENCE VIRUSES																											
A/Brisbane/10/2007	2007-02-06	E2/E4	>5120	80	320	320	80																				
A/Perth/16/2009	2009-07-04	E3/E4	320	1280	640	640	320																				
A/Victoria/208/2009	2009-06-02	E3/E2	320	640	2560	1280	640	*	*					*	*										*		
A/Alabama/5/2010	2010-07-13	MK1/MDCK2/SIAT2	80	160	80	1280	640	*	*					*	*					*	*	*	*	*	*		R208K, 216X, N225D
TEST VIRUSES																											
A/Rhode Island/1/2010 (X199)		Ex/E1/E1	40	320	80	320	160	*	*					*				*		*					*		T48A, K92R, D188Y
A/Gothenburg/1/2010	2010-09-10	C2/SIAT1	40	160	80	640	320	*	*					*	*					*	*	*	*	*	*	*	L157S
A/Stockholm/10/2010	2010-11-19	C1/SIAT1	40	640	80	320	160						*	*	*	*					*	*	*	*	*	*	G5E, E50K, P162S
A/Poitiers/1071/2010	2010-11-19	SIAT3	40	320	80	1280	640						*	*	*	*											
A/England/270/2010	2010-12-06	SIAT1/SIAT1	80	320	160	1280	320																				
A/Lyon/1014/2010	2010-12-06	MDCK2/SIAT1	40	640	160	1280	640																				
A/Lyon/1058/2010	2010-12-10	MDCK2/SIAT1	80	320	160	1280	640																				
A/Lyon/1135/2010	2010-12-19	MDCK2/SIAT1	80	320	160	1280	640																				
A/Stockholm/17/2010	2011-01-03	C0/SIAT1	40	320	160	1280	640																				
A/Fes/158/2010	2011-01-04	SIATx/SIAT3	40	640	160	2560	640																				
A/Athens/108/2010	2011-01-16	SIAT3	80	320	160	1280	640																				

Sequence included in HA phylogeny

Vaccine Virus

¹ Based on 50% plaque reduction compared to serum negative controls

Table 14. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 25/01/2011

Viruses	Collection Date	Passage History	Haemagglutination inhibition titre ¹							Amino acid substitutions compared to A/Perth/16/2009																									
			Post infection ferret sera							Defining genetic groups														Virus specific											
			A/Wis	A/Bris	A/Perth	A/Wis	A/Vic	A/Vic	A/Ala	K62E	K144N	N133D	R142G	V213A	I260M	R261Q	L183H	S214I	N145S	V223I	S45N	N312S	D53N		Y94H	I230V	E280A	S199A	T212A						
			67/05 F18/08	10/07 F29/08	16/09 F30/09	15/09 F24/09	208/09 F7/10	210/09 F11/10	5/10 F27/10	Perth/16 clade				Victoria/208 clade																					
REFERENCE VIRUSES																																			
A/Wisconsin/67/2005	2005-08-31	SpfCk3E3/E10	2560	1280	<	<	320	40	<																										
A/Brisbane/10/2007	2007-02-06	E2/E4	5120	5120	40	80	320	320	40																										
A/Perth/16/2009	2009-07-04	E3/E4	40	<	1280	640	1280	2560	640																										
A/Wisconsin/15/2009	2009-07-06	E2/E3	<	<	640	320	40	160	1280					*	*									L194P, V213A											
A/Victoria/208/2009	2009-06-02	E3/E1	160	160	640	640	5120	5120	1280	*	*					*	*									*									
A/Victoria/210/2009	2009-06-02	E2/E1	160	80	1280	1280	2560	5120	320					*	*					*									G186V, 222X, 227X						
A/Alabama/5/2010	2010-07-13	MK1/M2/S3	<	<	<	80	40	40	160	*	*					*	*					*	*	*	*	*	*	*	*					R208K, 216X, N225D	
TEST VIRUSES																																			
A/Norway/1330/2010	2010-12-03	MDCK1/SIAT1	<	<	80	320	80	80	80					*	*	*	*													*					
A/Hessen/5/2010	2010-12-13	C3/SIAT1	<	<	80	320	80	80	80							*	*	*	*													E50K, P162S			
A/Yokohama/96/2010	2010-12-06	MDCK2+2/SIAT1	<	<	80	160	80	80	80	*	*					*	*					K	*	*	*	*	*	*	*	*					N31S, I192T, P227H, I242T
A/Paris/1844/2010	2010-11-29	C1/SIAT1	40	<	640	1280	320	640	640							*	*	*	*													E50K, P162S			

Sequences included in HA phylogeny

Vaccine virus

1. < = <40

Table 15. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 01/02/2011

Viruses	Collection Date	Passage History	Haemagglutination inhibition titre ¹								Amino acid substitutions compared to A/Perth/16/2009																			
			Post infection ferret sera								Defining genetic groups																Virus specific			
			A/Wis	A/Bris	A/Perth	A/Wis	A/Vic	A/Vic	A/Alab	X-199	K62E	K144N	N133D	R142G	V213A	I260M	P261Q	L183H	S214I	N145S	V223I	S45N	N312S	D53N	Y94H	I230V		E280A	S199A	T212A
			67/05 F1/06	10/07 F29/08	16/09 F25/09	15/09 F24/09	208/09 F7/10	210/09 F11/10	5/10 CDC F38/11	(RI/1/10) CDC F36/11	Perth/16 clade						Victoria/208 clade													
REFERENCE VIRUSES																														
A/Wisconsin/67/2005	2005-08-31	SpfCk3E3/E10	2560	640	<	<	160	40	<	<																				
A/Brisbane/10/2007	2007-02-06	E2/E4	2560	5120	40	40	320	320	160	80																				
A/Perth/16/2009	2009-07-04	E3/E4	80	80	1280	640	2560	2560	1280	1280																				
A/Wisconsin/15/2009	2009-07-06	E2/E3	<	<	640	320	40	160	320	320	L194P, V213A																			
A/Victoria/208/2009	2009-06-02	E3/E1	320	320	1280	1280	5120	5120	5120	1280	*	*											*	G186V, 222X, 227X						
A/Victoria/210/2009	2009-06-02	E2/E1	320	320	1280	1280	2560	5120	5120	1280	*	*											*	R208K, 216X, N225D						
A/Alabama/5/2010	2010-07-13	MK1/M2/S3	<	40	80	40	<	<	640	640	*	*											*	T48A, K92R, D188Y						
A/Rhode Island/1/2010 (X-199)		Ex/E1/E1	<	<	320	160	160	80	1280	1280	*	*											*							
TEST VIRUSES																														
A/Gothenburg/1/2010	2010-09-10	C2/SIAT1	<	40	80	160	160	160	640	320	*	*											*	L157S						
A/Georgia/15/2010	2010-09-28	C1/C1/SIAT1	<	<	1280	320	320	640	1280	1280																	No sequence available			
A/Stockholm/6/2010	2010-09-29	C2/SIAT1	40	<	1280	160	160	640	1280	640																	I25V, E50K, P162S			
A/Brussels/G0515/2010	2010-10-18	SIAT3	<	<	320	320	160	160	640	320																	No sequence available			
A/Wisconsin/13/2010	2010-10-23	C1/C2/SIAT1	<	<	160	<	<	<	640	160	*	*											*							
A/Brussels/G0550/2010	2010-10-27	SIAT3	<	80	80	80	80	80	640	320																	No sequence available			
A/Stockholm/7/2010	2010-11-12	C0/SIAT1	40	<	1280	320	320	640	1280	1280																	G5E, E50K, P162S			
A/Stockholm10//2010	2010-11-19	C1/SIAT1	<	<	640	80	80	320	640	320																	G5E, E50K, P162S			
A/Denmark/105/2010	2010-11-22	MDCK4/SIAT1	<	80	320	160	160	160	640	640																	No sequence available			
A/Paris/2031/2010	2010-11-25	C2/SIAT2	<	160	160	320	320	320	1280	640	*	D											*							
A/Paris/1838/2010	2010-11-29	C1/SIAT2	<	160	320	160	320	320	1280	640	*	D											*							
A/Pays de Loire/1883/2010	2010-12-01	C2/SIAT2	<	80	160	160	160	160	640	640	*	*											*							
A/Paris/1896/2010	2010-12-01	C1/SIAT2	<	80	160	160	160	160	640	640																	No sequence available			
A/Paris/2119/2010	2010-12-10	C1/SIAT2	<	160	160	320	320	320	640	640	*	D											*							
A/Paris/2120/2010	2010-12-10	C1/SIAT2	<	80	80	160	160	160	640	640	*	D											*							
A/Paris/2106/2010	2010-12-17	C1/SIAT2	<	160	160	320	320	320	640	640	*	D											*							
A/Paris/2260/2010	2010-12-17	C1/SIAT2	<	80	160	320	320	320	640	640	*	*											*	T48A, K92R						
A/Brussels/G0801/2010	2010-12-20	SIAT3	<	40	80	80	80	160	640	320																	No sequence available			
A/Paris/2174/2010	2010-12-21	C1/SIAT2	<	80	160	320	320	320	640	640	*	*											*	14X, T48A, K92R						
A/Finland/40/2010	2010-12-22	MDCK2/SIAT1	<	160	640	320	320	320	1280	1280																	No sequence available			
A/Paris/2291/2010	2010-12-29	SIAT1/SIAT2	<	80	160	320	160	160	640	640	*	*											*	T48A, K92R, D291G						
A/Stockholm/17/2010	2011-01-03	C0/SIAT1	<	40	80	80	80	160	640	320																	No sequence available			
A/Fes/158/2011	2011-01-04	SIATx/SIAT3	<	80	640	320	320	320	1280	1280																	No sequence available			
A/Paris/1843/2010	2010-11-27	C1/SIAT1	<	40	160	320	320	320	640	640	*	*											*	T48A, K92R						

Sequences included in HA phylogeny

Vaccine virus

1. < = <40

Table 16. Antigenic analyses of influenza A(H3N2) viruses (Human RBC with 20nM Oseltamivir) 08/02/2011

Viruses	Collection Date	Passage History	Haemagglutination inhibition titre ¹								Amino acid substitutions compared to A/Perth/16/2009																								
			Post infection ferret sera								Defining genetic groups												Virus specific												
			A/Wis	A/Bris	A/Perth	A/Wis	A/Vic	A/Vic	A/Alabama	A/Perth	K62E	K144N	N133D	R142G	V213A	I260M	R261Q	L183H	S214I	N146S	V223I	S45N	N312S	D53N	Y94H	I230V	E280A	S199A	T212A						
			67/05 F18/08	10/07 F29/08	16/09 F30/09	15/09 F24/09	208/09 F7/10	210/09 F11/10	CDC F38/11	5/10 F03/11	Perth/16 clade				Victoria/208 clade																				
REFERENCE VIRUSES																																			
A/Wisconsin/67/2005	2005-08-31	SpfCk3E3/E10	2560	2560	40	<	320	40	<	160																									
A/Brisbane/10/2007	2007-02-06	E2/E4	2560	5120	40	40	320	320	160	160																									
A/Perth/16/2009	2009-07-04	E3/E4	40	160	1280	640	640	1280	1280	1280																									
A/Wisconsin/15/2009	2009-07-06	E2/E3	<	<	640	320	80	160	320	640																									
A/Victoria/208/2009	2009-06-02	E3/E1	160	160	1280	1280	5120	5120	2560	5120	*	*								*	*														
A/Victoria/210/2009	2009-06-02	E2/E1	160	160	1280	1280	2560	5120	2560	2560										*	*														
A/Alabama5/2010	2010-07-13	MK2/M2/S3	<	<	80	40	40	40	640	320	*	*								*	*				*	*	*	*	*						
A/Perth/10/2010	2010-05-25	E2/E2	<	<	320	80	80	80	640	320	*	*							X	*				*	*	*	*	*							
TEST VIRUSES																																			
A/Taiwan/851/2010	2010-09-15	E2/E4/E1	<	<	1280	320	160	320	1280	640																									
A/Brussels/SO066/2010	2010-11-09	SIAT3	<	<	160	160	160	160	320	160																									
A/Poitiers/1071/2010	2010-11-19	MDCKx+1/SIAT1	40	80	320	640	320	320	1280	1280																									
A/Lyon/1014/2010	2010-12-06	MDCK2/SIAT1	40	40	640	1280	640	640	2560	1280																									
A/England/270/2010	2010-12-06	SIAT1/SIAT1	<	80	160	160	80	80	1280	640																									
A/Lyon/1058/2010	2010-12-10	MDCK2/SIAT1	<	80	320	640	320	320	1280	1280																									
A/Lyon/1135/2010	2010-12-19	MDCK2/SIAT1	<	80	320	320	320	320	1280	1280																									
A/Netherlands/34/2010	2010-12-23	xMDCK2/SIAT1	<	40	160	80	80	80	1280	320																									
A/Mures/47467/2011	2010-12-27	MDCK1	<	160	640	1280	640	640	1280	2560																									
A/Galati/47519/2011	2010-12-29	MDCK2/SIAT3	<	160	1280	1280	1280	2560	1280	5120																									
A/Netherlands/63/2011	2011-01-05	xMDCK2/SIAT1	<	40	640	640	160	640	1280	640																									
A/Athens/108/2011	2011-01-16	SIAT3	<	80	320	320	320	320	1280	640																									

Sequences included in HA phylogeny

Vaccine virus

1. < = <40

Figure 12.
Phylogenetic comparison of influenza A (H3N2) HA genes (HA1 region)

Vaccine strain

Reference strains

Collection date

Oct 2010

Nov 2010

Dec 2010 - Jan 2011

@ Antisera used in VN assays

**See HI/VN Tables for
antigenic characterisation**

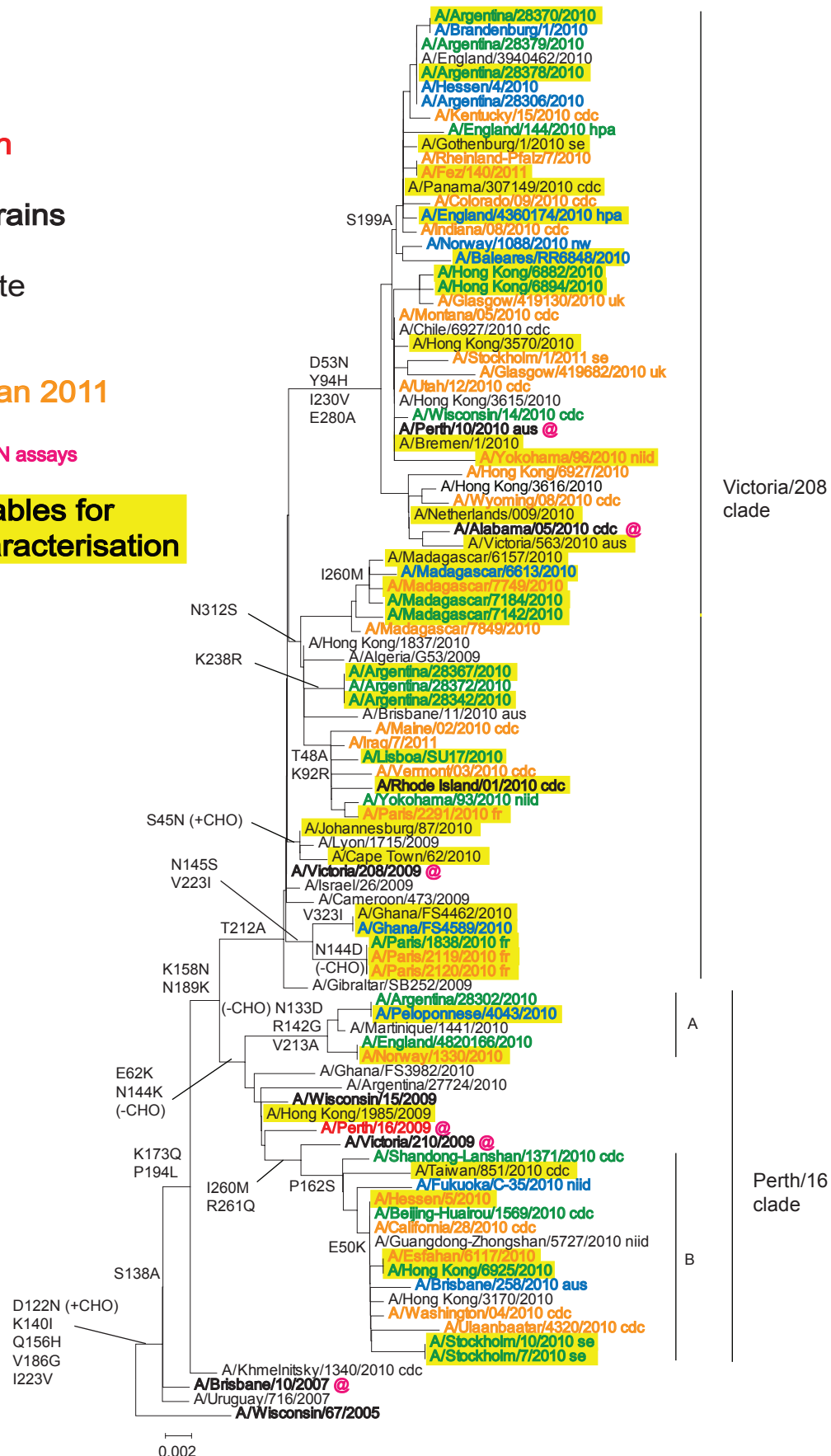


Figure 13. HA1 protein alignment for A(H3N2) viruses.

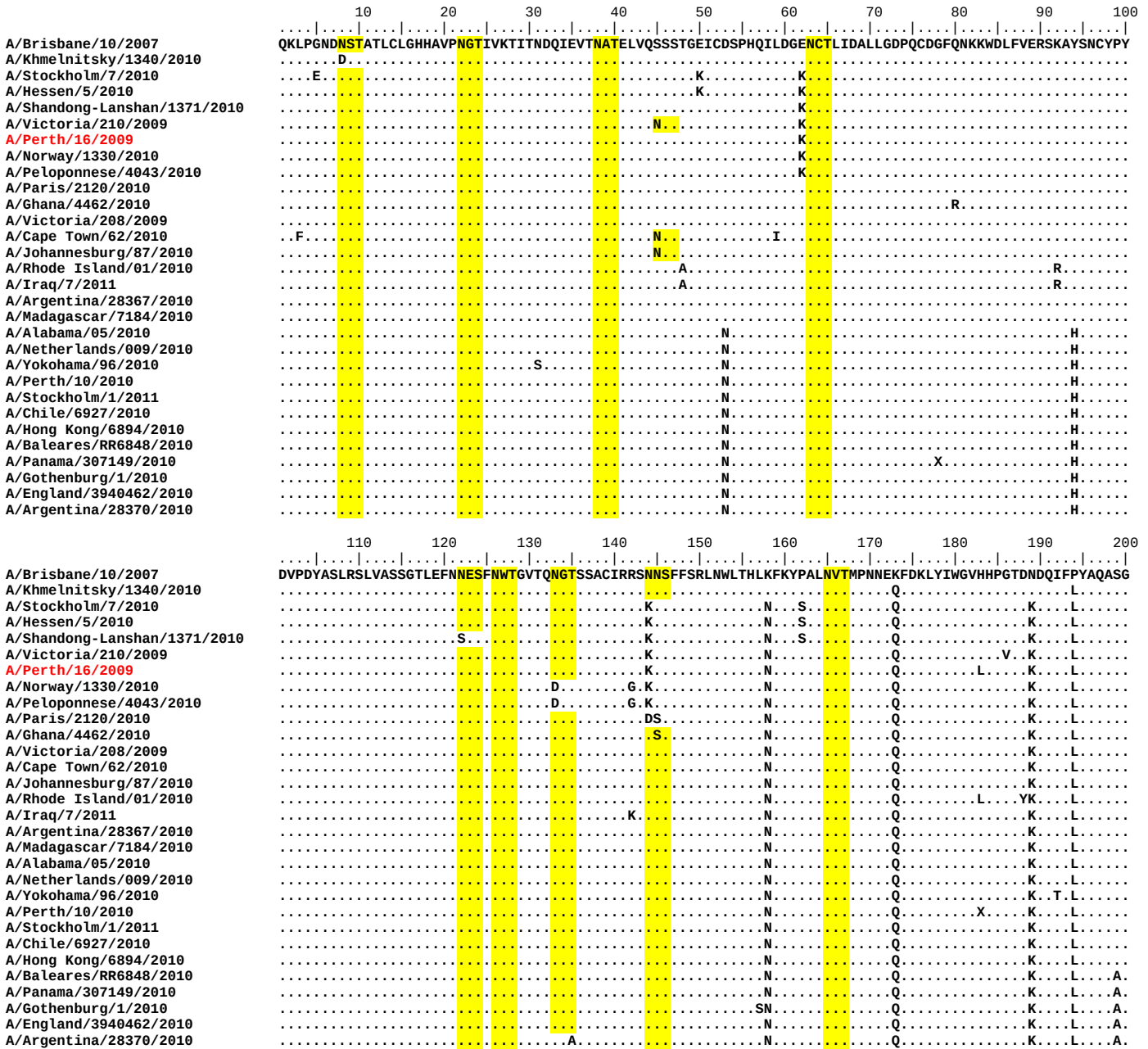


Figure 14.
Location of amino acid substitutions observed in H3 HA of the Perth/16
clade (LHS) and the Victoria/208 clade (RHS)

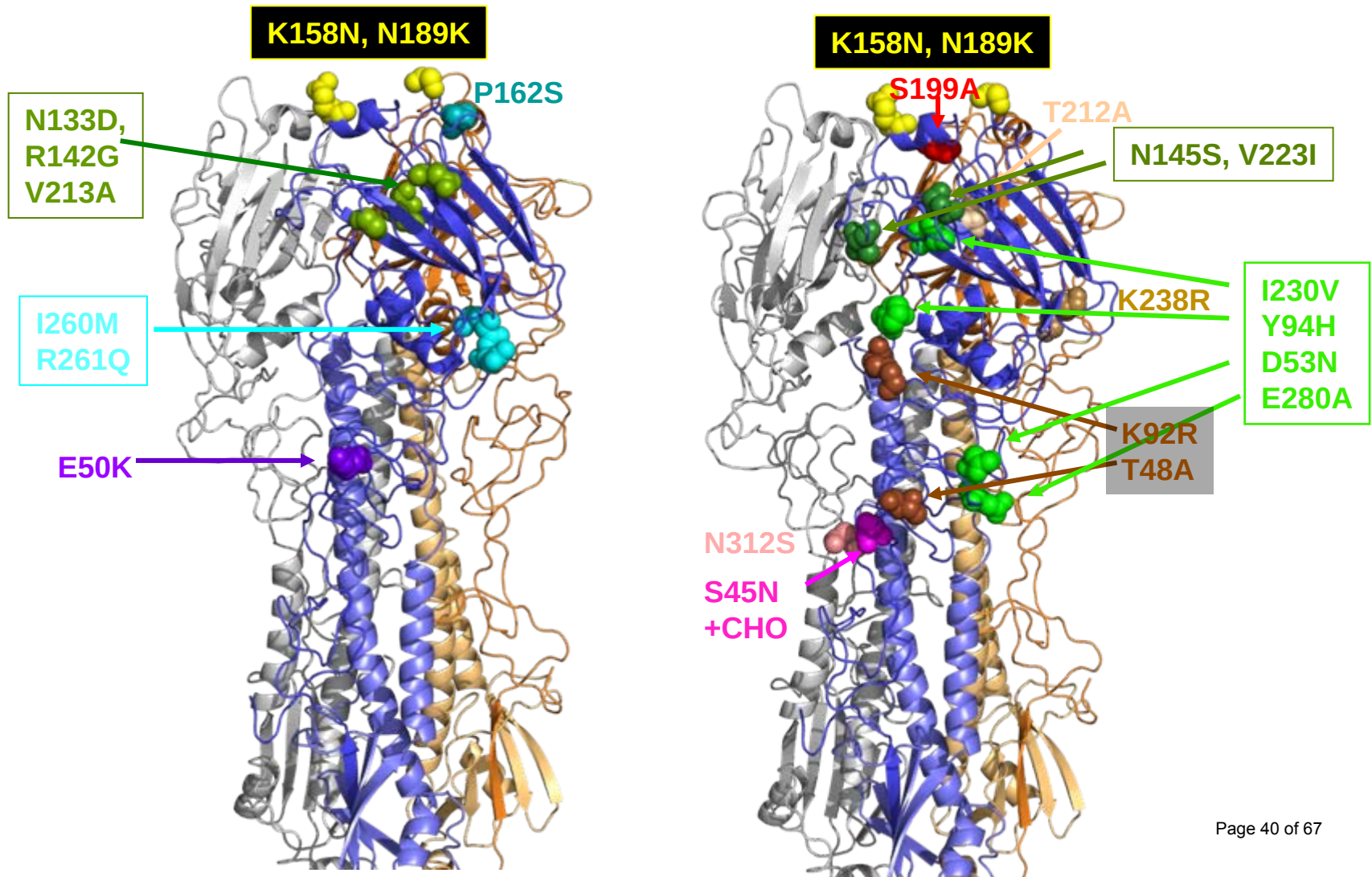
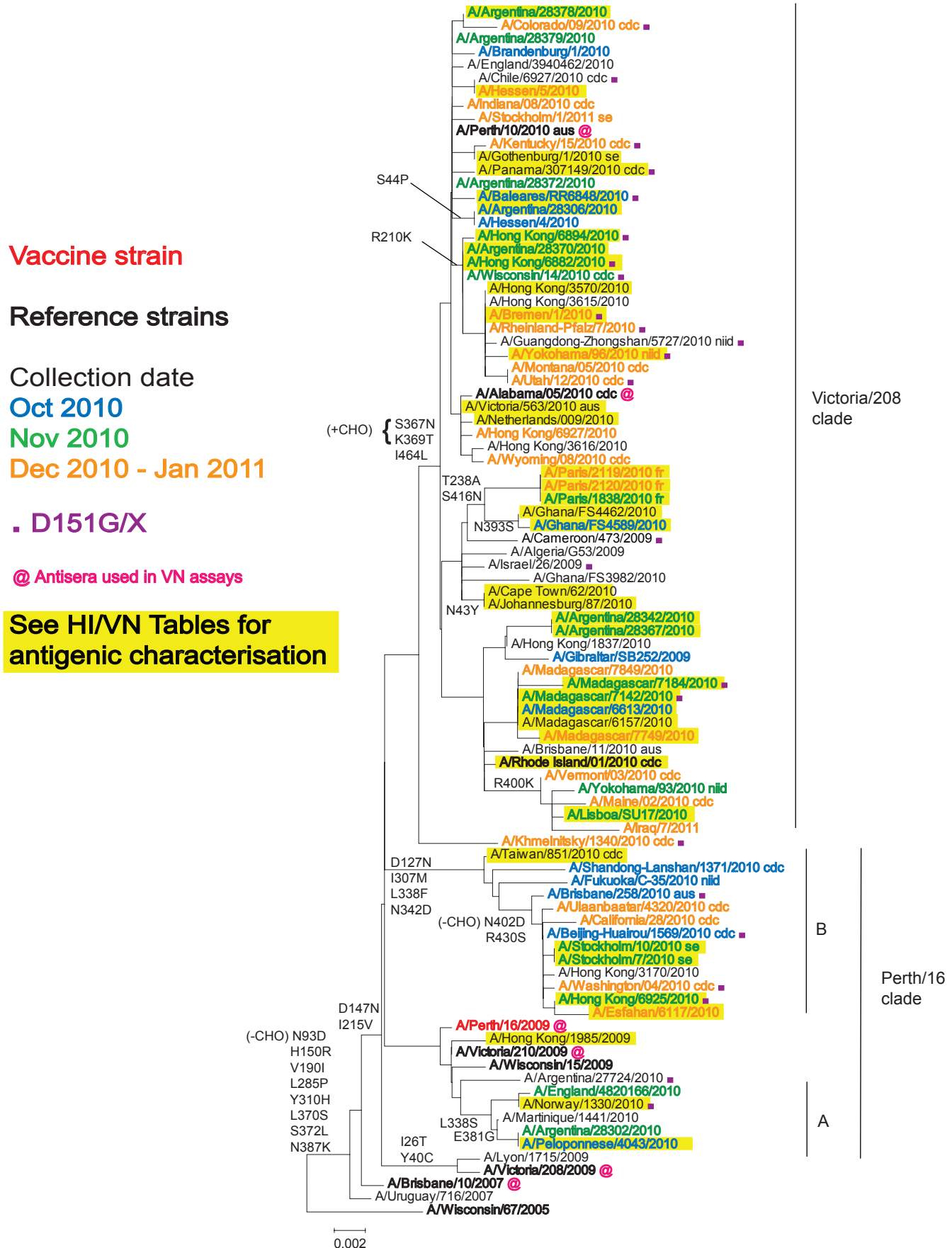
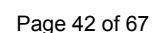


Figure 15.
Phylogenetic comparison of influenza A (H3N2) NA genes



Potential N-linked glycosylation sites



	210	220	230	240	250	260	270	280	290	300
A/Brisbane/10/2007	ATASFIYNGRLVDSIVSWSKEILLRTQSECV	INGT	CTVVMTDGSASGKADTKILFIEEGKIVHTSTLSGSAQHVEECSCYPRYPGVRCVCRDNWKG	SNR						
A/Khmelnitsky/1340/2010		V								
A/Stockholm/7/2010		V								
A/Hessen/5/2010		V								
A/Shandong-Lanshan/1371/2010		V					V			
A/Victoria/210/2009		V								
A/Perth/16/2009		V								
A/Norway/1330/2010		V								
A/Peloponnese/4043/2010		V								
A/Paris/2120/2010		V		A						
A/Ghana/FS4462/2010		V		A						
A/Victoria/208/2009		V								
A/Cape Town/62/2010		V								
A/Johannesburg/87/2010		V								
A/Rhode Island/01/2010		V								
A/Iraq/7/2011		V								
A/Argentina/28367/2010		V								
A/Madagascar/7184/2010		V					A			
A/Alabama/05/2010		V								
A/Netherlands/009/2010		V								
A/Yokohama/96/2010	K	V								
A/Perth/10/2010		V								
A/Stockholm/1/2011		V								
A/Chile/6927/2010		V								
A/Hong Kong/6894/2010	K	V								
A/Balears/RR6848/2010		V								
A/Panama/307149/2010		V								
A/Göteborg/1/2010		V								
A/England/3940462/2010		V								
A/Argentina/28370/2010	K	V								

	310	320	330	340	350	360	370	380	390	400
A/Brisbane/10/2007	PIVDINIKDHSTVSSVVCGLVGDTPRKNDSSSSSHCLDPNNEEGGHGVKGWAFDDGNDVVMGRTISEKSRLGYETFKVIEGWSNP	SKLQINRQVIVDR								
A/Khmelnitsky/1340/2010	I								H	
A/Stockholm/7/2010	M	I		F	D					
A/Hessen/5/2010	I						N	T		
A/Shandong-Lanshan/1371/2010	M	I		F	D					
A/Victoria/210/2009	I									
A/Perth/16/2009	I									
A/Norway/1330/2010	I			S						
A/Peloponnese/4043/2010	I			S				G		
A/Paris/2120/2010	I						N	T		
A/Ghana/FS4462/2010	I						N	T		S
A/Victoria/208/2009	I									
A/Cape Town/62/2010	I						N	T		
A/Johannesburg/87/2010	I						N	T		
A/Rhode Island/01/2010	I						N	T		
A/Iraq/7/2011	I						N	T		
A/Argentina/28367/2010	I		R	G			N	T		K
A/Madagascar/7184/2010	I						N	T		
A/Alabama/05/2010	I						N	T		
A/Netherlands/009/2010	I						N	T		
A/Yokohama/96/2010	I						N	T		
A/Perth/10/2010	I						N	T		
A/Stockholm/1/2011	I						N	T		
A/Chile/6927/2010	I						N	T		
A/Hong Kong/6894/2010	M	I					N	T		
A/Balears/RR6848/2010	I						N	T		
A/Panama/307149/2010	I						N	T		
A/Göteborg/1/2010 se	I						N	T		
A/England/3940462/2010	I						N	T		
A/Argentina/28370/2010	I						N	T		

	410	420	430	440	450	460	470
A/Brisbane/10/2007	GNRRSGYSGIFSVEGKSCINRCFYVELIRGRKEETVLWTSNSIVVFCGTSGYTGSGWPDGADINLMPI	*					
A/Khmelnitsky/1340/2010		N					*
A/Stockholm/7/2010	D		S				*
A/Hessen/5/2010						L	*
A/Shandong-Lanshan/1371/2010							*
A/Victoria/210/2009							*
A/Perth/16/2009							*
A/Norway/1330/2010							*
A/Peloponnese/4043/2010							*
A/Paris/2120/2010		N				L	*
A/Ghana/FS4462/2010		N				L	*
A/Victoria/208/2009							*
A/Cape Town/62/2010						L	*
A/Johannesburg/87/2010						L	*
A/Rhode Island/01/2010						L	*
A/Iraq/7/2011						L	*
A/Argentina/28367/2010						L	*
A/Madagascar/7184/2010						L	*
A/Alabama/05/2010						L	*
A/Netherlands/009/2010						L	*
A/Yokohama/96/2010						L	*
A/Perth/10/2010						L	*
A/Stockholm/1/2011						L	*
A/Chile/6927/2010						L	*
A/Hong Kong/6894/2010						L	*
A/Balears/RR6848/2010						L	*
A/Panama/307149/2010						L	*
A/Göteborg/1/2010						L	*
A/England/3940462/2010	S					L	*
A/Argentina/28370/2010						L	*

Figure 17.
Amino acid substitutions in the N2 NA

119 and 274 shaded
147 and 151 shaded

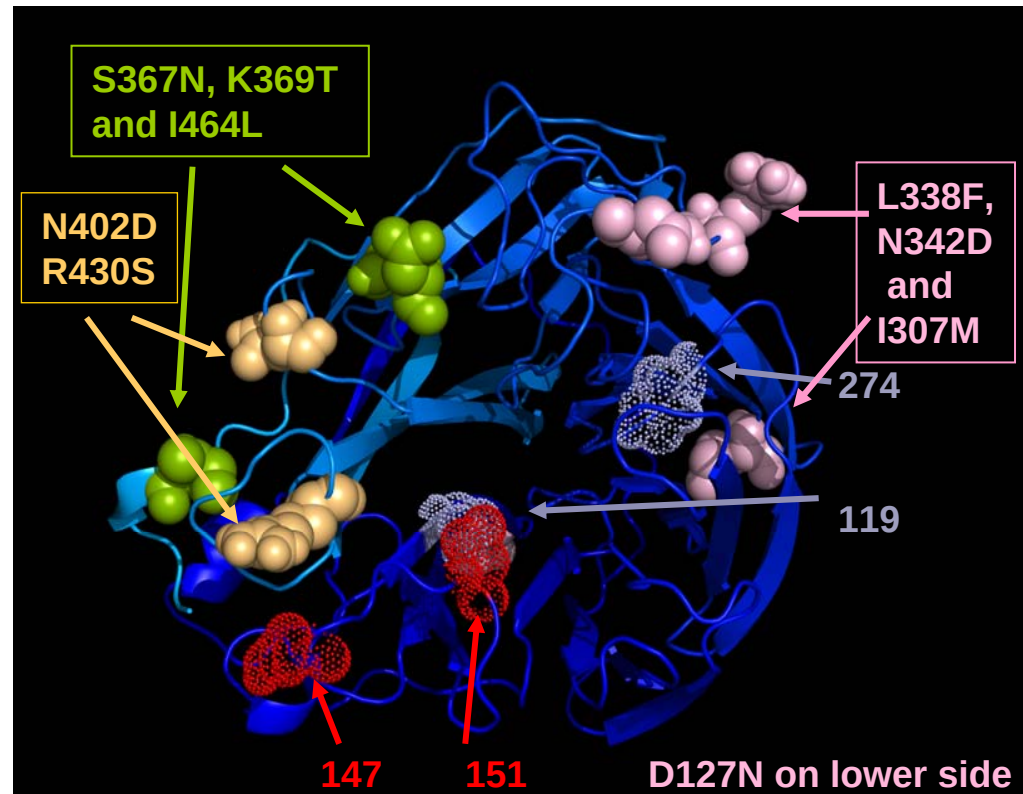
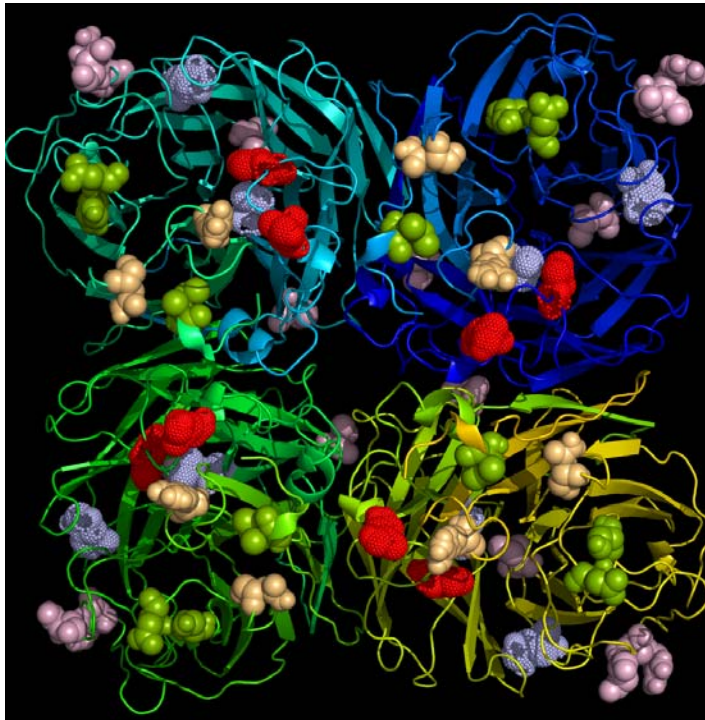


Table 17. Antigenic analyses of influenza B viruses (Victoria lineage) 30/11/2010 and 29/12/2010

Haemagglutination inhibition titre ¹											
Viruses	Collection date	Passage History	Post infection ferret sera								
			B/Mal ²	B/Bris ²	B/Mal	B/Vic	B/England	B/Bris	B/Paris	B/HK	B/Odessa
			2506/04 SH456	60/08 SH524	2506/04 F28/05	304/06 F16/06	393/08 F31/08	60/08 F2/09	1762/08 F11/09	514/09 F13/10	3886/10 F17/10
REFERENCE VIRUSES											
B/Malaysia/2506/2004	2004-12-06	E3/E5	2560	640	640	320	80	160	<	<	<
B/Victoria/304/2006	2006-06-06	E2/E4	640	640	160	320	80	160	<	<	<
B/England/393/2008	2008-08-29	E1/E6	640	640	160	160	160	640	40	20	40
B/Brisbane/60/2008	2008-08-04	E4/E4	640	640	160	160	160	640	40	20	40
B/Paris/1762/2008		C2/MDCK2	<	1280	20	10	40	40	80	160	160
B/Hong Kong/514/2009	2009-10-11	MDCK1/MDCK2	<	1280	20	10	40	80	160	160	160
B/Odessa/3886/2010	2010-03-19	MDCK2/MDCK1	<	1280	20	10	80	80	80	160	160
TEST VIRUSES											
B/England/81/2010	2010-10-19	SIAT1/MDCK1	160	40	20	10	<	<	<	<	<
B/England/121/2010	2010-11-19	SIAT1/MDCK1	320	160	40	40	<	<	<	<	<
B/La Reunion/845/2010		MDCK2/MDCK1	<	1280	20	<	40	20	40	160	80
B/Slovakia/1731/2010	2010-07-27	Cx/MDCK1	<	2560	40	<	40	40	80	320	320
B/Netherlands/19/2010	2010-08-11	MDCK2/MDCK1	<	1280	40	<	80	20	80	320	320
B/England/78/2010	2010-10-12	SIAT1/MDCK1	<	640	10	10	20	10	80	80	80
B/Lyon/809/2010	2010-10-22	MDCK2/MDCK1	<	1280	20	<	40	20	80	320	160
B/Marseille/889/2020	2010-10-26	MDCK2/MDCK1	<	2560	40	<	40	20	80	320	160
B/Marseille/890/2020	2010-10-26	MDCK2/MDCK1	<	1280	20	<	40	20	40	160	80
B/Marseille/891/2020	2010-10-28	MDCK2/MDCK1	<	640	20	<	40	20	40	160	160
B/England/96/2010	2010-11-02	SIAT1/MDCK1	<	640	20	10	40	40	80	160	80
B/England/123/2010	2010-11-13	SIAT1/MDCK1	<	1280	20	10	40	40	80	160	160
B/England/105/2010	2010-11-15	SIAT1/MDCK1	<	640	20	10	40	80	160	160	160
B/England/128/2010	2010-11-23	SIAT1/MDCK1	<	1280	20	10	40	40	80	160	160
B/England/137/2010	2010-11-29	SIAT1/MDCK1	<	1280	20	10	40	40	80	160	160
B/England/149/2010	2010-11-30	SIAT1/MDCK1	<	1280	20	10	40	40	160	160	160
B/England/150/2010	2010-11-30	SIAT1/MDCK1	<	1280	20	10	80	80	160	160	160

1. < = <10; 2. hyperimmune sheep serum

Vaccine strain

Sequences included in HA phylogeny

Table 18. Antigenic analyses of influenza B viruses (Victoria lineage) 13/01/2011 and 19/01/2011

Viruses	Collection date	Passage History	Haemagglutination inhibition titre ¹							
			Post infection ferret sera							
			B/Bris ² 60/08 SH524	B/Mal 2506/04 F28/05	B/Vic 304/06 F16/06	B/England 393/08 F31/08	B/Bris 60/08 F5/09	B/Paris 1762/08 F11/09	B/HK 514/09 F3/10	B/Odessa 3886/10 F17/10
REFERENCE VIRUSES										
B/Malaysia/2506/2004	2004-12-06	E3/E5	1280	640	160	80	160	<	<	10
B/Victoria/304/2006	2006-06-06	E2/E4	1280	320	320	160	320	<	<	<
B/England/393/2008	2008-08-29	E1/E6	2560	640	320	320	640	40	160	160
B/Brisbane/60/2008	2008-08-04	E4/E4	1280	320	320	320	640	40	80	80
B/Paris/1762/2008	2009-02-09	C2/MDCK2	2560	40	10	40	80	160	640	320
B/Hong Kong/514/2009	2009-10-11	MDCK1/MDCK2	2560	20	10	40	80	80	640	320
B/Odessa/3886/2010	2010-03-19	MDCK2/MDCK1	2560	20	10	80	80	160	640	320
TEST VIRUSES										
B/Madagascar/6555/2010	2010-09-27	MDCK1/MDCK2	1280	20	<	40	40	80	320	160
B/Tehran/5277/2010	2010-11-01	MDCK2/MDCK1	320	10	<	10	40	80	320	160
B/Madagascar/7151/2010	2010-11-03	MDCK1/MDCK1	1280	10	<	20	40	80	320	320
B/Madagascar/7146/2010	2010-11-08	MDCK1/MDCK1	640	10	<	10	40	80	320	320
B/Lisboa/3/2010	2010-11-19	MDCK1/MDCK2	1280	20	<	40	80	80	640	320
B/Roma/2/2010	2010-11-26	MDCK1/MDCK1	1280	20	<	20	20	40	ND	80
B/Roma/1/2010	2010-11-27	MDCK1/MDCK1	1280	20	<	20	40	80	ND	160
B/Lisboa/11/2010	2010-11-29	MDCK2/MDCK2	1280	20	<	40	40	80	ND	160
B/Madagascar/7505/2010	2010-11-29	MDCK1/MDCK1	1280	10	<	20	40	80	ND	80
B/Madagascar/7506/2010	2010-11-30	MDCK1/MDCK1	640	10	<	10	40	80	ND	160
B/Madagascar/7556/2010	2010-12-01	MDCK1/MDCK1	1280	10	<	20	40	80	ND	160
B/Zamora/52/2010	2010-12-01	MDCK1/MDCK2	320	10	<	10	20	40	ND	80
B/Genova/2/2010	2010-12-01	MDCK1/MDCK1	640	20	<	20	40	40	ND	160
B/Lisboa/8/2010	2010-12-02	MDCK2/MDCK1	640	10	<	10	20	40	ND	80
B/Ghana/FS4716/2010	2010-12-02	MDCK3	1280	20	<	20	40	80	ND	80
B/Lisboa/9/2010	2010-12-05	MDCK1/MDCK1	640	10	<	20	20	40	ND	160
B/Lisboa/13/2010	2010-12-07	MDCK1/MDCK1	640	10	<	10	40	80	ND	160
B/Madagascar/7747/2010	2010-12-08	MDCK1/MDCK1	640	20	<	10	20	40	ND	160
B/Madagascar/7756/2010	2010-12-10	MDCK1/MDCK1	640	10	<	10	20	40	ND	160
B/Madagascar/7766/2010	2010-12-13	MDCK1/MDCK1	640	10	<	10	20	40	ND	160
B/Madagascar/7807/2010	2010-12-16	MDCK2	2560	20	10	40	20	40	ND	80
B/Madagascar/7797/2010	2010-12-20	MDCK2	1280	20	<	20	20	40	ND	160
B/Luxembourg/558/2010	2010-12-29	C1/MDCK1	320	10	<	10	20	40	ND	80

1. < = <10; 2. hyperimmune sheep serum; ND = not done

Vaccine virus

Sequences included in HA phylogeny

Table 19. Antigenic analyses of influenza B viruses (Victoria lineage) 26/01/2011 and 01/02/2011

Viruses	Collection date	Passage History	Haemagglutination inhibition titre ¹							
			Post infection ferret sera							
			B/Bris ² 60/08 SH524	B/Mal 2506/04 F28/05	B/Vic 304/06 F16/06	B/England 393/08 F31/08	B/Bris 60/08 F5/09	B/Paris 1762/08 F11/09	B/HK 514/09 F3/10	B/Odessa 3886/10 F17/10
REFERENCE VIRUSES										
B/Malaysia/2506/2004	2004-12-06	E3/E5	1280	640	160	80	160	<	<	<
B/Victoria/304/2006	2006-06-06	E2/E4	640	320	160	80	160	<	<	<
B/England/393/2008	2008-08-29	E1/E6	1280	320	320	640	640	40	80	80
B/Brisbane/60/2008	2008-08-04	E4/E4	640	320	640	320	640	40	80	80
B/Paris/1762/2008	2009-02-09	C2/MDCK2	320	10	<	20	20	80	160	80
B/Hong Kong/514/2009	2009-10-11	MDCK1/MDCK2	1280	10	<	40	20	80	320	160
B/Odessa/3886/2010	2010-03-19	MDCK2/MDCK1	640	10	<	40	20	80	320	160
TEST VIRUSES										
B/Bolivia/1526/2010		E3/E1	640	320	320	160	320	160	80	80
B/Egypt/46/2010	2010-02-12	E3/E1	1280	160	160	320	20	40	160	160
B/Egypt/65/2010	2010-08-12	E2/E1	640	160	80	80	320	20	80	40
B/Hiroshima/9/2010	2010-09-03	E1+1/E1	640	320	160	160	40	40	320	160
B/Hiroshima/9/2010	2010-09-03	MDCK1+1/MDCK1	1280	20	10	40	80	80	320	320
B/Centre/1719/2010	2010-11-04	C1/MDCK1	640	10	<	20	20	40	160	80
B/Stockholm/7/2010	2010-11-10	C2/MDCK1	1280	320	80	160	20	40	320	160
B/Pays de Loire/1767/2010	2010-11-15	C1/MDCK1	640	20	<	80	20	40	160	80
B/Bretagne/1775/2010	2010-11-16	C1/MDCK1	320	10	<	20	320	20	40	40
B/Norway/1270/2010	2010-11-20	MDCK1/MDCK1	640	10	<	20	20	40	160	80
B/Paris/1812/2010	2010-11-23	C1/MDCK1	640	10	<	20	40	80	320	160
B/Georgia/28/2010	2010-11-24	E3/E1	160	160	80	160	20	40	160	80
B/Norway/1298/2010	2010-11-29	MDCK1/MDCK1	320	10	<	20	40	80	320	160
B/Brussels/G0650/2010	2010-11-29	MDCK2	1280	20	20	20	20	80	160	160
B/Centre/1858/2010	2010-11-30	C1/MDCK1	640	10	<	20	40	80	320	160
B/Brussels/G0655/2010	2010-11-30	MDCK3	1280	20	10	40	20	80	160	80
B/Egypt/27/2010	2010-11-30	C1/MDCK1	1280	20	<	40	40	80	320	320
B/Centre/1876/2010	2010-12-01	C1/MDCK1	1280	10	<	20	40	40	320	160
B/Brussels/G0665/2010	2010-12-01	MDCK2	1280	20	20	20	40	80	320	160
B/Paris/1873/2010	2010-12-02	C1/MDCK1	640	10	<	20	10	40	160	80
B/Lisboa/17/2010	2010-12-02	MDCK2/MDCK1	1280	20	<	40	40	40	320	160
B/Hong Kong/2200/2010	2010-12-02	MDCK3/MDCK2	1280	20	<	40	320	160	640	320
B/Brussels/G0667/2010	2010-12-03	MDCK2	1280	40	20	40	20	40	160	160
B/Lisboa/15/2010	2010-12-03	MDCK1/MDCK2	1280	20	<	40	80	80	640	320
B/Brussels/G0674/2010	2010-12-06	MDCK2	640	40	20	20	640	20	80	40
B/Paris/1924/2010	2010-12-07	C1/MDCK1	640	10	<	20	40	80	640	160
B/Lisboa/14/2010	2010-12-07	MDCK1/MDCK2	1280	20	<	40	640	40	160	80
B/Lisboa/20/2010	2010-12-07	MDCK1/MDCK2	640	20	<	40	80	80	320	160
B/Norway/1359/2010	2010-12-08	MDCK1/MDCK1	640	10	<	20	40	40	320	160
B/Norway/1379/2010	2010-12-08	MDCK1/MDCK2	1280	20	<	40	640	20	80	80
B/Brussels/G0695/2010	2010-12-08	MDCK2	1280	80	20	80	40	80	320	160
B/Lisboa/21/2010	2010-12-08	MDCK1/MDCK2	1280	20	<	40	40	80	320	80
B/Norway/1360/2010	2010-12-09	MDCK1/MDCK1	640	10	<	20	20	40	160	80
B/Brussels/G0707/2010	2010-12-09	MDCK2	1280	40	10	40	20	40	160	80
B/Stockholm/9/2010	2010-12-10	C0/MDCK1	1280	320	80	160	40	40	320	160
B/Stockholm/10/2010	2010-12-10	C0/MDCK1	640	20	10	40	20	80	320	160
B/Norway/1388/2010	2010-12-13	MDCK1/MDCK2	1280	320	<	160	40	80	320	80
B/Tanger/28/2011	2010-12-13	MDCKx/MDCK1	1280	10	<	40	40	40	320	80
B/Hong Kong/2207/2010	2010-12-13	MDCK3/MDCK1	1280	20	<	40	20	40	160	80
B/Norway/1393/2010	2010-12-14	MDCK1/MDCK2	1280	10	<	40	40	80	320	160
B/Norway/1394/2010	2010-12-15	MDCK1/MDCK2	1280	320	80	160	40	80	320	160
B/Paris/2087/2010	2010-12-17	C1/MDCK1	1280	20	<	20	40	40	320	160
B/Rheinland-Pfalz/1/2010	2010-12-20	C2/MDCK1	640	20	<	20	40	80	320	160
B/Latvia/12-41046/2010	2010-12-23	MDCK1/MDCK1	640	20	<	40	80	160	320	320
B/Nordrhein-Westfalen/2/2010	2010-12-26	C2/MDCK1	640	10	<	10	640	80	80	80
B/Latvia/12-43110/2010	2010-12-27	MDCKx/MDCK2	640	10	<	20	160	40	40	40
B/Bremen/3/2010	2010-12-27	C2/MDCK1	1280	20	<	20	640	40	80	80
B/Bremen/4/2010	2010-12-27	C2/MDCK1	640	20	<	20	40	80	160	320
B/Nord Pas de Calais/2314/2010	2010-12-28	C1/MDCK1	640	10	<	20	40	80	160	320
B/Nordrhein-Westfalen/4/2010	2010-12-29	C1/MDCK1	640	10	<	10	40	160	320	640
B/Nordrhein-Westfalen/5/2010	2010-12-29	C2/MDCK1	640	10	<	10	40	80	160	320
B/Niedersachsen/3/2010	2010-12-29	C2/MDCK1	640	20	<	40	40	80	160	320
B/Paris/2312/2010	2010-12-30	C1/MDCK1	640	10	<	20	40	80	160	320
B/Agadir/156/2011	2011-01-03	MDCKx/MDCK1	640	10	80	40	40	80	160	320
B/Pays de Loire/2382/2010	2011-01-03	C1/MDCK1	1280	10	<	20	40	80	160	320
B/Nordrhein-Westfalen/1/2011	2011-01-03	C2/MDCK1	1280	40	10	40	40	80	160	320

1. < = <10; 2. hyperimmune sheep serum

Vaccine virus

Sequences included in HA phylogeny

Table 20. Antigenic analyses of influenza B viruses (Victoria lineage) 09/02/2011

Viruses	Collection date	Passage History	Haemagglutination inhibition titre ¹						
			Post infection ferret sera						
			B/Bris ² 60/08 SH524	B/Mal 2506/04 F28/05	B/Vic 304/06 F16/06	B/Bris 60/08 F5/09	B/Paris 1762/08 F11/09	B/HK 514/09 F3/10	B/Odessa 3886/10 F17/10
REFERENCE VIRUSES									
B/Malaysia/2506/2004	2004-12-06	E3/E5	1280	1280	320	160	<	<	10
B/Victoria/304/2006	2006-06-06	E2/E4	1280	640	640	320	<	<	10
B/England/393/2008	2008-08-29	E1/E6	2560	320	320	1280	160	160	160
B/Brisbane/60/2008	2008-08-04	E4/E4	2560	320	320	1280	160	160	80
B/Paris/1762/2008	2009-02-09	C2/MDCK4	1280	40	<	80	320	320	320
B/Hong Kong/514/2009	2009-10-11	MDCK1/MDCK2	1280	40	<	80	320	640	320
B/Odessa/3886/2010	2010-03-19	MDCK2/MDCK1	2560	40	<	40	320	640	320
TEST VIRUSES									
B/Lyon/1152/2010	unknown	MDCK2/MDCK1	1280	20	<	40	160	640	160
B/Lyon-CHU/1.271/2011	unknown	MDCK2/MDCK1	2560	20	<	40	160	640	160
B/Slovenia/961/2010	2010-11-29	MDCK1/MDCK1	1280	40	<	40	160	160	320
B/Brussels/S0123/2010	2010-12-02	MDCK3	1280	20	<	20	160	320	160
B/Slovenia/998/2010	2010-12-06	MDCK1/MDCK1	1280	40	<	40	160	160	320
B/Brussels/G0727/2010	2010-12-11	MDCK3	1280	20	<	20	160	320	320
B/Brussels/G0716/2010	2010-12-13	MDCK3	1280	40	<	20	320	640	160
B/Brussels/G0719/2010	2010-12-14	MDCK2	1280	20	<	20	160	320	160
B/Brussels/G0725/2010	2010-12-14	MDCK3	1280	20	<	20	160	320	160
B/Brussels/S0155/2010	2010-12-14	MDCK2	1280	40	<	20	160	320	160
B/Brussels/G0733/2010	2010-12-15	MDCK3	1280	20	<	20	160	320	160
B/England/349/2010	2010-12-16	SIAT1/MDCK1	1280	20	<	40	160	320	160
B/England/355/2010	2010-12-16	MDCK1/MDCK1	2560	40	<	40	160	640	320
B/Brussels/S0183/2010	2010-12-18	MDCK2	1280	20	<	20	320	320	160
B/Lyon/1140/2010	2010-12-18	MDCK2/MDCK1	2560	20	<	40	160	640	160
B/England/511/2010	2010-12-19	MDCK1/MDCK1	2560	20	<	40	160	160	320
B/Brussels/G0788/2010	2010-12-20	MDCK2	1280	20	<	20	160	320	160
B/Brussels/G0796/2010	2010-12-20	MDCK2	1280	20	<	20	80	320	160
B/Valladolid/57/2010	2010-12-20	MDCK1/MDCK2	1280	20	<	40	160	160	320
B/Poitiers/36/2011	2010-12-21	MDCKx+1/MDCK1	2560	40	<	40	160	640	160
B/England/505/2010	2010-12-21	SIAT1/MDCK1	2560	40	<	80	320	320	640
B/England/493/2010	2010-12-22	SIAT1/MDCK1	1280	40	<	40	160	160	640
B/England/497/2010	2010-12-22	MDCK1/MDCK1	1280	20	<	40	160	160	320
B/England/494/2010	2010-12-23	MDCK1/MDCK1	2560	40	<	80	160	160	640
B/Brussels/S0190/2010	2010-12-24	MDCK2	1280	20	<	20	160	320	160
B/Limoges/1160/2010	2010-12-24	MDCK2/MDCK1	1280	20	<	40	160	320	160
B/Slovenia/1166/2010	2010-12-27	MDCK2/MDCK1	2560	40	<	40	160	160	320
B/England/421/2010	2010-12-31	MDCK1/MDCK1	1280	40	<	40	160	160	640
B/Brussels/S0012/2011	2011-01-01	MDCK2	1280	20	<	20	160	640	320
B/Slovenia/3/2011	2011-01-02	MDCK2/MDCK1	1280	40	<	40	160	160	320
B/Lyon/74/2011	2011-01-03	MDCK2/MDCK1	1280	20	<	40	160	320	160
B/Slovenia/31/2011	2011-01-03	MDCK2/MDCK1	1280	20	<	40	160	640	160
B/Slovenia/46/2011	2011-01-04	MDCK2/MDCK1	1280	40	<	80	160	640	160
B/Georgia/47/2011	2011-01-06	MDCK3	1280	20	<	40	160	160	320
B/Georgia/85/2011	2011-01-12	MDCK3	1280	40	<	40	160	320	320
B/Georgia/97/2011	2011-01-13	MDCK3	1280	20	<	40	160	320	320
B/Georgia/112/2011	2011-01-14	MDCK2	1280	20	<	40	320	320	320

1. < = <10; 2. hyperimmune sheep serum

Vaccine virus

Phylogenetic comparison of influenza B (Victoria-lineage) HA genes (HA1 region)

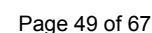
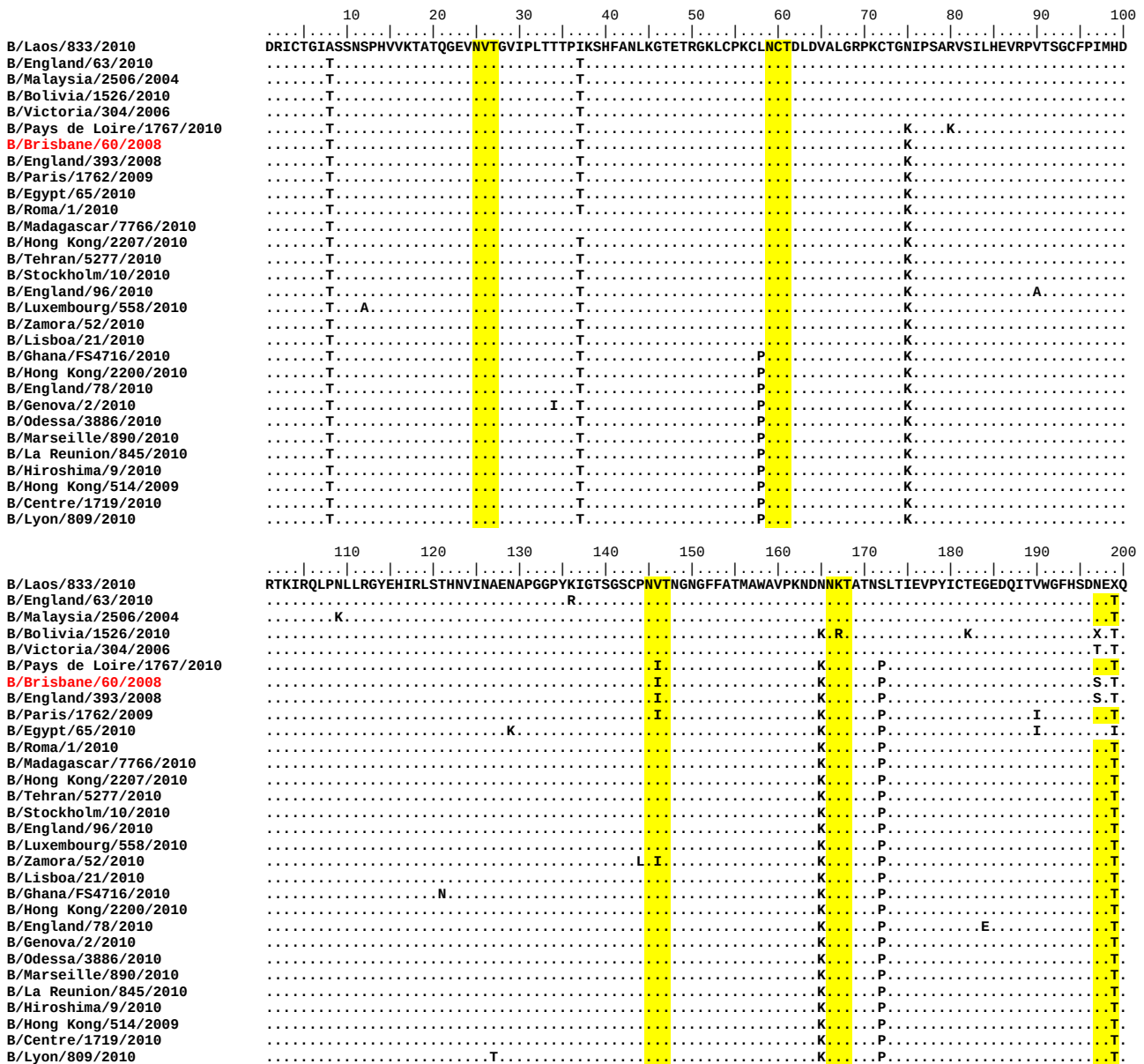


Figure 19. HA1 protein alignment for B-Victoria lineage viruses.



	210	220	230	240	250	260	270	280	290	300
B/Laos/833/2010	MAKLYGDSKPQKFTSSANGVTTHYVSQIGGFPNQT	EDGGLPQSGRIVVDYMQKSGKTGTITYQRGILLPQKVMCASGRSKVIKGSPLIGEADCLHEKY								
B/England/63/2010	.V.									
B/Malaysia/2506/2004										
B/Bolivia/1526/2010										
B/Victoria/304/2006			D.							
B/Pays de Loire/1767/2010										
B/Brisbane/60/2008										
B/England/393/2008										
B/Paris/1762/2009			I.				P.			
B/Egypt/65/2010										
B/Roma/1/2010										
B/Madagascar/7766/2010										
B/Hong Kong/2207/2010										
B/Tehran/5277/2010										
B/Stockholm/10/2010										
B/England/96/2010										
B/Luxembourg/558/2010										
B/Zamora/52/2010										
B/Lisboa/21/2010										
B/Ghana/FS4716/2010							R.			
B/Hong Kong/2200/2010										
B/England/78/2010										
B/Genova/2/2010							R.			
B/Odessa/3886/2010										
B/Marseille/890/2010										
B/La Reunion/845/2010										
B/Hiroshima/9/2010								K.		
B/Hong Kong/514/2009								K.		
B/Centre/1719/2010										
B/Lyon/809/2010										

	310	320	330	340
B/Laos/833/2010	GGLNKS	KPYTGEHAKAIGNCPIWVKTLKLANGTKYRPPAKLLKER		
B/England/63/2010				
B/Malaysia/2506/2004				
B/Bolivia/1526/2010				
B/Victoria/304/2006				
B/Pays de Loire/1767/2010				
B/Brisbane/60/2008				
B/England/393/2008				
B/Paris/1762/2009				
B/Egypt/65/2010				
B/Roma/1/2010				
B/Madagascar/7766/2010				
B/Hong Kong/2207/2010				
B/Tehran/5277/2010				
B/Stockholm/10/2010				
B/England/96/2010		V.		
B/Luxembourg/558/2010		V.		
B/Zamora/52/2010				
B/Lisboa/21/2010				
B/Ghana/FS4716/2010				
B/Hong Kong/2200/2010				
B/England/78/2010				
B/Genova/2/2010				
B/Odessa/3886/2010				
B/Marseille/890/2010				
B/La Reunion/845/2010				
B/Hiroshima/9/2010				
B/Hong Kong/514/2009				
B/Centre/1719/2010				
B/Lyon/809/2010				

Potential N-linked glycosylation sites

Figure 20.
Phylogenetic comparison of influenza B (Victoria-lineage) NA genes

Vaccine strain

Reference strains

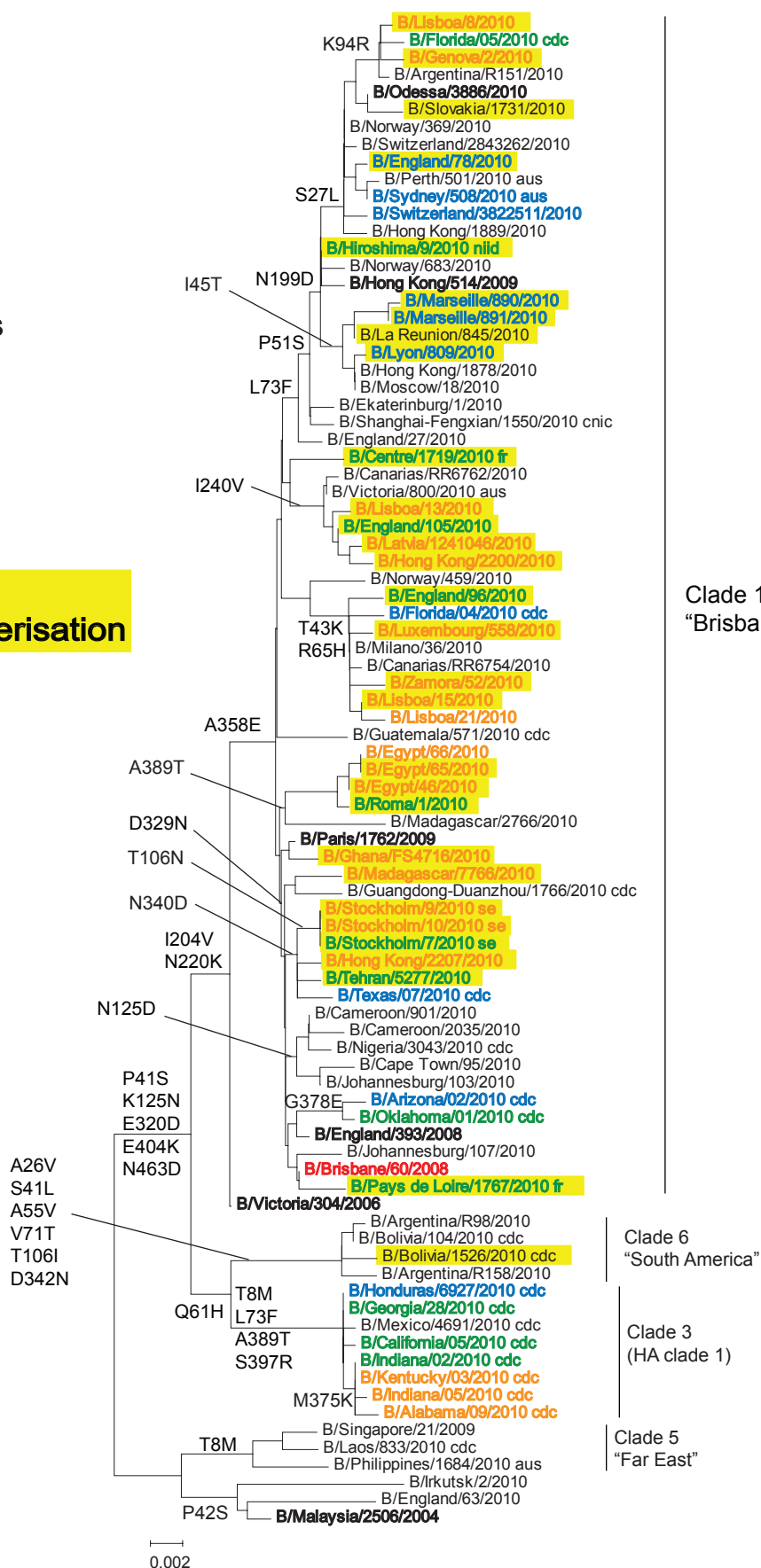
Collection date

Oct 2010

Nov 2010

Dec 2010

See HI Tables for
antigenic characterisation



Potential N-linked glycosylation sites

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	210	220	230	240	250	260	270	280	290	300
B/Laos/833/2010	LLKIKYGEAYTD	TYHSYANNILRT	QESACNCIGGNCYL	MITDGSASGVSECR	FLKIREGRIIKEIF	PPTGRVKHTEECT	CGFASNK	TIECACR	DN	SYTAKR
B/England/63/2010							T			
B/Malaysia/2506/2004							I			
B/Bolivia/1526/2010										
B/Victoria/304/2006			K							
B/Pays de Loire/1767/2010	V		K							
B/Brisbane/60/2008	V		K							
B/England/393/2008	V		K							
B/Paris/1762/2009	V		K							
B/Egypt/65/2010	V		K			V				
B/Roma/1/2010	V		K							
B/Madagascar/7766/2010	V		K							
B/Hong Kong/2207/2010	V		K			S				
B/Tehran/5277/2010	V		K							
B/Stockholm/10/2010	V		K							
B/England/96/2010	V		K							
B/Luxembourg/558/2010	V		K							
B/Zamora/52/2010	V		K							
B/Lisboa/21/2010	V		K							
B/Ghana/FS4716/2010	V		K							
B/Hong Kong/2200/2010	V		K		V					
B/England/78/2010	V		K							
B/Genova/2/2010	V		K							
B/Odessa/3886/2010	V		K							
B/Marseille/890/2010	V		K		I					
B/La Reunion/845/2010	V		K							
B/Hiroshima/9/2010	V		K							
B/Hong Kong/514/2009	V		K							
B/Centre/1719/2010	V		K							
B/Lyon/809/2010	V		K							

	310	320	330	340	350	360	370	380	390	400
B/Laos/833/2010	PFVKLVN	ETDTAEIRLMCT	ETYLDTPRPDG	SITGPCESNGDK	GSGIKGGFVHQ	RMASKIGRWYSRT	ISKTERMG	MG	LVKYD	GDPWADSDALAFSGVM
B/England/63/2010				N			M			
B/Malaysia/2506/2004							M	K		
B/Bolivia/1526/2010		D		N			M			
B/Victoria/304/2006		D					M			
B/Pays de Loire/1767/2010		D	N			E	M			
B/Brisbane/60/2008		D	N			E	M			
B/England/393/2008		D	N			E	M			
B/Paris/1762/2009		D	N			E	M			
B/Egypt/65/2010		D	N			E	M		T	
B/Roma/1/2010		D	N			E	M		T	
B/Madagascar/7766/2010		D	N			E	M			
B/Hong Kong/2207/2010		D	N	D		E	M			
B/Tehran/5277/2010		D	N	D		E	M	G		
B/Stockholm/10/2010		D	N	D		E	M			
B/England/96/2010		D				E	M			
B/Luxembourg/558/2010		D				E	M	E		
B/Zamora/52/2010	I	D				E	M			
B/Lisboa/21/2010		D				E	M			
B/Ghana/FS4716/2010		D	N			E	M			
B/Hong Kong/2200/2010		D				E	M			
B/England/78/2010		D				E	M			
B/Genova/2/2010		D				E	M			
B/Odessa/3886/2010		D				E	M			
B/Marseille/890/2010		D				E	M	R		
B/La Reunion/845/2010		D				E	M			
B/Hiroshima/9/2010		D				E	M			
B/Hong Kong/514/2009		D				E	M			
B/Centre/1719/2010		D				E	M		T	
B/Lyon/809/2010		D				E	M			

	410	420	430	440	450	460
B/Laos/833/2010	VSMKEPGWYSFG	FEIKDKKCDVPC	IGIEMVHDG	GKETWHS	AATAIYCLMG	SGQLLWDVTGVNMAL*
B/England/63/2010	E					*
B/Malaysia/2506/2004	E					*
B/Bolivia/1526/2010						D*
B/Victoria/304/2006						D*
B/Pays de Loire/1767/2010						D*
B/Brisbane/60/2008						D*
B/England/393/2008						D*
B/Paris/1762/2009						D*
B/Egypt/65/2010						D*
B/Roma/1/2010						D*
B/Madagascar/7766/2010						D*
B/Hong Kong/2207/2010						DI*
B/Tehran/5277/2010						D*
B/Stockholm/10/2010						D*
B/England/96/2010						D*
B/Luxembourg/558/2010	P					D*
B/Zamora/52/2010						D*
B/Lisboa/21/2010						D*
B/Ghana/FS4716/2010						D*
B/Hong Kong/2200/2010						D*
B/England/78/2010						D*
B/Genova/2/2010						D*
B/Odessa/3886/2010						D*
B/Marseille/890/2010						D*
B/La Reunion/845/2010						D*
B/Hiroshima/9/2010						D*
B/Hong Kong/514/2009						D*
B/Centre/1719/2010						D*
B/Lyon/809/2010						D*

Table 21. Antigenic analyses of influenza B viruses (Yamagata lineage) 23/12/2010

Viruses	Collection date	Passage History	Haemagglutination inhibition titre							
			Post infection ferret sera							
			B/Fi ³ 4/06 SH479	B/Eg ¹ 144/05 F3/04	B/Fi ¹ 4/06 F1/10	B/Bris ¹ 3/07 F24/07	B/Eng ² 145/08 F9/08	B/Bang ² 3333/07 F25/08	B/Alg ² G-486/10 F15/10	B/Wis ² 1/10 F23/10
REFERENCE VIRUSES										
B/Egypt/144/2005	2005-05-01	E3/E6	5120	80	640	640	40	160	20	80
B/Florida/4/2006	2006-12-15	E3/E4	5120	320	1280	1280	160	320	80	320
B/Brisbane/3/2007	2007-09-03	E2/E1	5120	160	640	640	80	160	20	80
B/England/145/2008		Ex/E4	320	<	80	40	80	10	10	10
B/Bangladesh/3333/2007	2007-08-07	E3/E4	2560	80	320	160	40	160	40	80
B/Algeria/G-486/2010	2010-06-06	SIAT0/MDCK3	5120	160	320	160	320	160	1280	160
B/Wisconsin/1/2010	2010-02-20	E3/E2	1280	40	160	160	20	80	20	80
TEST VIRUSES										
B/Johannesburg/23/2010	2010-06-28	MDCK2/MDCK1	5120	160	160	320	320	320	1280	ND
B/Hong Kong/1832/2010	2010-07-03	MDCK2	640	<	160	<	40	20	320	ND
B/Johannesburg/40/2010	2010-07-12	MDCK1/MDCK1	5120	160	320	320	320	320	1280	ND
B/Hong Kong/1895/2010	2010-07-16	MDCK2	320	<	80	<	80	10	160	ND
B/Niedersachsen/1/2010	2010-10-18	SIAT2/MDCK1	5120	160	2560	2560	640	640	1280	160
B/Tehran/5246/2010	2010-11-01	MDCK2/MDCK1	5120	80	320	640	640	320	2560	160
B/England/110/2010	2010-11-11	SIAT1/MDCK1	5120	160	320	160	160	160	1280	160

1. < = <40; 2. < = <10 ; 3. hyperimmune sheep serum; ND = not done

Sequences included in the HA phylogeny

Table 22. Antigenic analyses of influenza B viruses (Yamagata lineage) 26/01/2011

Viruses	Collection date	Passage History	Haemagglutination inhibition titre							
			Post infection ferret sera							
			B/Fi ³ 4/06 SH479	B/Eg ¹ 144/05 F3/04	B/Fi ¹ 4/06 F1/10	B/Bris ¹ 3/07 F24/07	B/Eng ² 145/08 F9/08	B/Bang ² 3333/07 F25/08	B/Alg ² G-486/10 F15/10	B/Wis ² 1/10 F23/10
REFERENCE VIRUSES										
B/Egypt/144/2005	2005-05-01	E3/E6	2560	80	320	320	40	80	20	40
B/Florida/4/2006	2006-12-15	E3/E4	5120	320	1280	1280	160	160	320	320
B/Brisbane/3/2007	2007-09-03	E2/E1	2560	160	640	640	80	80	40	80
B/England/145/2008		Ex/E4	320	<	40	<	80	10	20	10
B/Bangladesh/3333/2007	2007-08-07	E3/E4	1280	80	320	160	40	80	40	40
B/Algeria/G-486/2010	2010-06-06	SIAT0/MDCK3	5120	160	320	320	160	160	1280	80
B/Wisconsin/1/2010	2010-02-20	E3/E2	640	40	160	160	20	40	20	80
TEST VIRUSES										
B/Hong Kong/2188/2010	2010-11-24	MDCK3/MDCK2	5120	80	160	320	160	80	640	160
B/Hong Kong/2190/2010	2010-11-30	MDCK2/MDCK2	5120	80	160	320	160	160	640	160
B/Niedersachsen/2/2010	2010-12-16	C2/MDCK1	1280	40	80	40	80	80	160	40
B/Nordrhein-Westfalen/1/2010	2010-12-22	C1/MDCK2	5120	160	320	640	320	320	1280	320
B/Berlin/1/2010	2010-12-27	C2/MDCK1	5120	80	160	160	160	160	640	160
B/Berlin/2/2010	2010-12-28	C2/MDCK1	5120	160	160	160	320	160	640	160

1. < = <40; 2. < = <10 ; 3. hyperimmune sheep serum

Sequences included in the HA phylogeny

Table 23. Antigenic analyses of influenza B viruses (Yamagata lineage) 02/02/2011

Viruses	Collection date	Passage History	Haemagglutination inhibition titre							
			B/FI ³ 4/06 SH479	Post infection ferret sera						
				B/Eg ¹ 144/05 F3/04	B/FI ¹ 4/06 F1/10	B/Bris ¹ 3/07 F24/07	B/Eng ² 145/08 F9/08	B/Bang ² 3333/07 F25/08	B/Alg ² G-486/10 F15/10	B/Wis ² 1/10 F23/10
REFERENCE VIRUSES										
B/Egypt/144/2005	2005-05-01	E3/E6	5120	160	640	640	80	80	20	80
B/Florida/4/2006	2006-12-15	E3/E4	5120	320	2560	1280	80	160	40	160
B/Brisbane/3/2007	2007-09-03	E2/E1	5120	160	1280	640	80	160	20	160
B/England/145/2008		Ex/E4	320	<	80	40	80	20	20	10
B/Bangladesh/3333/2007	2007-08-07	E3/E4	2560	80	640	160	40	160	40	80
B/Algeria/G-486/2010	2010-06-06	SIAT0/MDCK3	5120	80	320	160	160	160	640	80
B/Wisconsin/1/2010	2010-02-20	E3/E2	2560	160	640	160	40	160	40	160
TEST VIRUSES										
B/Norway1181/2010	2010-11-02	MDCK1/MDCK1	5120	160	640	320	320	160	640	160
B/Paris/1850/2010	2010-11-29	C1/MDCK1	2560	80	160	40	80	80	160	40
B/Bretagne/2278/2010	2010-12-27	C1/MDCK1	1280	80	160	80	160	80	320	80
B/Brussels/G0710/2010	2010-12-13	MDCK2	5120	80	160	80	160	80	320	40
B/Stockholm/6/2010	2010-11-09	C1/MDCK1	5120	320	640	320	320	320	640	160
B/Finland/33/2010	2010-12-16	MDCK1/MDCK1	2560	80	160	80	80	40	80	40
B/Finland/34/2010	2010-12-30	MDCK1/MDCK1	2560	80	320	80	40	20	80	40
B/Finland/39/2010	2010-11-30	MDCK1/MDCK1	5120	160	320	160	80	40	160	80

1. < = <40; 2. < = <10 ; 3. hyperimmune sheep serum

Sequences included in the HA phylogeny

Table 24. Antigenic analyses of influenza B viruses (Yamagata lineage) 09/02/2011

Viruses	Collection date	Passage History	Haemagglutination inhibition titre							
			B/FI ³ 4/06 SH479	Post infection ferret sera						
				B/Eg ¹ 144/05 F7/05	B/FI ¹ 4/06 F10/07	B/Bris ¹ 3/07 F24/07	B/Eng ² 145/08 F9/08	B/Bang ² 3333/07 F24/10	B/Wis ² 1/10 F23/10	B/Nieder ² 1/10 F4/11 New serum
REFERENCE VIRUSES										
B/Egypt/144/2005	2005-05-01	E3/E6	5120	320	1280	640	80	160	160	40
B/Florida/4/2006	2006-12-15	E3/E4	5120	320	2560	1280	160	320	320	80
B/Brisbane/3/2007	2007-09-03	E2/E1	5120	320	2560	1280	160	320	80	40
B/England/145/2008		Ex/E4	640	40	160	80	320	40	40	<
B/Bangladesh/3333/2007	2007-08-07	E3/E4	5120	320	1280	320	160	640	320	40
B/Wisconsin/1/2010	2010-02-20	E3/E2	5120	160	1280	640	160	320	320	160
B/Niedersachsen/1/2010	2010-10-18	SIAT2/MDCK2	5120	320	640	320	320	320	320	1280
TEST VIRUSES										
B/England/170/2010	2010-12-01	SIAT1/MDCK1	5120	320	640	80	320	320	320	640
B/England/499/2010	2010-12-14	SIAT1/MDCK1	5120	320	640	320	320	320	320	1280
B/Brussels/G0724/2010	2010-12-15	MDCK3	5120	160	320	80	160	160	160	640
B/Brussels/G0786/2010	2010-12-18	MDCK2/MDCK1	5120	320	640	160	320	160	160	640
B/England/512/2010	2010-12-20	SIAT1/MDCK1	5120	640	1280	320	320	320	320	1280
B/Poitiers/128/2011	2010-12-29	MDCKx+1/MDCK1	5120	320	1280	320	320	320	320	1280
B/Lyon/68/2011	2011-01-05	MDCK2/MDCK1	5120	320	640	160	320	320	320	1280

1. < = <40; 2. < = <10 ; 3. hyperimmune sheep serum

Figure 22.
Phylogenetic comparison of influenza B (Yamagata-lineage) HA genes (HA1 region)

Reference strains

Collection dates

Sep 2010

Oct 2010

Nov - Dec 2010

See HI Tables for
 antigenic characterisation

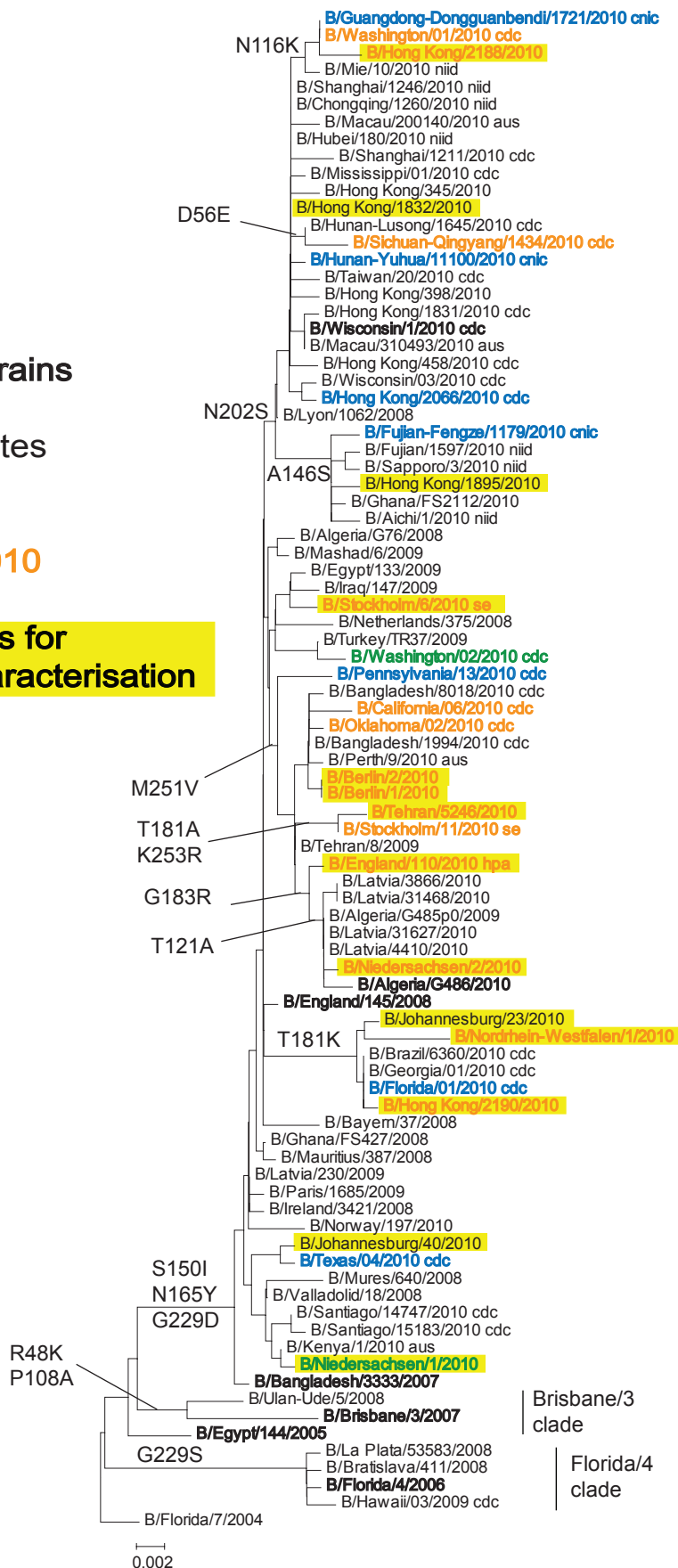
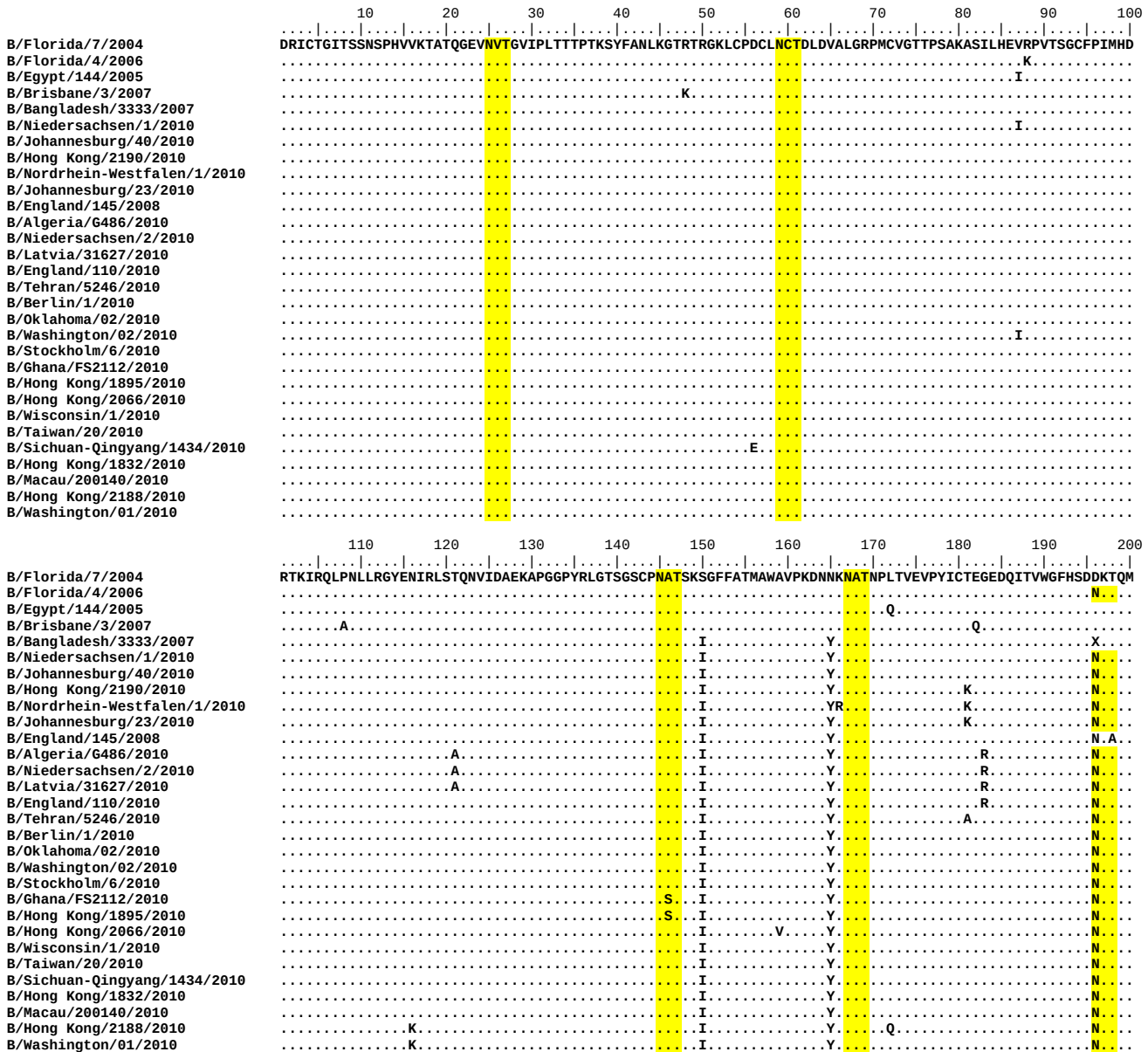


Figure 23. HA1 protein alignment for B-Yamagata lineage viruses.



	210	220	230	240	250	260	270	280	290	300
B/Florida/7/2004	KNLYGDSNPQKFTSSANGVTTHYVSQIGGFPDQTEDGGLPQSGRIVVDYMMQKPKGTGTIVYQRGVLLPQKVVWCASGRSKVIKGSPLPIGEADCLHEKYG									
B/Florida/4/2006	S.....									
B/Egypt/144/2005										
B/Brisbane/3/2007										
B/Bangladesh/3333/2007	D.....									
B/Niedersachsen/1/2010	D.....									
B/Johannesburg/40/2010	D.....									
B/Hong Kong/2190/2010	D.....									
B/Nordrhein-Westfalen/1/2010	D.....									
B/Johannesburg/23/2010	D.....									
B/England/145/2008	D.....									
B/Algeria/G486/2010	D.....V.....L.....									
B/Niedersachsen/2/2010	D.....V.....									
B/Latvia/31627/2010	D.....V.....									
B/England/110/2010	D.....V.....									
B/Tehran/5246/2010	D.....V.R.....									
B/Berlin/1/2010	D.....V.....									
B/Oklahoma/02/2010	D.....V.....									
B/Washington/02/2010	D.....V.....									
B/Stockholm/6/2010	D.....									
B/Ghana/FS2112/2010	.S.....D.....									
B/Hong Kong/1895/2010	.S.....D.....									
B/Hong Kong/2066/2010	.S.....D.....									
B/Wisconsin/1/2010	.S.....D.....									
B/Taiwan/20/2010	.S.....D.....									
B/Sichuan-Qingyang/1434/2010	.S.....D.....									
B/Hong Kong/1832/2010	.S.....D.....									
B/Macau/200140/2010	.S.....D.V.....									
B/Hong Kong/2188/2010	.S.....D.....									
B/Washington/01/2010	.S.....D.....									

	310	320	330	340
B/Florida/7/2004	GLNKSKPYTGEHAKAIGNCPIWVKPLKLANGTKYRPPAKLLKER			
B/Florida/4/2006				
B/Egypt/144/2005				
B/Brisbane/3/2007	PMT.....			
B/Bangladesh/3333/2007				
B/Niedersachsen/1/2010				
B/Johannesburg/40/2010				
B/Hong Kong/2190/2010				
B/Nordrhein-Westfalen/1/2010				
B/Johannesburg/23/2010				
B/England/145/2008				
B/Algeria/G486/2010				
B/Niedersachsen/2/2010				
B/Latvia/31627/2010				
B/England/110/2010				
B/Tehran/5246/2010				
B/Berlin/1/2010				
B/Oklahoma/02/2010	D.....			
B/Washington/02/2010				
B/Stockholm/6/2010				
B/Ghana/FS2112/2010				
B/Hong Kong/1895/2010				
B/Hong Kong/2066/2010				
B/Wisconsin/1/2010				
B/Taiwan/20/2010				
B/Sichuan-Qingyang/1434/2010				
B/Hong Kong/1832/2010				
B/Macau/200140/2010				
B/Hong Kong/2188/2010				
B/Washington/01/2010				

Potential N-linked glycosylation sites

Figure 24.
Phylogenetic comparison of influenza B (Yamagata-lineage) NA genes

Reference strains

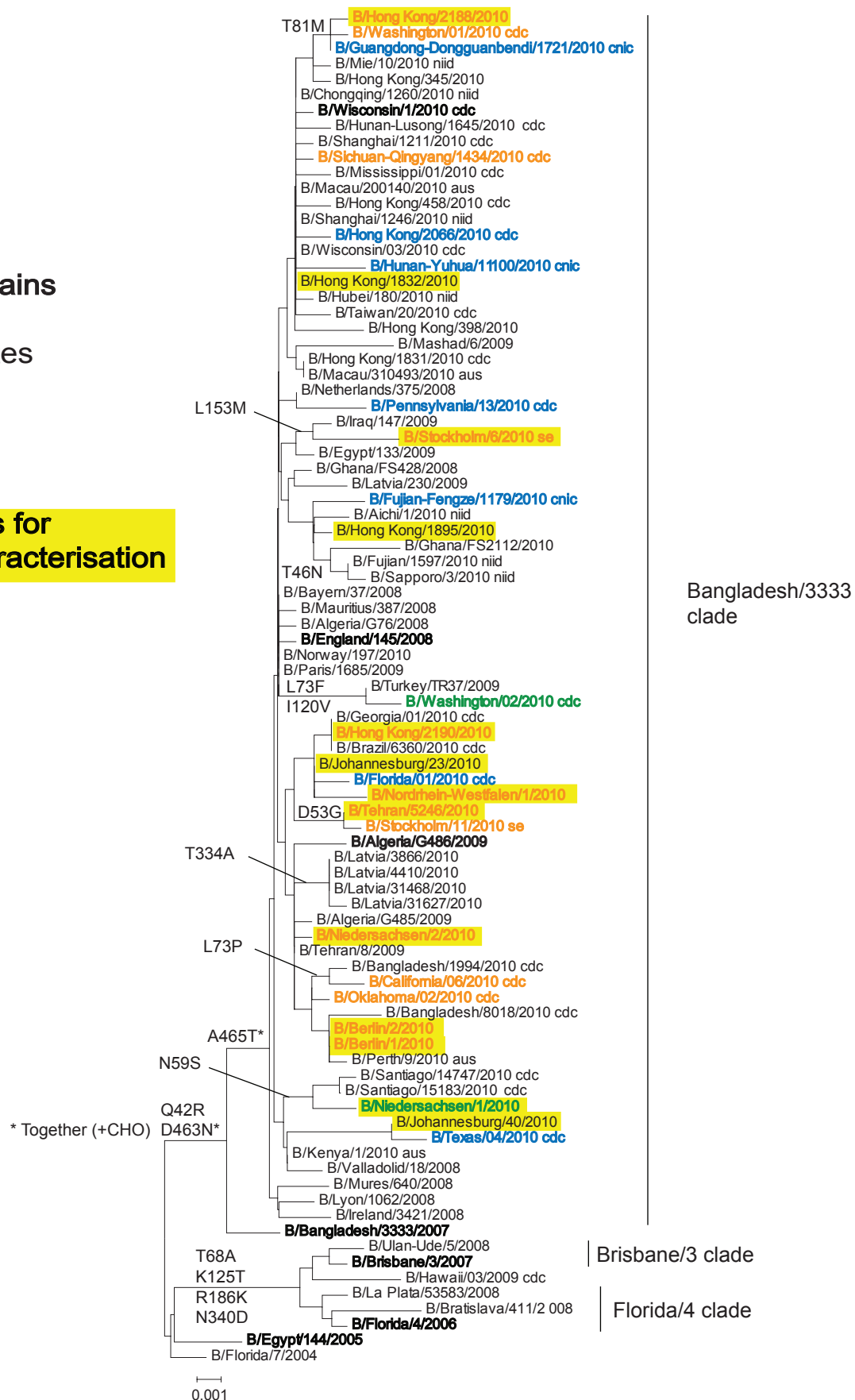
Collection dates

Sep 2010

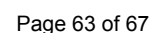
Oct 2010

Nov 2010

See HI Tables for
 antigenic characterisation



Potential N-linked glycosylation sites



	210	220	230	240	250	260	270	280	290	300
B/Florida/7/2004	LLKIKYGEAY	TDITYHSYAKN	ILRTQESACNC	IGGDCYLMIT	DGPASGIS	ECRFLKIREG	RIKEIFPTGR	VKHTEECTG	FASNKTIE	CACRDN
B/Florida/4/2006										SYTAKR
B/Egypt/144/2005										
B/Brisbane/3/2007					V					
B/Bangladesh/3333/2007										
B/Niedersachsen/1/2010										
B/Johannesburg/40/2010										
B/Hong Kong/2190/2010										
B/Nordrhein Westfalen/1/2010										
B/Johannesburg/23/2010										
B/England/145/2008										
B/Algeria/G486/2009										
B/Niedersachsen/2/2010										
B/Latvia/31627/2010										
B/Tehran/5246/2010										
B/Berlin/1/2010										
B/Oklahoma/02/2010										
B/Washington/02/2010										
B/Stockholm/6/2010										
B/Ghana/FS2112/2010										
B/Hong Kong/1895/2010										
B/Hong Kong/2066/2010										
B/Wisconsin/1/2010										
B/Taiwan/20/2010										
B/Sichuan-Qingyang/1434/2010										
B/Hong Kong/1832/2010										
B/Macau/200140/2010										
B/Hong Kong/2188/2010										
B/Washington/01/2010										
	310	320	330	340	350	360	370	380	390	400
B/Florida/7/2004	PFVKLN	VETDTAEIR	LMCTETYL	DTPRPNDG	SITGPCE	SNKGSGG	IKGGFVH	QRMASKI	GRWYSRT	MSKTKRM
B/Florida/4/2006					D					
B/Egypt/144/2005								R		
B/Brisbane/3/2007				D						
B/Bangladesh/3333/2007										
B/Niedersachsen/1/2010										
B/Johannesburg/40/2010										
B/Hong Kong/2190/2010										
B/Nordrhein-Westfalen/1/2010										
B/Johannesburg/23/2010										
B/England/145/2008										
B/Algeria/G486/2009										
B/Niedersachsen/2/2010										
B/Latvia/31627/2010			A							
B/Tehran/5246/2010										
B/Berlin/1/2010										
B/Oklahoma/02/2010										
B/Washington/02/2010					W					
B/Stockholm/6/2010										
B/Ghana/FS2112/2010										
B/Hong Kong/1895/2010										
B/Hong Kong/2066/2010									D	
B/Wisconsin/1/2010										
B/Taiwan/20/2010										
B/Sichuan-Qingyang/1434/2010										
B/Hong Kong/1832/2010										
B/Macau/200140/2010										
B/Hong Kong/2188/2010										
B/Washington/01/2010										
	410	420	430	440	450	460				
B/Florida/7/2004	VSMEEP	GWYSFG	FEIKDKK	CDVPCIGI	EMVHDG	GKTTW	HSAATAI	YCLMGS	QLLDW	TVTG
B/Florida/4/2006										
B/Egypt/144/2005										
B/Brisbane/3/2007										
B/Bangladesh/3333/2007										
B/Niedersachsen/1/2010									N	T
B/Johannesburg/40/2010									N	T
B/Hong Kong/2190/2010									N	T
B/Nordrhein-Westfalen/1/2010									N	T
B/Johannesburg/23/2010									N	T
B/England/145/2008									N	T
B/Algeria/G486/2009									N	T
B/Niedersachsen/2/2010									N	T
B/Latvia/31627/2010									N	T
B/Tehran/5246/2010									N	T
B/Berlin/1/2010									N	T
B/Oklahoma/02/2010									N	T
B/Washington/02/2010									N	T
B/Stockholm/6/2010									N	T
B/Ghana/FS2112/2010									N	T
B/Hong Kong/1895/2010									N	T
B/Hong Kong/2066/2010									N	T
B/Wisconsin/1/2010									N	T
B/Taiwan/20/2010									N	T
B/Sichuan-Qingyang/1434/2010									N	T
B/Hong Kong/1832/2010									N	T
B/Macau/200140/2010									N	T
B/Hong Kong/2188/2010									N	T
B/Washington/01/2010									N	T

Table 25. Influenza antiviral testing: September 2010 to present

WHO Regional Office	Country	Influenza type		
		No. of H1N1pdm tested	No. of H3N2 tested	No. of Flu B tested
AFRO	Cameroon	23	2	26
	Egypt	4		
	Ghana	33	8	
	La Reunion (France)	4		
	Madagascar	2	2	
	Niger (France)	4		
	South Africa	7	2	7
	Senegal	12		
	Tunisia	3		
AMRO	Argentina		11	7
EMRO	Iran	5	1	
EURO	Serbia	21(2)*		
	Switzerland			1
(ECDC)	Austria	32		
	Finland	1		
	France	1		
	Germany	8		
	Italy	3		
	Latvia	3		
	Netherlands			1
	Ireland		1	
	Slovakia			1
	Spain	17		
WPRO	Hong Kong	2		
Viruses sent to HPA		1	61	118
Total Tested		186	88	161

*The number in brackets indicates outlier viruses. The two viruses from Serbia were sensitive to oseltamivir but had reduced sensitivity to zanamivir (IC50's of 6.8nM & 3.7nM) which correlated with G147R substitution in the neuraminidases. Routinely, H1N1pdm viruses sensitive to drugs have IC50's of <2.0nM for oseltamivir and <1.0nM with zanamivir.

Figure 26

**Phylogenetic comparison of European swine influenza A(H1N1) HA genes
(in relation to recent human infections)**

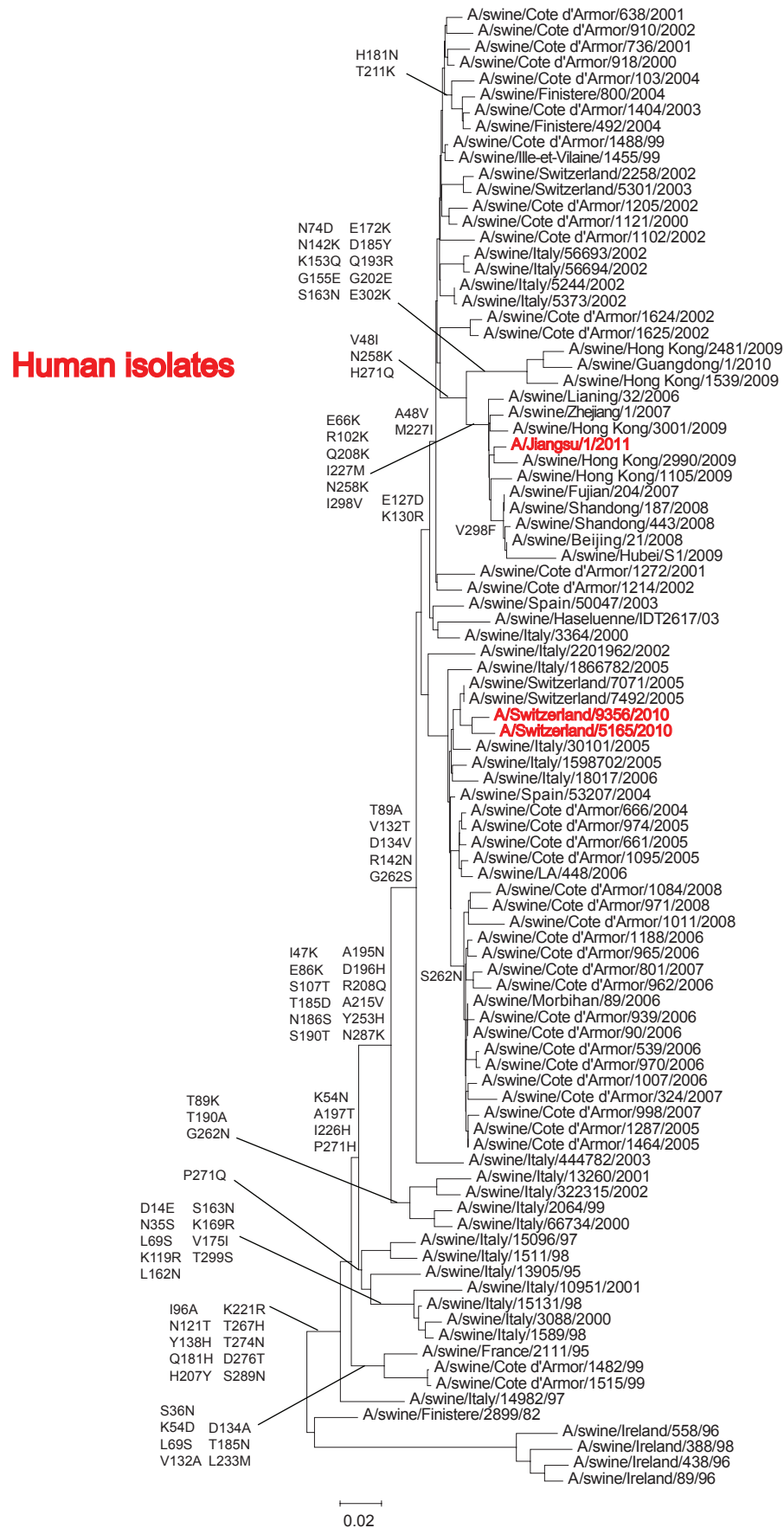


Table 26. Antigenic analyses of swine influenza A(H1N1) viruses

			Haemagglutination inhibition titre ¹							
			Post infection ferret sera							
Viruses	Collection Date	Passage History	sw/Fin/ ² 2899/82 R10/82	sw/Fin/ 2899/82 F1/97	sw/Ireland/ 89/96 F2/96	sw/Italy/ 1513/98 F22/98	sw/Italy/ 1533/99 F19/99	sw/IV/ 1455/99 F35/99	sw/Switz/ 7492/05 F35/05	sw/CA/ 324/07 F14/08
REFERENCE VIRUSES										
A/swine/Finisterre/2899/82		Ex	≥5120	640	160	<	<	40	<	160
A/swine/Ireland/89/96		Ex	≥5120	640	1280	40	<	40	40	80
A/swine/Italy/1513-1/98		Ex	≥5120	1280	320	160	320	80	<	160
A/swine/Italy/1533/99		Ex	≥5120	640	80	80	640	<	<	<
A/swine/IV/1455/99		Ex	1280	80	<	<	<	320	<	<
A/swine/Switzerland/7070/2005		Ex	≥5120	320	<	<	<	<	80	<
A/swine/CA/324/2007		Ex	≥5120	640	160	40	<	40	<	640
TEST VIRUS										
A/Switzerland/3885165/2010	2010-10-21	Cx/MDCK1	≥5120	1280	320	80	40	80	80	320
A/Switzerland/2649356/2009	2009-12-13	Cx/MDCK1	≥5120	1280	320	40	40	80	80	320

1. < = 40

2. Rabbit hyperimmune serum